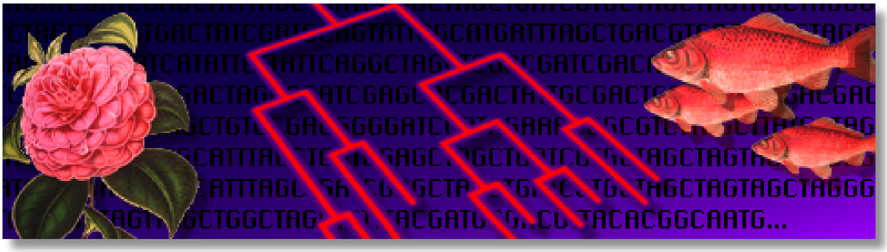
Diversity Database™



User's Guide for Version 1.1

PC and Macintosh

Catalog Number 170-7921

Software Tools for all Types of Fragment Analysis and Databasing

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P/N 4000121 RevA

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Table of Contents

Preface

1. About This Document

Throughout this document, we use certain fonts to indicate specific meanings:

This Font indicates the name of a menu or function.

This font indicates text found in dialog boxes or the names of dialog boxes.

`This font` indicates an entry that you type.

This font indicates the name of a *pdi* software product.

The following chapters contain instructions for scanning and analyzing X-ray films, wet and dry gels, blots and photographs. For the sake of simplicity, the aforementioned are all referred to as “gels.”

Whenever a function or a submenu is mentioned in this manual the full menu path is frequently given as well. The submenu name is preceded by another menu name and an arrow (i.e., menu → submenu). For example, **File** → **Export** → **TIFF Image** indicates that the **TIFF Image** function is located in the **Export** submenu which is found under the **File** menu.

Some of the illustrations of menus and dialog boxes found in this manual are taken from the PC version of the software. Some are taken from the Macintosh® version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

1. Introduction

1.1. A Brief Introduction to Diversity Database

Diversity Database is a software package for the analysis and databasing of 1-D electrophoresis gels, dot blots, and slot blots.

The software runs in a DOS™ or Macintosh environment. A graphical interface with function palettes, dialog boxes, mouse-assignable, and keyboard functions facilitates operation of the software.

Diversity Database can quantitate 1-D electrophoretic images, TLC plates, blotting membranes, and any other object in biology that can be scanned with a densitometer.

As a gel is scanned, its image will appear on your computer monitor. Subsequent operations are performed via commands available from the menus.

The software can be used to determine molecular weights and isoelectric points. It can also make quantitative measurements such as integrated volumes, relative amounts, and actual amounts of protein (in milligrams).

The software can cope with distortions in both lane and band shape. Lanes can be adjusted along their lengths to compensate for any curvature or smiling of the gel. Irregularly shaped bands can be accurately quantitated with the use of **Diversity Database**'s **Contour** features.

Data captured by the scanner is stored in a file under a user-defined name. Files can be transferred into or exported out of **The Discovery Series** software. Scans can be converted into TIFF format for easy compatibility with other Macintosh and PC applications.

1.2. About Gel Quality

Diversity Database is very tolerant of an assortment of electrophoretic artifacts. Lanes need not be perfectly straight or parallel. Bands need not be perfectly resolved.

However, for accurate quantitation, we suggest that bands be reasonably flat and horizontal. Quantitation involves calculating the average density of pixels across the band width and integrating over the band height. For the automatic band finder to function optimally, bands should be well resolved.

Dots that appear as halos, rings or craters, or that are of unequal diameter may be incorrectly quantified using the automatic functions.

1.3. Steps Involved in Diversity Database

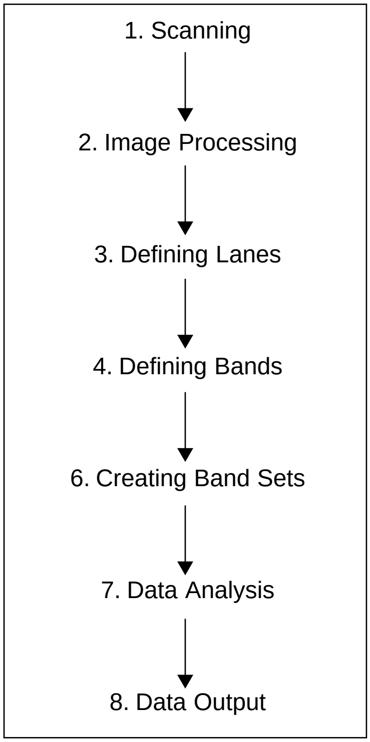


Fig.1-1. Steps in the Analysis of 1-D Gels.

Scanning

Before you can use **Diversity Database** to analyze a gel, you need to capture its image into your computer. Typically, this is done with a scanning densitometer. A densitometer can be used to scan films, wet gels or blots. The resulting images are stored on your hard disk.

Image Processing

Once you have an image of your gel with which to work, you may need to reduce any noise or background density caused by film fogging or the opacity of your carrier medium (film, gel matrix, or blot matrix). A variety of functions exist to minimize background density while maintaining data integrity.

Defining Lanes

Once a “clean” image is available, you will need to specify the position of the lanes on your gel. Lanes can be placed singly or all at once by creating a matrix of lanes over the gel image.

Defining Bands

Once lanes have been positioned, bands can be defined and quantitated. Both automatic and manual methods are available. Automatic band detection places brackets around each band it locates. The manual tools allow you to create new bands or edit existing ones.

Assigning Band Sets

Once bands have been defined for your gel, you will need to add the gel to a database and assign it a Band Set. Band Sets combine traditional allele calling with positional standards (usually molecular weight) to create a standard of comparison that can be applied to all the similarly detected gels in your database.

Data Analysis

Once your gel has a Band Set and is a member of a database file, a variety of searching and population comparison tools exist to provide detailed analysis of your gel. **Diversity Database** supports both single- and multi-lane sample definition as well as phylogenetic tree and 3-D principle component analysis display.

Data Output

When your analysis is complete, flexible reporting features allow you to generate customized reports that can be sent to a PostScript® printer or exported to another system for further analysis.

1.4. Installing Diversity Database on a PC

1.4.a. System Requirements

To run **Diversity Database** on a PC, you'll need the following:

Architecture:	ISA or EISA bus (not IBM microchannel)
Processor:	486 DX 33 Mhz or higher
RAM Memory:	16 MB Minimum
PC OS:	DOS 6.2 or higher
Adapter & Display:	SVGA: 800 x 600 x 8-bit resolution or better
Monitor:	15" monitor, 256 colors
Mouse:	2 or 3 button
Printer:	PostScript compatible printer

Alternate video boards include: Tseng 4000 (Diamond Speedstar), 8514 (ATI Graphics Ultra), S3 (#9 GXE), VESA (#9 GXE64pro, ATI VGA Wonder XL).

1.4.b. How to Install Diversity Database

See Release Notes for installation instructions.

1.4.c. Where to Install Diversity Database

The default directory for **Diversity Database** is **`c:\pdi`**. By following the installation instructions you will create the **`c:\pdi`** directory on your hard drive and automatically update your **autoexec.bat** file to include the new directory location. If you choose to install **Diversity Database** in a location other than the default directory, simply type in the name of the new directory when prompted to do so by the installation program. The new directory will be installed on your hard drive and its location will be added to the **path** statement of your **autoexec.bat** file.

1.4.d. Starting Diversity Database

Next to the `c:\` prompt, type: `dvdb  <return>` to start **Diversity Database**.

1.5. Installing Diversity Database on a Mac

1.5.a. System Requirements

To run **Diversity Database** on a Macintosh, you'll need the following:

Processor:	68040 or any PowerMac
RAM Memory:	16 MB Minimum
Hard Disk:	3-4 MB to install
Mac OS:	System 7.5 or later
Monitor:	16" color monitor (832 x 624, 256 colors)
Printer:	PostScript printer

Note: Systems running virtual memory are *not* currently supported. We suggest that you turn virtual memory *off* while running **Diversity Database**.

1.5.b. How to Install Diversity Database

See Release Notes for installation instructions.

1.5.c. HASP Protection Key

A HASP protection key has been provided with your system. You will need to insert the key into your Apple Desktop Bus (ADB) path prior to operation of **Diversity Database**.

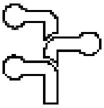


Fig.1-2. Apple Desktop Bus icon.

The HASP key can be inserted at any point in the ADB path (e.g., between the keyboard and mouse, between the keyboard and monitor, on the back of the computer, etc.) depending on the specific configuration of your Macintosh.

1.5.d. To Start Diversity Database

To begin operation, double-click on the **Diversity Database** application icon inside the **Diversity Database** folder.

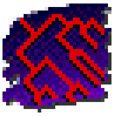


Fig.1-3. **Diversity Database** application icon.

2. General Operation

2.1. The Graphical Interface

2.1.a. Menus

The menus in **Diversity Database** contain several unique elements that provide extra functionality to the software. Each element is represented in the menu by its own icon. The three icons are listed below:

Pop-Menu

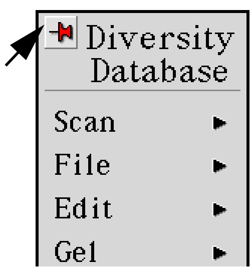
Mouse-Assignable

Function Palette

Fig.2-1. Unique menu icons.

The Pop-Menu Icon (PC Only)

Clicking on the Pop-menu icon in the upper left corner of a PC menu will cause a floating version of that menu to appear on-screen. Once displayed, floating menu boxes can be moved to any location and will not disappear until the Close button in the upper left corner of the menu is pressed. Floating menu boxes function identically to stacking menus.



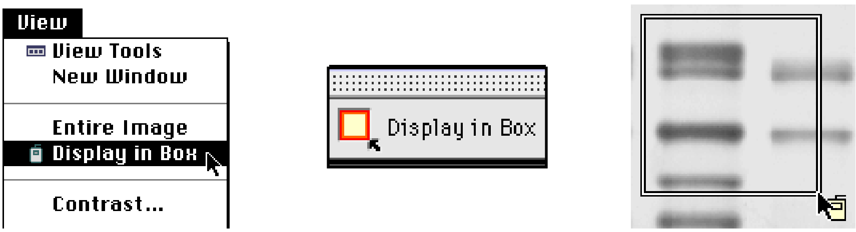
Click on the push-pin icon.

Pop-up menu box will appear.

Fig.2-2. How to access pop-up menus.

The Mouse-Assignable Icon

Mouse-assignable functions are designated in the menus with a mouse-assignable icon (see Figure 2-1, above). A mouse-assignable function is a command that can be assigned to the mouse and executed by clicking the mouse in a particular window or on a particular region of a image. Once assigned, the function’s icon will appear in the status bar at the top of the screen. A good example of a mouse-assignable command is **Display in Box** (in the **View** menu).



Select mouse-assignable command from the menu.   Icon appears in status bar.   Command is executed in gel.

Fig.2-3. How to use mouse-assignable commands.

Function Palettes

Diversity Database supports the use of function palettes. Function palettes are designated in the menus with a function palette icon (see Figure 2-1, above). Function palettes are similar to the “tool bars” commonly available in other software packages. **Diversity Database’s** palettes contain groups of associated commands that are used in gel analysis. Clicking on an icon from a function palette will either perform the command immediately or assign the function to the mouse.

Each palette can be toggled between a vertical and horizontal format by clicking on the resize button in the upper right corner of the palette. If you allow the mouse to linger over a palette icon, the name of the icon will appear, and a description of the command will appear in the status bar. This utility is called “Tool Help.” Tool Help works on a time delay basis that can be specified on the **Miscellaneous Preferences** form (see Section 2.3.d). First-time users may wish to specify a short delay (< 1 sec.)

in order to learn which functions are available on the palettes. Experienced users may prefer to specify a longer delay (> 5 sec.) once they are familiar with the icons.

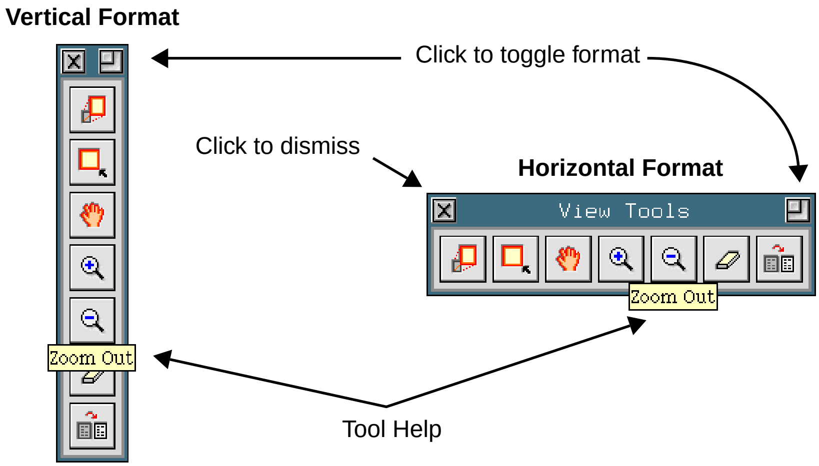


Fig.2-4. The **View** palette.

2.1.b. Dialog Boxes & Windows (PC Only)

The PC version of **Diversity Database** supports an extra feature in its dialog boxes and image windows: a Back button.

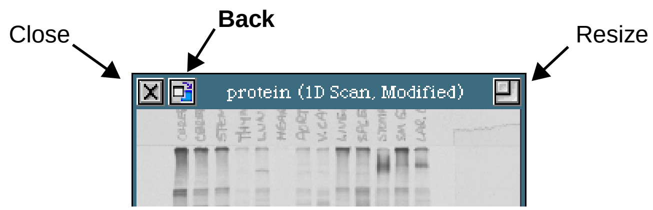


Fig.2-5. Window control buttons.

The Back button allows the user to move the current window or dialog box behind all other open windows. This is often useful when analyzing large numbers of gels. It enables the user to temporarily ignore a gel or dialog box without needing to close it. If a gel or dialog box is obscuring your view, simply send it to the back.

2.1.c. On-Line Help

Diversity Database provides several levels of Help features to answer the most common questions regarding the proper usage and application of functions.

On a PC

Answers to the most common questions regarding the proper usage and application of functions can be found in the **Help** files. To see a **Help** file, select the function in question with the cursor and press <**Control**> <**H**>. Note: using <**Control**> <**H**> in an image window will call the **Get Info** function (see below).

On a Mac

Frequently useful is **Diversity Databases**'s Balloon Help, which can be accessed through the **Guide** menu (represented by a '**?**' symbol in the upper right corner of your screen).

On Both Platforms

The **Get Info** function is available on both platforms. This feature allows you to enter a description and/or comments regarding your scan and to display information such as file name and description, scan date, scan color, memory size, etc. **Get Info** can be called from the keyboard with a number of different key combinations (<**Control**> <**H**> or <**Alt**> <**I**> on a PC; <**Control**> <**H**> or <> <**I**> on a Mac) or from the **File** menu.

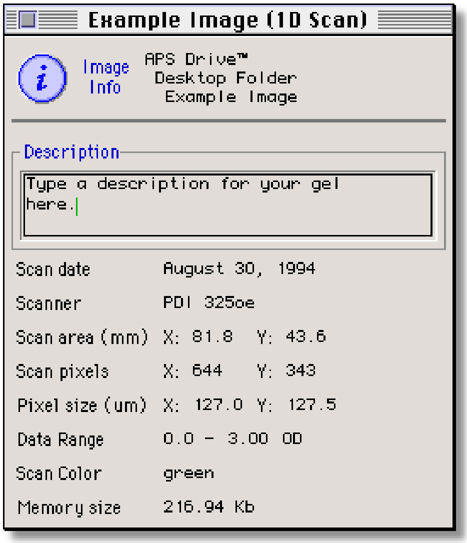


Fig.2-6. The **Get Info** command.

2.2. File Features

2.2.a. PC File Commands

A working knowledge of these functions is fundamental to the smooth and efficient operation of **Diversity Database**. This chapter contains brief descriptions and explanations of each of the **File** features in the order in which they appear in the menu.

Fig.2-7. The **File** Menu.

The New Database Command

The **New Database** command creates a new database file and displays a new **Database Summary** form on-screen. The new database file will be unsaved and will not contain any gels (see Section 7 for more information on database files).

Opening a File

Use the **Open** command to recall the image of a gel that was captured and saved during a previous session, or a previously saved database. Selecting **Open** from the menu will display the following form:

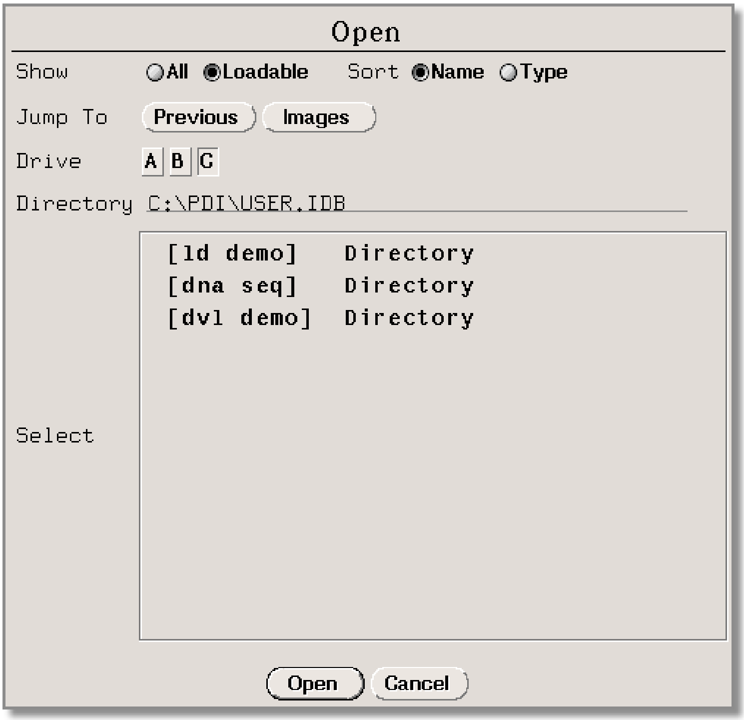


Fig.2-8. **Open** dialog box.

The current directory is listed next to the Directory prompt. Preceding it is the path that identifies its location within the directory "tree."

In the Select field is a list of the contents of that directory. Both file name and type are listed.

If the images you want to load are stored in the Image directory, click on the Images button.

If the images you want to load are in a subdirectory of your current directory, you can access that subdirectory by clicking on its name from the list. Once you have accessed the directory that contains your images, you can begin to load them.

To select a file in a different drive, click on the letter of the drive next to the Drive prompt.

Once you have selected all the images to be loaded from a directory, click on the Open button at the bottom of the dialog box.

Opening TIFF Images

The **Open** command can also be used to import TIFF images from other software applications. TIFF files can be loaded as if they were files generated by **The Discovery Series** software.

There are many types of TIFF formats that exist on the market. Not all are supported by **The Discovery Series**. There are two broad categories of TIFF files that are supported:

1: **8 Bit Grayscale:** Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with **The Discovery Series** software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard™, Microtek™, and Sharp™ each make scanners that produce compatible 8 bit grayscale images.

2: **16 Bit Grayscale:** Molecular Dynamics™ scanners use 16 bit pixel values and custom formatting to describe intensity of scale. **The Discovery Series** correctly understands these formats and can interpret images from Molecular Dynamics scanners.

Note: **Diversity Database** is capable of importing 8 and 16 bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1: **1 Bit Line Art:** This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.

2: **24 Bit Full Color**: This format is frequently used for retouching photographs, and is currently unsupported in **The Discovery Series**, although most scanners that are capable of producing 24 bit color images will be able to produce grayscale scans as well.

3: **Compressed Files:** The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with **The Discovery Series** should be formatted with the compression option turned off.

Closing an Image

The **Close** function removes a displayed file from active use. If you have made changes to the file since the last time you saved it, closing it will cause a pop-up box to appear, allowing you to save the file before unloading, to unload without saving, or to cancel out of the operation. If you unload a file that has never been saved, that file will effectively be deleted. If you unload a previously saved file without saving changes, those changes will be lost and the file will revert to its last saved version.

The Close All Function

The **Close All** function unloads every file currently loaded. A pop-up box will appear for each file, prior to the execution of the function, allowing you to save the files before unloading, or to cancel out of the operation.

The Save Function

This feature allows you to save a new file that you scanned or an old file to which you made changes.

The Save As Command

This function displays a dialog box that allows you to change the file name of an image. You will also be given the option of entering a new description and version number.

Renaming a file to a new directory is equivalent to moving that file within your directory system, because a file's location is encoded in its path name (e.g., **`c:\pdi\user.idb\1d_scan`**).

The Save All Function

The **Save All** function saves every file currently loaded. **Save All** is a useful tool for ensuring that *all* your scans, even those that you are not currently working on, are saved.

The Delete Command

To permanently remove a file from your computer, assign **Delete** to a mouse button and click on the image of the file that you want to erase. This function will only work on a file that is currently open.

When you click on the image that you wish to delete, a pop-up box will appear, confirming that the file you selected is the file you wish to delete.

Click on the Cancel button to escape the operation.

The Delete File Command

Use **Delete File** to remove a file that is not currently loaded. **Delete File** is intended to be used to remove files from your disk if there is insufficient room for **Diversity Database** to save an open image.

Delete File will not remove an open image.

When you select this function, the following dialog box will be displayed:

Fig.2-9. **Delete File** dialog box.

The directory currently being accessed is listed at the end of the text next to the Directory prompt. Preceding it is the directory path leading to this directory. Beneath the name of the directory is a list of its contents.

Access the directory containing the files you want to delete by clicking on the appropriate button next to the Jump To prompt. The text next to the directory prompt will change to indicate the directory that you are currently accessing. You can delete some, or all of the files in a directory.

To select a file in a different drive, click on the letter of the drive next to the Drive prompt.

Once you have selected all the files you wish to delete, click on the Delete Selected Files button at the bottom of the dialog box.

If you decide not to delete files at this time, you can dismiss the dialog box by clicking on the Cancel button.

Disk Space

To find out how much space is available to store image files, use the function **Disk Space**.

Selecting this function displays a dialog box which lists the file systems and the directories found in them. The amount of space available is given for each file system. Removing files from any one of the directories in a file system creates free space that can be used by any or all of the directories in that file system.

Fig.2-10. **Disk Space** pop-up box.

2.2.b. Mac File Commands

A working knowledge of these functions is fundamental to the smooth and efficient operation of **Diversity Database**. This chapter contains brief descriptions and explanations of each of the **File** features in the order in which they appear in the menu.

Fig.2-11. **File** menu.

The New Database Command

The **New Database** command creates a new database file and displays a new **Database Summary** form on-screen. The new database file will be unsaved and will not contain any gels (see Section 7 for more information on database files).

Opening an Image

If you want to examine the image of a gel that was scanned during a previous session and the gel is not currently loaded, you need to recall it from the disk.

You can do this using the **Open** function.

If your file has been copied from a PC or Sun and is not visible to the **Open** dialog, check the Show All Files box to display all the files on your disk, including non-**Discovery Series** typed files. Opening a non-Macintosh **Diversity Database** file with this option will change the file document type so that it will thereafter be visible to the **Open** dialog.

Opening TIFF Images

The **Open** feature can also be used to import TIFF images from other software applications. TIFF files can be loaded as if they were files generated by **The Discovery Series** software.

TIFF images do not contain all of the information that would normally be included in a **Discovery Series** image file (e.g., scanner type, date, color, etc.). For this reason, scan information features such as **Scan Report** and **Help** will not display all such tagged data from TIFF files.

There are many types of TIFF formats that exist on the market. Not all the formats that exist are supported by **The Discovery Series** (see Section 2.2.a for more information on supported TIFF formats).

The Close Function

The **Close** function removes a displayed file from active use. If you have made changes to the file since the last time you saved it, closing will cause a pop-up box to appear, allowing you to save the file before unloading it, to unload without saving, or to cancel out of the operation. If you unload a file that has never been saved, that file will effectively be deleted. If you unload a previously saved file without saving changes, those changes will be lost and the file will revert to its last saved version.

The Close All Function

The **Close All** function unloads every file currently loaded. A pop-up box will appear for each file, prior to the execution of the function, allowing you to save the files before unloading, or to cancel out of the operation.

The Save Function

This feature allows you to save a new file that you scanned or an old file to which you made changes.

The Save As Function

This function allows you to save a file under a different name or in a different location on the hard drive.

The Save All Function

The **Save All** function saves every file currently loaded. **Save All** is a useful tool for ensuring that *all* your scans, even those that you are not currently working on, are saved.

2.3. Preferences

Mac Version **PC Version**

Fig.2-12. **Preferences** menus.

The **Preference** commands (in the **Edit** menu) access forms that control basic features of your system (i.e., displays, colors, printers, etc.), by allowing you to create default settings of your choice for various parameters that pertain to software functionality. Each of the available **Preference** commands is listed below:

2.3.a. Color Preferences (Mac & PC)

Selecting this command displays a dialog box in which you can specify the background color of the menus, windows, buttons, etc.

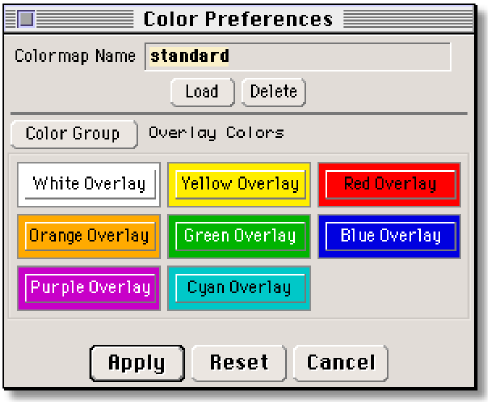


Fig.2-13. **Color Preferences** dialog box.

To specify a color, click on the color button of interest. A pop-up box will appear that allows you to adjust the RGB values of the color you selected. The color of the button will change accordingly with your adjustments.

When you find a set of colors that you like, you can save them by entering a file name for them and clicking on the Apply button.

To select a set of colors that you want to use, click on the Load button. From the list displayed, choose the set of colors that you wish to apply by clicking on the name of the file. Those colors will then be automatically applied to the menus, dialog boxes, buttons, etc.

To remove a colormap file, click on the Delete button. On the list displayed, click on the name of the file that should be removed from the list. A pop-up box will ask you to confirm that you want to remove it. Click on the Yes button to delete it. Click on No to cancel out of the Delete function.

The Color Group button allows you to select and change the colors of a particular group of functions. For example, you can select the colors used to describe pop-up boxes, image colors, etc., and adjust them to match your taste.

If you change your mind about applying *any* changes to a Color Preferences file, click on the Cancel button to dismiss the dialog box without altering the colormap. If you wish to return to the Standard colormap, click on the Reset button. All colors will be returned to their default values.

2.3.b. Display Preferences (Mac & PC)

In the Display Preferences dialog box, you can specify parameters related to the way **Diversity Database** displays its operations.

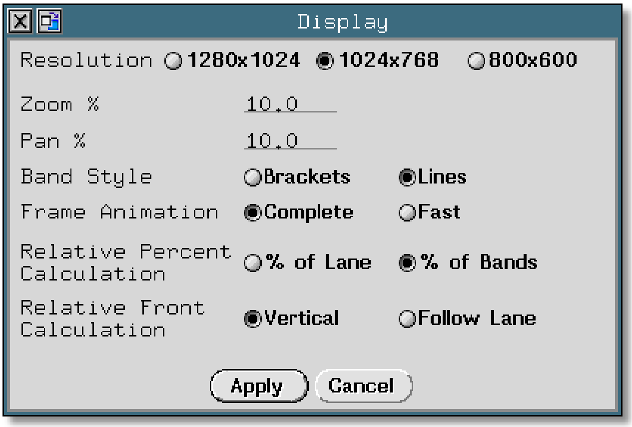


Fig.2-14. **Display Preferences** dialog box.

The Resolution prompt appears only on the PC version of **Diversity Database**. Select 1280x1024, 1024x768, or 800x600 depending on the resolution of your monitor.

The Zoom % sets the degree by which an image moves closer or further away when you use **Zoom** functions.

The Pan % determines the degree by which the image moves when you use **Pan** functions.

Bands on your gel can be designated in one of two ways: they can either be marked with brackets that define the top and bottom boundaries of the band, or they can be marked with a dash at the center of the band.

Indicate your preference by clicking on the Brackets or Lines button next to the Band Style prompt.

Frame Animation refers to the movement of the frame when you use **Adjust Lanes** to change the arrangement of a frame on your gel. Selecting Fast results in only the borders of the frame moving as you reposition the frame. This is faster but does not update the position of each lane until you stop moving the border of the frame. Clicking on Complete means that all the lanes will move as you reposition the frame border. Keep in mind that this process will require increasing amounts of time as the number of lanes on the gel increases.

The Relative Percent Calculation option allows the user to define how Relative % will be determined for all histograms, reports, and band information functions in **Diversity Database**: either as a percentage of the optical density of an entire lane, or as a percentage of the optical density of the defined bands in a lane.

Selecting % of Lane means that the total lane density will equal 100% and the band that you select will be reported as a fraction thereof. Selecting % of Bands means that the sum of the density of the defined bands in a lane will equal 100%, and that your band will be reported as a fraction of that sum. If you create, adjust, or remove bands with Relative % defined as % of Bands, the Relative % of all the other bands in the lane will update to reflect changes in the total density of defined bands.

The Relative Front Calculation option allows the user to define how relative fronts (Rf) will be calculated in **Diversity Database**. Rf is calculated by dividing the distance a band has migrated from the gel's origin by the total distance between the origin and the dye front. These distances can be determined in one of two ways: (**1**) you can drop a vertical line from the origin to the dye front (Vertical Projection) or (**2**) you can measure along the length of the lane (Follow Lane). If a lane is straight and vertical, these methods will give identical results. However, results will differ if a lane is curved or slanted. Choose your preferred method by clicking on one of the two buttons next to the Relative Front Calculation prompt.

2.3.c. File Paths (PC Only)

The **Paths Preferences** form lets you set the default paths (location on the hard disk) for the Images, Local, System, Temporary, and Log directories.

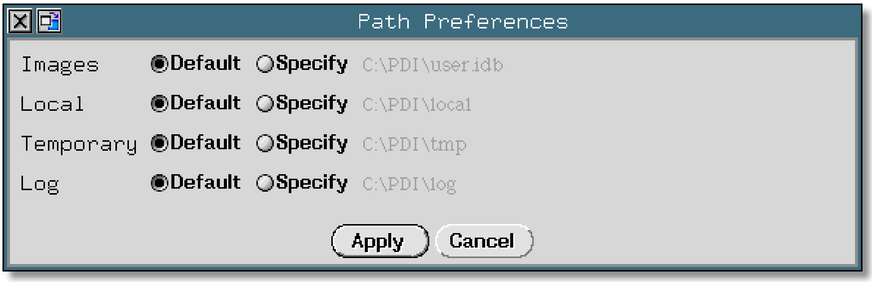


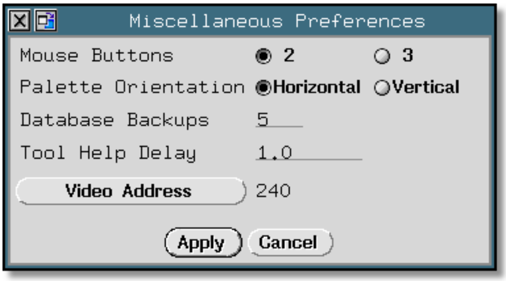
Fig.2-15. **Paths Preferences** dialog box.

To change a default path, click on the Specify button and type the desired path in the text box.

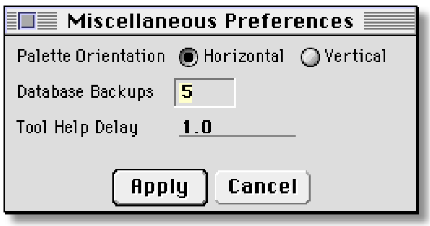
If you change the Images path, you will initially need to click on the Images button again on the **Open** dialog box to activate the new directory path.

2.3.d. Miscellaneous Preferences (Mac & PC)

The following items can be specified in the **Miscellaneous Preferences** dialog box:



PC Version



Mac Version

Fig.2-16. **Miscellaneous Preferences** dialog box.

Next to the Mouse Buttons prompt you can select between the 2 or 3 button mouse settings (based on your mouse; PC version only).

The Palette Orientation option allows you to specify whether function palettes appear in their vertical or horizontal format the first time they are displayed in a session.

The Database Backups option allows you to specify the number of backup database files (0-9 copies possible) that will be maintained in your Backup directory. Each time you save a database file, a backup will be generated and stored on disk. Backups allow you to recover from mistakes or inadvertent changes to your database file. Specifying a Database Backups value of zero means that no backup copies will be maintained. We recommend that you maintain at least 3 backup copies during normal software operation.

The Tool Help Delay option allows you to specify the amount of time that the cursor must linger over an icon before calling Tool Help. First-time users may wish to specify a short delay (< 1 sec.) in order to learn which functions are available on the palettes. Experienced users may prefer to specify a longer delay (> 5 sec.) once they are familiar with the icons.

The Video Address button (PC version only) allows you to change the video address in the software for the video board included with the **NightHawk** imaging system. This feature allows users to resolve any addressing conflicts that might arise between the **NightHawk** system and other peripheral devices operating on the same PC.

2.3.e. Print Preferences (PC Only)

The **Printer Preferences** box allows you to alter various parameters pertaining to your printer.

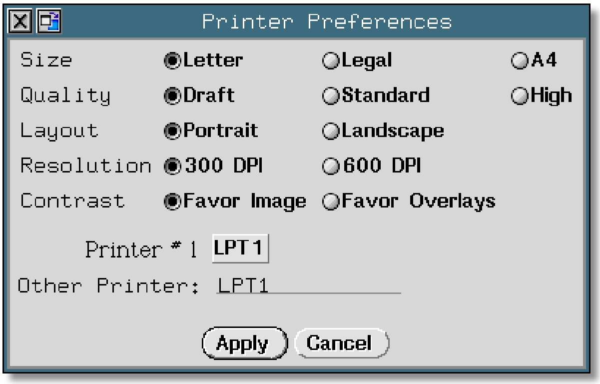


Fig.2-17. **Printer Preferences** dialog box.

Indicate the size of the paper in the printer by clicking on either the Letter, Legal or A4 button next to the Size prompt.

Quality determines how much fine detail can be seen in a printed image. Draft produces an image in less time but with lower quality. High produces prints with better contrast and detail at the expense of extra printing time. Standard produces an intermediate quality image and requires an intermediate print time. Indicate your preference by clicking on the appropriate button next to the Quality prompt.

Select the orientation of the paper by clicking on either the Portrait or Landscape button next to the Layout prompt.

You can specify the resolution at which you wish to print by clicking on either the 300 DPI or 600 DPI button next to the Resolution prompt.

If, when printing images, you do not want to display overlays as prominently, click on the Favor Image button next to the Contrast prompt. This allows the full range (black to white) to be used to print the image and will generate the clearest, sharpest image possible. If you do want to display enhanced overlays on the printout of an image, click on the Favor Overlays button. This will optimize the use of the printing range so you can read overlays even on the darkest regions of the image.

Specify the name of the printer(s) accessible to your computer by clicking on the button next to the Printer prompt. If there is a printer name that does not appear on the button, type it in the text box next to the Other Printer prompt.

After entering all the information in the **Printer Preferences** dialog box, click on the Apply button.

If you change your mind about editing the **Printer Preferences**, you can press Cancel to dismiss the form without applying any changes.

2.3.f. Scan Preferences (Mac & PC)

Before you scan a gel, the software allows you to specify certain parameters related to scanner performance. You can enter this information in the **Scan Preferences** dialog box.

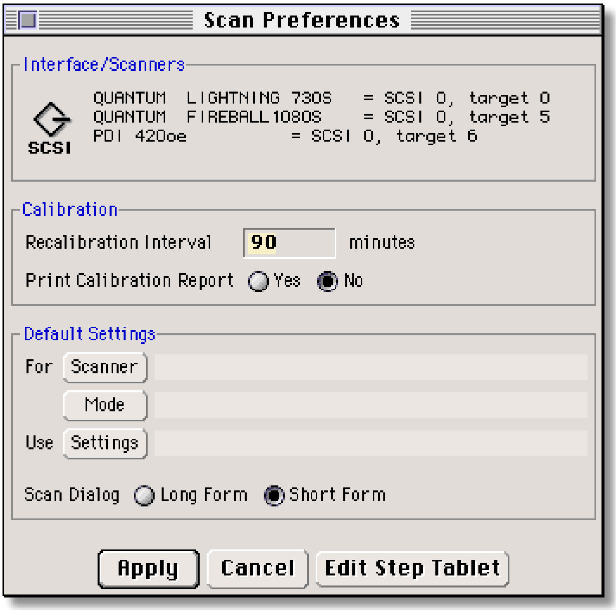


Fig.2-18. **Scan Preferences** dialog box.

Under the Interface/Scanners heading at the top of the form, you will see a list of all the devices currently connected to your computer. SCSI target addresses are printed next to the name of each device, and you can use this information to determine available SCSI addresses before connecting your scanner to your system (and thereby avoid potential SCSI conflicts) by opening the **Scan Preferences** dialog box *before* turning on the power to your scanner.

Under the Calibration header, the Recalibration Interval option allows you to specify the number of minutes, from 0 to 99,999, that will pass before an OE scanner automatically recalibrates. Specifying an interval of 0 minutes will cause a scanner to auto-calibrate every time you scan. If you plan to scan several gels over the course of a day, 90 or 120 minute recalibration intervals may be appropriate. See Section 3.1.a for more information on *Recalibration Intervals*.

The Print Calibration Report option is intended for use in labs where it is necessary to document the accuracy of the scanner each time that it is calibrated. If you select Yes, you will be given the option of generating a report every time you calibrate. If you do *not* wish to document each calibration, we suggest that you leave this option turned *off*.

Under the Default Settings heading are two options that allow you to save Scan Parameter settings for use with specific scanner models.

Clicking on the Scanner button accesses a list of all the scanner models with which **Diversity Database** is compatible. Select the model of scanner that you'll be using in future scanning sessions.

The Mode button allows you to change the scan mode once a scanner is selected. Scanner settings files are mode as well as scanner specific.

Clicking on the Settings button brings up a list of all the available Scan Settings files. Select the Settings file that you want to associate with a particular type, or mode, of scanner. In the future, when you scan with that model and mode of scanner, the Scan Parameter settings that you indicated will automatically be applied to the **Scanner Settings** dialog box when you begin scanning (see Section 3.3.a for information on how to save Scan Settings.)

There are two forms of the **Scanner Settings** dialog box. The "Long Form" displays all the scanning parameters that can be adjusted. The "Short Form" displays only the most frequently adjusted scanning parameters. Specify which of these two forms you prefer to display by default by clicking on either the Long Form or the Short Form button next to the Scan Dialog prompt.

If you are satisfied with the changes you have made and want to save them, click on the Apply button. The changes you made will be recorded and the **Scan Preferences** dialog box will disappear.

2.3.g. Enter Passwords

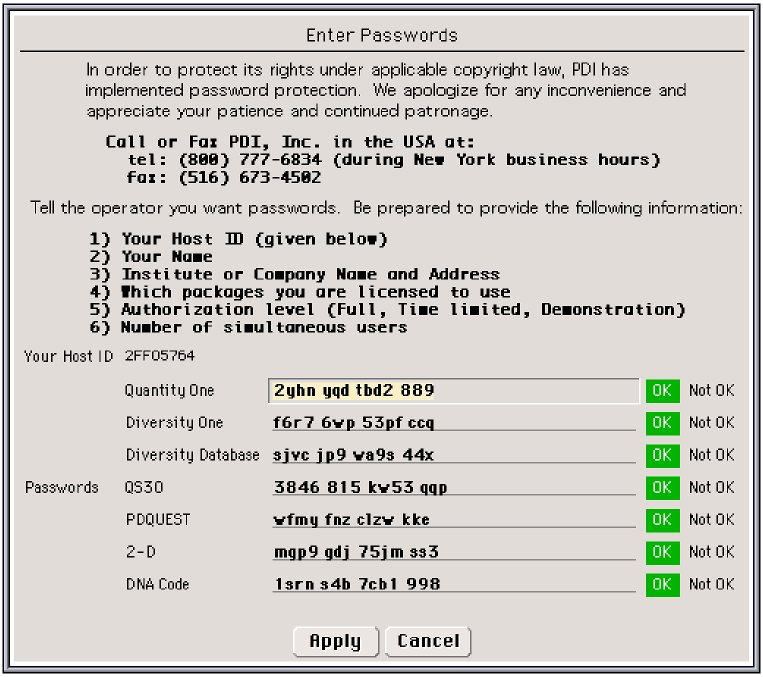


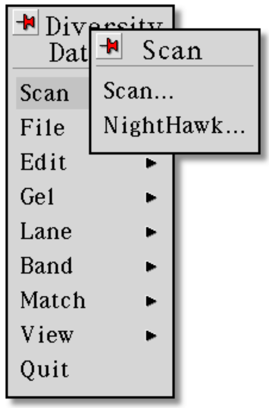
Fig.2-19. **Enter Passwords** dialog box.

The **Enter Passwords** form allows you to review or edit the passwords for all **Discovery Series** applications currently loaded on your machine. If you change the password for an application, that application will *not* run the next time you launch it unless you re-enter the correct passwords. Because of this, we strongly recommend that you do not change your passwords once they have been entered.

3. Scanning

This section describes a basic protocol for scanning gels with **The Discovery Series** software. A *pdi* 420oe™ ("oe" stands for Optically Enhanced™) scanner will be used as an example of a generic scanning platform, although many other imaging devices are supported by **The Discovery Series** software.

Mac Menus



PC Menus

Fig.3-1. **Scan** command.

Scanning is accomplished in three steps:

1: Calibration

2: Specifying scan parameters

3: Executing the scan

3.1. Calibrating Your Densitometer

Calibration is achieved automatically with a *pdi* Optically Enhanced scanner. The Optically Enhanced housing on the surface of your scanner contains internal calibration standards which are specific to your scanner. In transmissive scanning (gels or films), the scanner is calibrated against a transmissive step tablet, which is a strip of celluloid foil with an embedded grayscale. In reflective scanning (blots or polaroids), the scanner is calibrated against a reflective step tablet, which is a printed grayscale.

If you are scanning for the first time, your Optically Enhanced scanner will auto-calibrate before you scan. This is necessary to ensure the accuracy of your scans.

3.1.a. Recalibration Intervals

Since it is not necessary to calibrate before every scan, **Diversity Database** allows you can specify a "Recalibration Interval" on the **Scan Preferences** form (**Edit** → **Preferences** → **Scan**). The Recalibration Interval is a user-specified period of time for which the Optically Enhanced scanner will wait before recalibrating in a given color (or combination of colors).

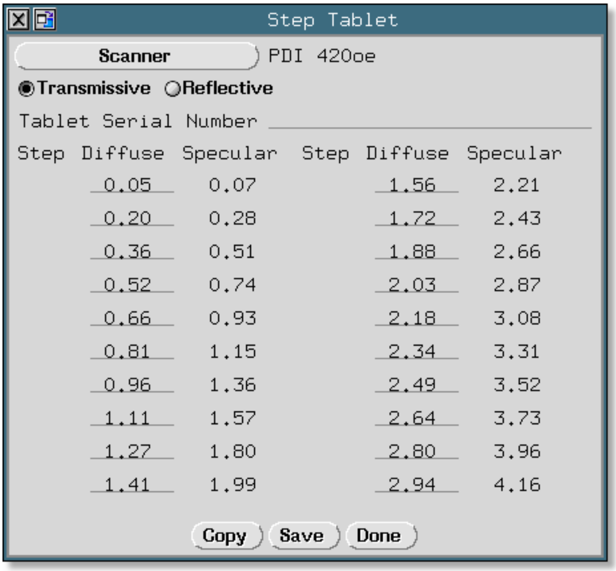
For example, if you scan a gel in Green light, and have entered a Recalibration Interval of 90 minutes, **Diversity Database** will wait for 90 minutes before auto-calibrating. If you switch to scan in Red, **Diversity Database** will auto-calibrate because you are scanning in an uncalibrated color.

The scanner must be calibrated in the same scan color that you scan your sample (see the *Scanning Color Selection* chart on page 3-8).

3.1.b. Entering Step Tablet Data

In the package with your scanner, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into your computer in order for the software to associate a correct density value with each step on the step tablet. Scanning with incorrect density values entered into the **Edit Step Tablet** form can cause significant errors in the reported densities of your scans.

To enter the density values into your computer, click on the **Edit Step Tablet** button on the **Scan Preferences** form (see Section 2.3.f). If you are currently connected to a scanner, the **Edit Step Tablet** form for that scanner format will appear (in this case, the *pdi* 420oe), otherwise you will be allowed to choose the model of scanner for which you would like to enter values.



Step	Diffuse	Specular	Step	Diffuse	Specular
	0.05	0.07		1.56	2.21
	0.20	0.28		1.72	2.43
	0.36	0.51		1.88	2.66
	0.52	0.74		2.03	2.87
	0.66	0.93		2.18	3.08
	0.81	1.15		2.34	3.31
	0.96	1.36		2.49	3.52
	1.11	1.57		2.64	3.73
	1.27	1.80		2.80	3.96
	1.41	1.99		2.94	4.16

Fig.3-2. **Step Tablet** dialog box.

Updating Density Values

If you change the step tablet with which you calibrate, you will need to update the density values on this form. Calibrating and scanning with incorrect values entered into the **Step Tablet** form will cause errors in quantitation, leading to incorrect results.

If you use different step tablets with different scanners (for example, if one of your scanners has an embedded step tablet and another does not), then you can save different sets of density values for use with different machines. Click on the Scanner button at the top of the **Step Tablet** form to select the type of scanner to which you want to save a set of density values. Enter the serial number and density values for the step tablet associated with that machine and click on the Save button at the bottom of the box.

Use the Copy button to save the same set of density values to a different scanner model. When you click on this button, a pop-up box will appear, allowing you to specify the scanner to which the current set of density values will be saved.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the step tablet is scanned. You only need to enter new values if you use a new step tablet.

After scanning the step tablet, each density value is then associated internally with its corresponding segment on the step tablet.

3.1.c. Diffuse vs. Specular O.D.

You will notice that O.D. is listed in both specular and diffuse density units. Specular density is a measure of the light that passes directly through a medium. Diffuse density is a value that includes light which is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

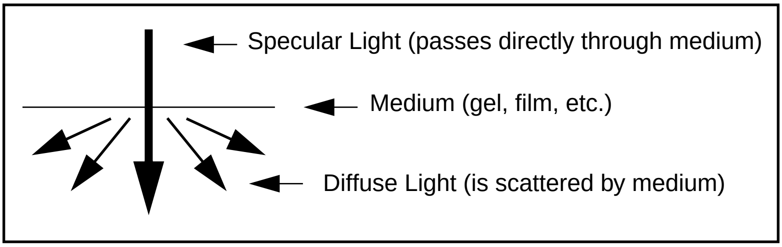


Fig.3-3. Specular vs. Diffuse Density

Diffuse density values are converted to specular optical density units according to the following formula:

$$Specular\ OD = 1.4 \cdot Diffuse\ OD$$

3.1.d. Printing Out Step Tablet Statistics

If you wish to print out a calibration report for every scan, you can indicate your preference in the **Scan Preferences** form (see Section 2.3.f) If you scan with the Print Calibration Report option selected Yes, the software will stop after every calibration and give you the option of printing out a report.

3.2. Placing Your Gel on the Scanner

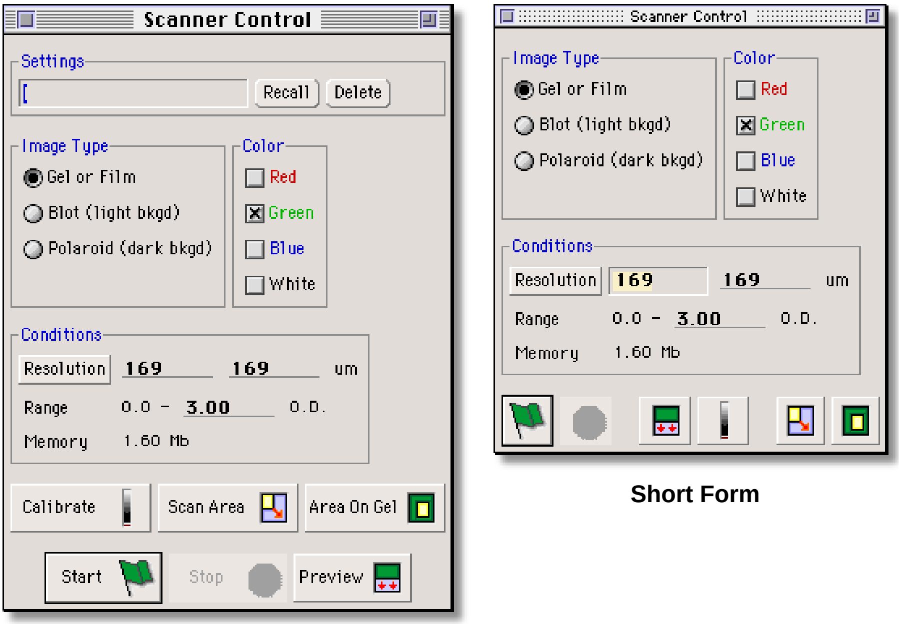
When placing the gel on the scanner, be sure that the top of the *image* (not necessarily the film) is nearest the front edge of the scanner. Place the gel face down on the scan bed with the top of the gel perpendicular to the direction of the scan.

3.3. Scanning Parameters: Long and Short Forms

Before you scan, there are parameters that you can set to optimize image quality and limit the memory size required for a given gel. If you do not change any parameter settings, default values will be used instead.

All adjustable scanning parameters can be accessed from the **Scanner Control** dialog box. To display this form, click on the **Scan** function (**File** → **Scan** on the Mac, or **Scan** → **Scan** on the PC). If you have just connected to and calibrated the scanner, the **Settings** form will automatically be displayed, along with a diagrammatic representation of the scanning surface.

Initially, the "Short Form" of the **Scanner Control** dialog box is displayed. This contains only those parameters that are most often adjusted before scanning.



Short Form

Long Form

Fig.3-4. **Scanner Control** form.

Click on the Resize button in the upper right corner of the **Scanner Control** box to toggle between the two versions.

Each of the available parameters is described below:

3.3.a. Settings

If you have a set of scan settings that you wish to save for use at a future time, enter the name under which they should be saved. Settings are automatically saved when you dismiss the form.

If you have previously saved scan settings, and you want to use them for this scanning session, click on the Recall button at the top of the page. A list of scan parameters will appear.

Select the desired set by highlighting it with the cursor and clicking. The parameters associated with that file will be displayed in the **Scanner Control** dialog box.

To delete a set of predefined scan parameters, use the Recall button to load the set, then press the Delete button to erase those settings from memory.

3.3.b. Image Type

Gel or Film

The Gel or Film setting is designed for scanning all biological samples that require transmission scanning, such as autoradiographs, Coomassie gels, silver stained gels, slides, etc.

Blot

The Blot setting is designed for scanning most biological samples that require reflective scanning, such as Southern, Northern and Western blots, dot or slot blots, TLC plates, photographs, etc.

Polaroid

The Polaroid setting is designed for scanning polaroid images of ethidium bromide gels. It is optimized for detecting faint bands on a dark background.

3.3.c. Color

Scanning can be performed at different wavelengths - red, green or blue. Color are available singly or any combination. Select the color to be used by pressing the buttons labeled red, green, blue and white.

The choice of color is determined by the staining method of your gel. The optimal wavelength of light used for detection is the wavelength at which the stain has maximum absorbance (the color complementary to the color that you can see with your eyes). Below is a list of some common stains and the colors we recommend.

Staining Method	***Color***
Coomassie Brilliant Blue	Red
Silver Staining	Blue
Autoradiograms	Green, White*
Blotting Membranes	
Alkaline Phosphatase	Blue
Horseradish Peroxidase	Blue
Colloidal Gold or Auro Dye	Blue
Immuno Gold	Blue
India Ink	White*
*White = Red+Blue+Green	

Fig.3-5. Scanning Color Selection chart.

3.3.d. Conditions

Resolution

The units of pixel size are micrometers. The pixel size, or resolution of the scan, should be set to be one tenth of the height of the smallest band.

Using too fine a resolution will generate too large a file and present problems related to insufficient memory.

If you scan at too low a resolution, you risk losing fine details and subtle features of your gel. It is to your advantage to use the minimum resolution possible without sacrificing details since the length of time it takes to complete a function is proportional to the size of the file.

To change the resolution from the default settings, click on the Pixel Size button and select a new size from the list of commonly used resolutions that appears, or type a new size next to the button.

If you do not have enough memory for the size of the scan file you specify, an error message will appear. If this happens, increase the pixel size or decrease the size of the area to be scanned (see below).

Density Range

To set the maximum O.D. value, go to the Density Range prompt on the **Scanner Control** form.

The minimum density value is 0.0 and cannot be changed.

Placing the cursor on the line in the text box, enter the value of the maximum O.D. Given that faint objects like Coomassie gels will typically have max O.D.'s of less than 1.0, it is appropriate to reduce the max O.D. value so that the scanner range matches that of the sample.

Memory Size

The file size is displayed at the bottom of the scan diagram as well as next to the Memory Size prompt on the **Scanner Control** form.

3.3.e. Scan Action Buttons

The following functions are available at the bottom of the **Scanner Control** dialog box. These commands allow you to calibrate the scanner, specify a specific region of a gel to scan, preview that region, and start or stop a scan.

Calibrate

If at any time you want to calibrate your scanner, regardless of the Recalibration Interval, click on the **Calibrate** button. This executes a calibration in the current scanning mode and color, and then offers you the option of printing out a calibration report. This is a particularly useful feature for users who wish to document the calibration of a specific scanning session, but do not want to leave the Print Calibration Report option on the **Scan Preferences** form turned On.

Scan Area

Use **Scan Area** to approximate the area of the scanner corresponding to your film or area of interest. Be sure to include the entire region of interest by being generous with the borders. This function is automatically assigned to the mouse when you open the **Scanner Control** form.

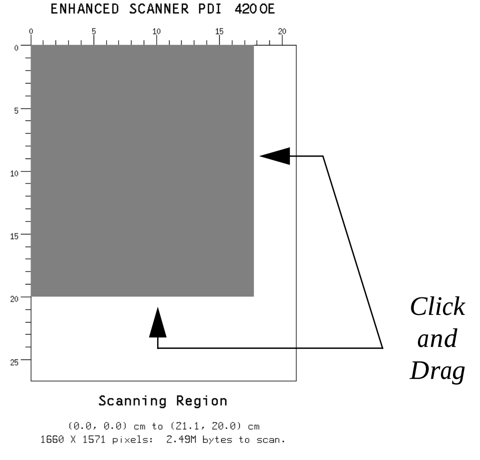


Fig.3-6. Specify the area to be scanned on the diagram.

Place the cursor on the borders one at a time, and while holding down the mouse button, move the border to the desired location, size and shape.

Area on Gel

Use **Area On Gel** to drag the edges of the box to cover the region you want to scan from your preview image.

Start (Scan)

Once you have set the scanning parameters for your gel, you can begin the scan by clicking the Start button.

The scanner will begin to “hum” and you will see the image of the gel appear on the screen one scan line at a time.

Stop (Scan)

To interrupt a scan, press the <**Esc**> key in the upper left of the keyboard or click on the Stop button. The area that has already been scanned is retained and the rest of the field is cleared. This feature can be useful if you overestimated the size of the area you asked to scan. Once you are satisfied with the image on-screen, proceed with cropping and saving the image as usual.

Preview

Use **Preview** on the selected area, to start a preview scan which is optimized for maximum scanner speed.

3.4. Manual Calibration

If you are working with a non-Optically Enhanced scanner, you will need to perform a manual calibration prior to scanning. Manual calibration consists of three steps:

1. Scanning a step tablet
2. Sampling the steps
3. Calculating a scale table

3.4.a. Scanning a Step Tablet

To calibrate by manually scanning a step tablet, position the step tablet on the scanner according to the instructions given in the **Step Tablet Calibration** dialog box, which will appear the first time that you connect to the scanner.

When the step tablet is in place and the scanning platform is properly covered, place the cursor on the Calibrate button and click the mouse.

The image of the step tablet will appear on-screen as the scan progresses. Press <**Esc**> to stop the scan early if the entire step tablet has been scanned.

3.4.b. Sampling the Steps

Diversity Database will automatically position sampling boxes over the image, one in each step. These boxes sample the O.D. within the individual steps of the image, enabling the computer to make a correspondence between the computer's average grey scale reading (which is an integer from 0 to 255) and the O.D. of each step in the tablet.

3.4.c. Calculating a Scale Table

A table of the measured grey scale values and their associated O.D. values will be displayed for your review.

Step Tablet Statistics

Step	OD	Min	Avg	Max	Std Dev
1	0.07	4	10	43	2.8
2	0.28	93	99	109	1.5
3	0.51	153	156	165	1.0
4	0.74	191	193	198	0.8
5	0.93	213	214	218	0.5
6	1.15	227	228	232	0.3
7	1.36	237	237	243	0.4
8	1.57	242	243	245	0.5
9	1.80	246	247	249	0.6
10	1.99	249	249	251	0.5
11	2.21	251	251	252	0.4
12	2.43	252	252	253	0.3
13	2.66	253	253	253	0.2
14	2.87	253	253	254	0.2
15	3.08	254	254	254	0.1

Accept | Move Boxes | Re-Scan | Cancel

Fig.3-7. **Step Tablet Statistics** form.

The grey scale value is listed in the Avg column. Each step that is below 2.0 O.D. should have an average pixel value of less than 255 and the values should increase as you look down the column.

The last column of the table gives you the standard deviations (Std Dev). Acceptable deviation values are less than or equal to 3.0.

If both types of data fit these criteria, then the statistics are acceptable. To proceed with these values, click on the Accept button.

If the statistics indicate a sampling problem, you may choose Move Boxes or Re-Scan and start over at the appropriate stage.

Move Step Boxes

If you click on Move Boxes, the function **Move Step Boxes** will be automatically assigned to the mouse. Use **Move Step Boxes** to drag the sampling boxes so that one is centered in each successive grey level of the step tablet. Be sure the first box on the left is positioned in the lightest region on the step tablet.

Sampling boxes should be within the boundaries of individual steps of the step tablet image (see below).

Note that the first segment of the calstrip is always larger than the rest. Be sure that one and only one sampling box is assigned to this segment.

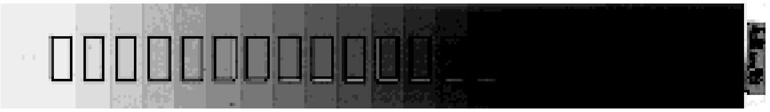


Fig.3-8. Correct position of the sampling boxes on the step tablet.

Review Step Tablet Statistics

If you wish to review step tablet statistics after manually positioning the sampling boxes, select **Calculate Scale Table** from the **Calibrate Scanner** menu. The **Step Tablet Statistics** box will re-appear.

If you are satisfied with the statistics...

If you are satisfied with the values in the statistics table, choose Accept by clicking on that button at the bottom of the dialog box.

If you are not satisfied with the statistics...

If the statistics still indicate a problem, check the step tablet itself. Sometimes dust or fingerprints can impair its accuracy. If necessary, gently wipe the step tablet with a soft, dry cloth. Check that sufficient warm up time has been used. If acceptable Std Dev (Standard deviation) values cannot be obtained due to problems with the step tablet, a new one must be ordered.

This completes the manual calibration step. Once you have accepted the step tablet statistics, you will be ready to start scanning gels.

3.5. The NightHawk™ System

Your version of **Diversity Database** is compatible with *pdi*'s **NightHawk** system. The **NightHawk** is a video camera and software package for imaging and documenting ethidium bromide gels, and is compatible with most **Discovery Series** software.

The **Video Camera** command in the **Scan** menu (PC) or **Acquire** menu (Mac) accesses the **NightHawk** camera controller. The function will remain inactive unless a **NightHawk** system is installed on your machine.

4. Image Processing

Frequently, your gel scan will require image processing prior to analysis. This section deals with the enhancement features that allow you to improve the clarity and readability of your images.

4.1. Image Editing Commands

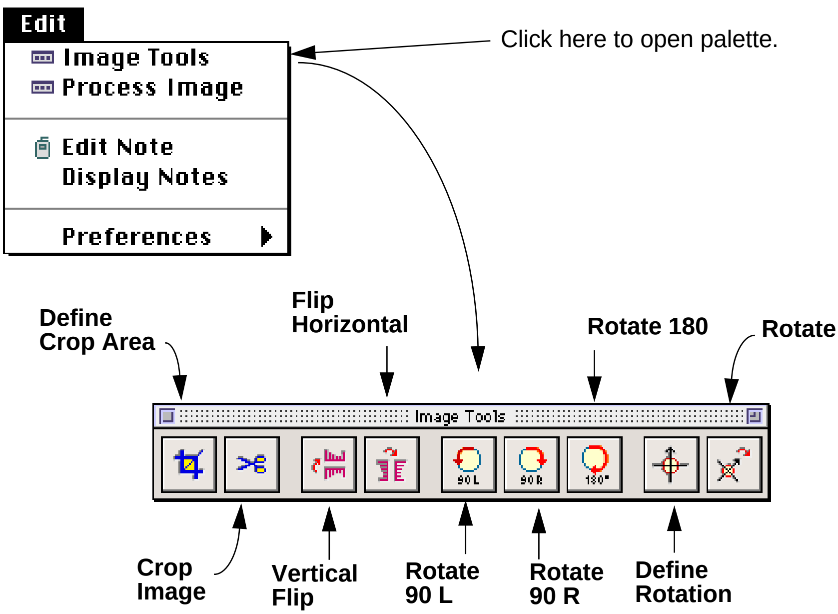


Fig.4-1. **Edit** menu and **Image Tools** palette.

4.1.a. Cropping

Once your scan is complete, the **Define Crop Area** function (on the **Image Tools** palette, in the **Edit** menu) will be automatically assigned to your mouse.

Place the cursor on the border of the area that you want to crop and click on the gel image with **Define Crop Area**. A crop box will appear at the point that you clicked on the gel. Drag the border of the crop area until it completely encompasses the area that you want cropped.

The area of the gel inside the borders of the crop box is the image that will be saved. Information on the physical dimensions of the crop area (given in millimeters and number of pixels) is listed at the bottom of the crop box, as is information on the memory size of the image inside the crop area. Cropping is a useful way to reduce the memory size of a particular image and thereby increase your overall storage capacity. The data displayed at the bottom of the crop box allows you to finely adjust the memory space required with each scan.

If the crop box does not exactly match the region of interest on your gel, you can move the entire box to its proper location, by clicking inside the crop area and dragging it across the gel.

Once you are satisfied with the placement of the crop box, you are ready to crop your image. To do so, select **Crop Image** (on the **Image Tools** palette).

A pop-up box will ask you whether you want to: (**1**) crop the original image, (**2**) keep the original intact and make a copy of the area inside the crop box (this will then be saved as a separate file) or (**3**) cancel out of the cropping operation.

Fig.4-2. Options for cropping an image.

If you select the Crop and Copy button, a dialog box will be displayed in which you can enter information such as the name of the image and its version number. Click on the Apply button once you have finished.

4.1.b. Flip and Rotate Features

If you find that your image is not in its proper orientation, there is no need to repeat scanning. You can simply flip and / or rotate the image.

To access these functions, open the **Image Tools** palette. Choose the appropriate command and invoke it on the image whose orientation you wish to change.

4.1.c. Non-90° Rotations

It is frequently useful to rotate an image in increments other than 90°. You can perform such rotations with the **Define Rotation** and **Rotate** commands.

Define Rotation allows you to define the degree to which you want an image rotated.

Rotate executes the rotation according to the **Define Rotation** command.

First select **Define Rotation**. Click on the image you wish to rotate. A circular diagram will appear over the gel image.

The diagram is composed of two elements: a circle representing the 360° of possible rotation, and an orange arrow that indicates actual rotation.

Point to the top of the gel and click **Define Rotation**. The arrow will rotate as you drag the mouse. When you release the button, the arrow will freeze, indicating the degree to which you want the image to be rotated. Upon executing **Rotate,** the image will be rotated so that the arrow becomes vertical again (90°).

Select **Rotate** and click on the image to execute the rotation.

Once the rotation is complete, a dialog box will appear. You will have the options of renaming your new image and changing the version number.

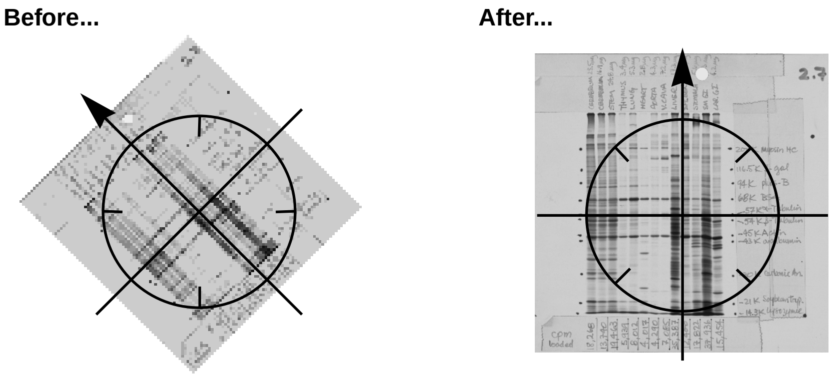


Fig.4-3. Gel images before and after a 45° rotation. An image rotated with the arrow set to the 135° position will turn 45° clockwise to the 90° position.

Note: **Rotate** automatically creates a new version of your image. Also, although the image may appear to shrink during the rotation process, there is no actual loss of information. The reduced image can be re-enlarged and any superfluous border removed with no loss of densitometric accuracy using the **Crop Image** function.

4.2. Process Image Features

Frequently, your gel scan will require image processing prior to analysis. This section deals with the enhancement features that allow you to improve the clarity and readability of your images.

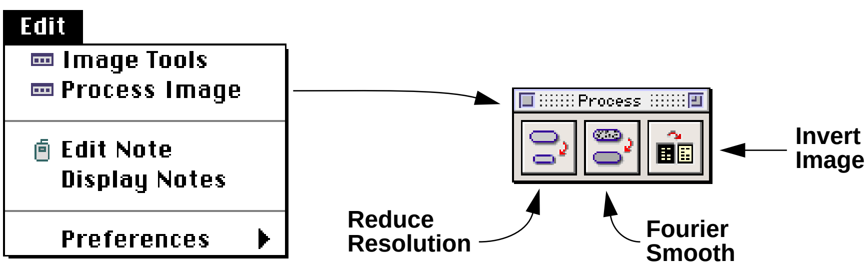


Fig.4-4. **Edit** menu and **Process Image** palette.

4.2.a. Reducing Resolution

Since computer systems do not have an unlimited amount of storage space available, **The Discovery Series** software offers several ways of minimizing the space that each image requires. One option is to decrease the resolution of an image. Lower resolution images require less storage space. Resolution can be expressed as the size of each pixel in microns, or in the number of pixels in the X and Y directions. You can lower the resolution to any size, including asymmetric pixels.

Note: Since with most 1-D gels you are more concerned with resolving bands in the vertical direction than the horizontal direction, you can lower the resolution by making rectangular pixels. That is, keeping the resolution in the Y-direction high, while lowering the resolution in the X-direction. This lets you decrease storage requirements without sacrificing detail.

Decreasing resolution is an *irreversible process*; therefore, we suggest that you change the resolution on a *copy* of the original image. This way, if you reduce the resolution too far, you can simply delete that copy and try again. If you decrease the resolution too far on the original, you will need to re-scan the gel. Of course, once you are happy with the reduced resolution gel, don't forget to delete the original. The goal is to save space!

To reduce the resolution of an image, first assign **Reduce Resolution** to the mouse.

Click in the window of the image whose resolution you want to change.

A dialog box will be displayed, asking you to enter the new pixel size. Remember, the larger the pixel size, the lower the resolution and thus, the lower the memory requirement.

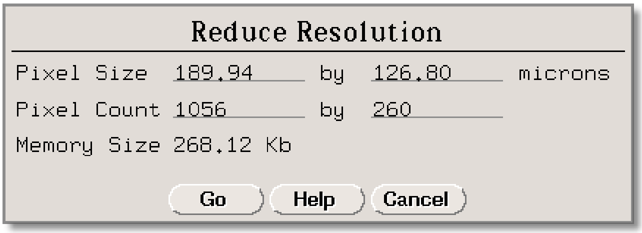


Fig.4-5. **Reduce Resolution** dialog box.

When you are finished, click on the Go button.

A new pop-up box will offer you the option of reducing the resolution of the image, making a copy of the image with a lower resolution, or cancelling out of this operation.

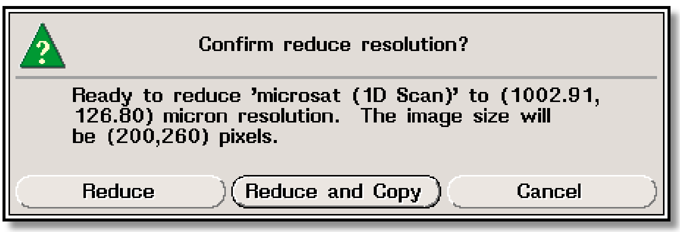


Fig.4-6. Pop-up box confirming new resolution.

If you decide to continue, a message in the upper left information bar will indicate the progress of the conversion process.

If you choose to change the resolution on a copy of the image, you will be asked to enter a name for the new copy as well as a different version number before the operation is performed.

4.2.b. Fourier Smooth

Fourier smoothing is a process which removes small features on a gel (like noise) with little effect on larger ones. To apply this operation, you need to enter a Window Size. You set the Window Size by entering its dimensions in terms of numbers of pixels in the X and Y directions. The larger the window, the greater the effect on the image. A typical window size is 2 or 3. A window size of less than 1 will essentially not change your image. A window size of 10 or greater is very likely to remove features that you do not want to lose.

Since Fourier smoothing is an *irreversible process*, we suggest that you perform it on a *copy* of the original image. If you Fourier smooth the original and do not like the results, you will need to go back and re-scan the gel.

To perform Fourier smoothing on your image, first assign **Fourier Smooth** to the mouse. Then, click the mouse in the window containing the image to be smoothed.

A dialog box will appear, giving a brief description of the function and asking you to enter the size of the smoothing window.

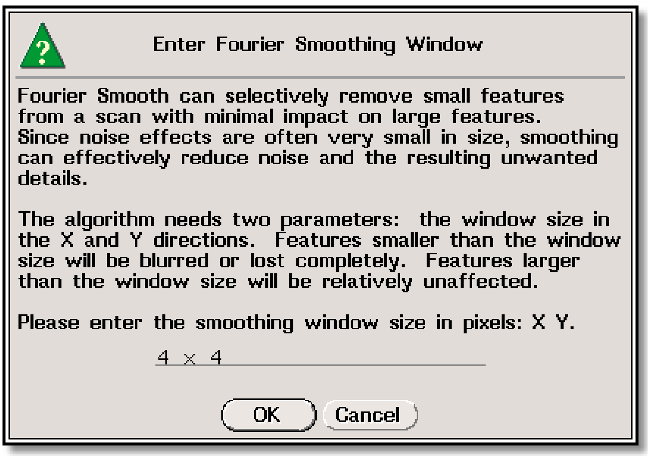


Fig.4-7. **Fourier Smooth** dialog box.

After entering the new window size, click on the OK button.

A new pop-up box will offer you the option of smoothing the image, smoothing a copy of the image, or cancelling out of this operation.

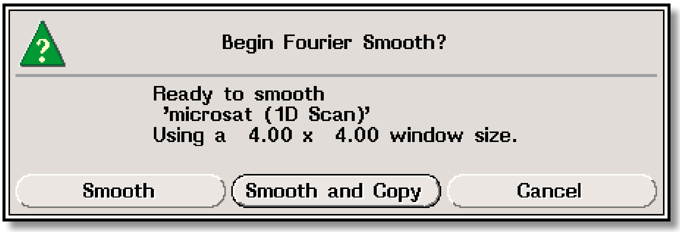


Fig.4-8. Pop-up box confirming Fourier smoothing of an image.

If you decide to continue, a message in the upper left information bar will indicate the progress of the smoothing process.

If you chose to smooth a copy of the image, you will be asked to enter a name for the new copy (as well as a version number) before the operation is performed. Once this information has been entered, the smoothing operation will be performed.

4.2.c. Invert Image

If the image you scanned has light bands or spots on a dark background, you will need to use the **Invert Image** function before you can proceed with analysis of the gel.

This function switches an image between positive and negative. In other words, light bands on the dark background of the photograph are switched to dark bands on a light background.

This process is reversible, so if you change your mind, you can always switch it back.

To invert an image, first select **Invert Image** from the **Process Image** palette.

A dialog box will appear, describing the function and giving you the option of proceeding or cancelling out of the inversion. This process is reversible, so if you change your mind, you can always switch it back.

If you click on the Invert button, the dialog box will disappear and the image will be inverted.

Messages in the status bar will indicate the progress of the inversion.

It may be necessary to adjust the contrast of an inverted image in order to produce a black-on-white image.

4.3. Background Subtraction

The following are descriptions of different methods of subtracting background noise from your gel to enhance the image of a gel and improve the accuracy of your results.

You may want to experiment with the different background subtraction options to see which works best for your images. Since whole gel background subtraction is a non-reversible process, we suggest that you experiment on copies of the original scan. This way, if you perform a background subtraction that is unsatisfactory, you can simply delete that copy of the image while the original remains unchanged. Background subtraction can be applied to either the image as a whole or to individual lanes.

You do not need to make a copy if you are going to perform lane-based background subtraction as the process does not alter the image and is, therefore, reversible. For more information on lane-based background subtraction, please refer to Section 5.4.

All the whole-gel background subtraction methods can be accessed through the **Subtract Background** form. To open this form, click on the image of the gel whose background you want to subtract with the **Background** function assigned to your mouse (**Gel** → **Background**).

The following box will appear:

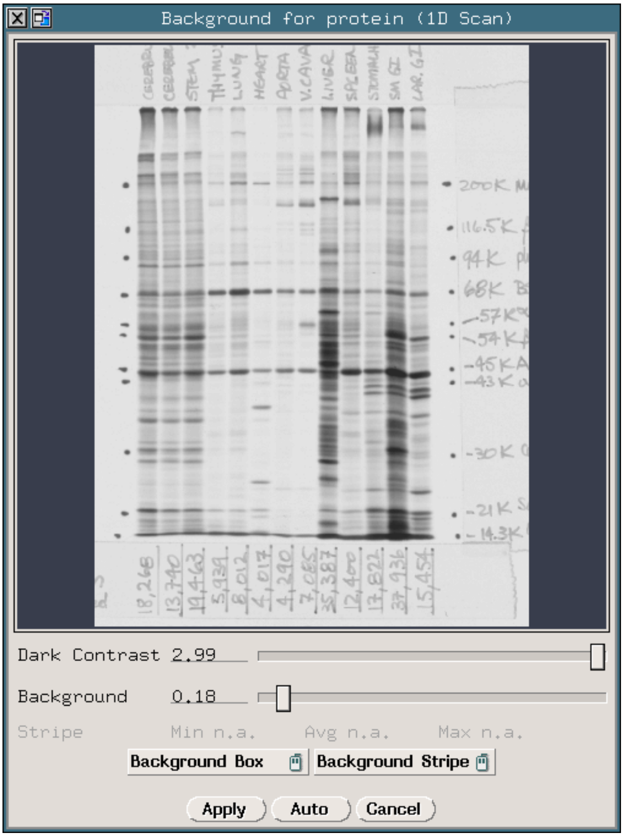


Fig.4-9. **Subtract Background** form.

Background is similar to the **Contrast** command (see Section 5.1.b) in that it allows you to preview changes to the gel image before performing any action. This preview ability allows you to experiment with different background subtraction strategies before finding one that suits your particular gel.

It is also similar in that you can use functions like **In Box** and **Zoom In** on the image in the dialog box if you are interested in improving the display of a smaller region (see Section 5.1.a).

Moreover, if you click on the gel in the window with one of the **Background** form's mouse-assignable functions (**Background Box** or **Background Stripe**) the preview image display will move to follow the position of the cursor on your original gel. This feature will work in conjunction with display features like **In Box** and **Zoom In**, such that you can move the cursor across the original gel, point at individual bands, and see them in magnified detail in the **Background** preview window.

When you are happy with the background subtraction shown in the preview image, click on the Apply button at the bottom of the form. You will be prompted by a pop-up box to make a copy of the image, prior to subtraction as the process is irreversible once performed. Click on the Subtract Background button to perform the subtraction on your original image, the Subtract and Copy button to make a duplicate of your image before subtracting, or Cancel to return to the **Background** form.

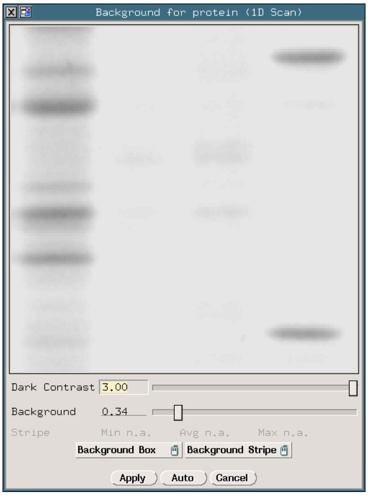
A description of each of the available features on the **Background** form is given below:

Dark Contrast

The Dark Contrast slider allows you to adjust the maximum O.D. displayed on the preview image (up to the max O.D. of the gel). Adjust the Dark Contrast setting as you would the Dark levels on the **Contrast** form: by dragging the slider box along the bar, clicking on the bar itself, or typing in new values.

Adjusting Dark Contrast levels prior to subtraction allows you to make fine distinctions in the amount of background noise present in your gel. Faint artifacts which may not be obvious to the unaided eye are made recognizable by sufficiently lowering the Dark Contrast levels (O.D. max) to within a few hundredths O.D. of the subtraction levels (O.D. min).

Example A: without adjustment



Example B: with adjustment

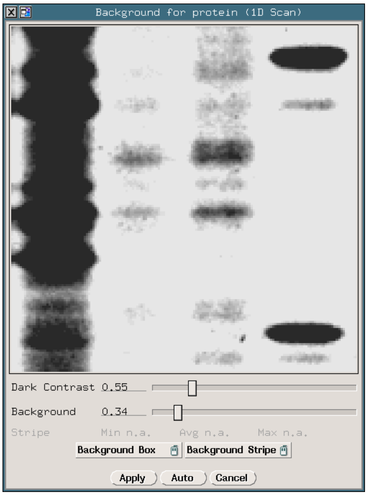


Fig.4-10. Gel images without and with Dark Contrast adjustment. Note that in Example A, with no Dark Contrast adjustment (3.00 O.D.), the surrounding medium appears free of background noise and gel artifacts, but in Example B, with Dark Contrast adjustment (0.59 O.D.), artifacts in the gel become immediately obvious, indicating that the subtraction levels were not sufficient to remove all background artifacts from the gel image.

Background Density

To adjust the subtraction threshold for your gel, click on the slider next to the Background prompt on the **Subtract Background** form and either drag the slider across the bar, or click on the bar itself. You can also type in a specific background value in the field next to the bar.

The value that you enter is a background noise threshold. Faint objects with optical densities lower the subtraction level will be eliminated from the image. Background noise can result from the opacity of your carrier medium (film, gel matrix or blot matrix) or film fogging.

Background Box

This style of subtraction is best suited for gels with uniform backgrounds.

Use **Background Box** to drag open a box that encompasses an area that is representative of the background for your entire gel. The average density of the pixels within the box will be used as the background density to be subtracted from your image.

To remove the background defined by this method, click on the Apply button.

Background Stripe

This function is useful when the background of the image is horizontally uniform but changes from the top to the bottom of the gel, such as on a gradient gel.

To subtract a background gradient, use **Background Stripe** to drag open a lane-like box down the length of the gel in a background region. The average density of the pixels in every horizontal scan line within the stripe will be subtracted from that entire scan line. This way, if your gel is darker at the bottom than at the top, more background will be removed from the lower regions of the gel.

To remove the background defined by this method, click on the Apply button.

4.4. Image Annotations

If you want to make a notation of a feature of interest directly on the gel image, use the **Edit Note** function (**Edit** → **Edit Note**).

Assign **Edit Note** to the mouse and click on the exact point on your gel to which you want the note to refer. The following dialog box will appear:

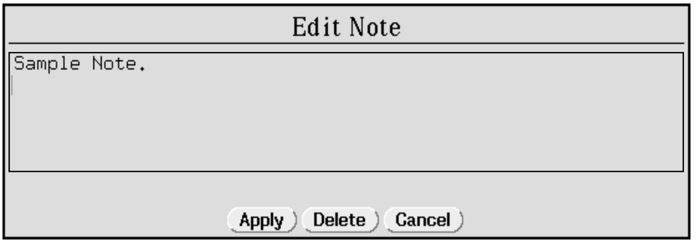
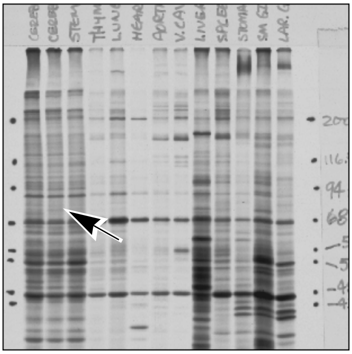


Fig.4-11. **Edit Note** dialog box.

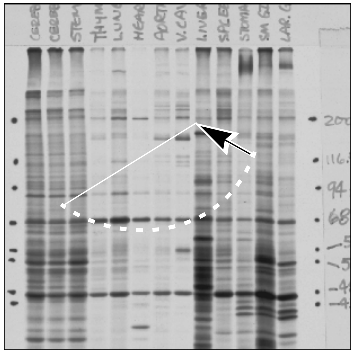
Type in the text that you would like to see displayed and press the Apply button. The dialog box will be replaced by your note, next to the point that you indicated on the gel.

Frequently, it is inconvenient to place a note directly on the feature that you want to annotate, because the note box will partially obscure your point of interest. To avoid this problem, drag the note box to a less obtrusive region of the gel (see Fig. 4-13, below).

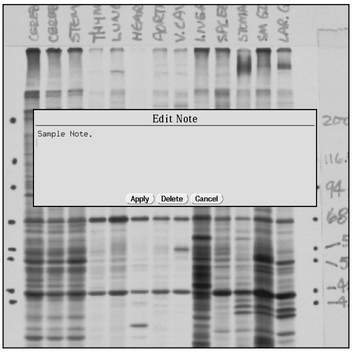
However, the note box can only be moved during its initial placement. The box can not be repositioned once it has been created.



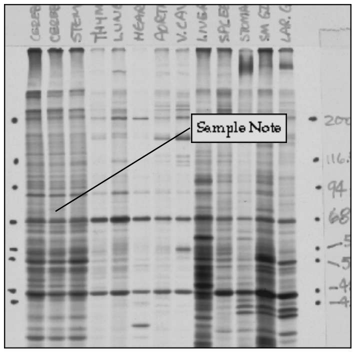
Step 1: Click on gel.



Step 2: Drag mouse to desired location and release button.



Step 3: Type note in dialog box and click on Apply button.



Step 4: Note will appear on image.

Fig.4-12. Positioning the **Edit Note** box.

If you want to change or remove a note after you have placed it on a gel, click on the note box with the **Edit Note** function. The **Edit Note** dialog box will re-appear, allowing you to change text, or delete the message.

If you copy an image file that has notes attached to it, the notes will be copied as well.

To remove notes from the display of an image without deleting them, use **Clear Overlays** (in the **View** menu). To re-display notes that have been cleared from your image, use **Display Notes** (**Edit** → **Display Notes**).

Note: Functions which remove all descriptive information from an image, such as the **Rotate** and **Flip** functions, will also delete notes from the image. Features which do not remove descriptive information, such as **Invert Image**, will not delete notes, although they may clear the display. Use the **Display Notes** function to restore cleared notes.

5. Defining Lanes

5.1. Displaying the Image

5.1.a. Getting a Closer Look

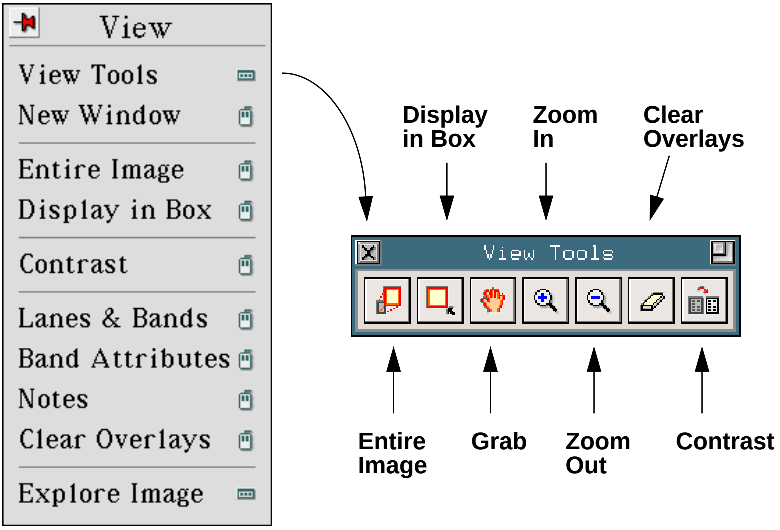


Fig.5-1. **View** menu and **View Tools** palette.

During the remainder of the analysis, you may want to increase the magnification of your gel and look at one section of the image at a time. A useful command for doing this is **Display in Box**. This function lets you draw a box around a section of the image and magnify that region so that it fills the entire window.

First, assign **Display In Box** to the mouse, then drag the cursor to enclose the area that you want to enlarge and release the mouse button.

The window will automatically display the area of the image that you selected.

You can then use **Zoom In** and **Zoom Out** to fine tune the size of the image. To return to your original image, use **Entire Image**.

Once you are satisfied with the image of your gel, the next step in data analysis is defining lanes.

5.1.b. Contrast Adjustment

To make the intensity of the displayed image correspond more closely to the original gel or to illustrate faint differences in optical density, use the **Contrast** function (in the **View** → **View Image** palette).

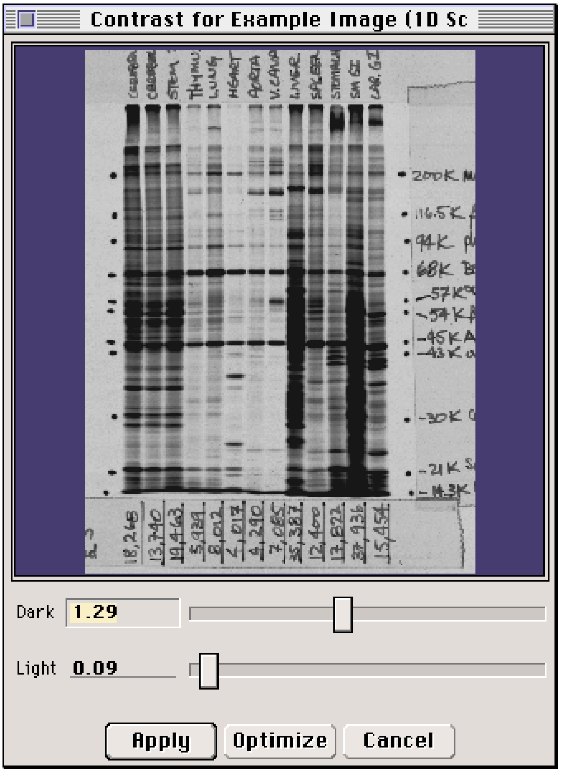


Fig.5-2. **Contrast** dialog box.

When you click this function a dialog box is displayed containing a miniature of the image in the window and controls for changing the display.

You can use functions like **Display In Box** and **Zoom In** (see Section 5.1 for more information) on the image in the dialog box if you are interested in improving the display of a smaller region.

The text boxes and sliders next to the Dark and Light prompts let you set the minimum and maximum density displayed. You can change the settings in several ways: (**1**) place the cursor on the box on a slider bar. Click and drag the box along the bar; the image in the dialog box changes, reflecting the density that you specify, (**2**) click anywhere on a slider bar to change the value in the text box by increments of approximately 0.01, (**3**) type an entry in the text box to set the desired density or to exceed the default range.

Clicking the Optimize button results in the lightest part of the image in the dialog box being set to the minimum density, and the darkest being set to the maximum density. This effectively enhances minor variations in the image, making fine details easier to see.

After selecting the density at which the image should be displayed (either with the sliders or with Optimize) click on the Apply button to update the display of the image.

If you change your mind about altering the contrast, click on the Cancel button. The **Contrast** dialog box is dismissed without applying any changes to the displayed images.

5.2. Defining Multiple Lanes

5.2.a. Creating a Matrix of Lanes

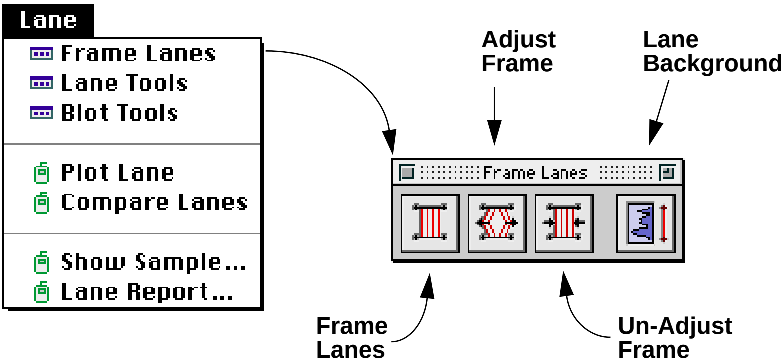


Fig.5-3. **Lane** menu and **Frame Lanes** palette.

Use the function **Frame Lanes** (on the **Frame Lanes** palette) to open a dialog box in which you can enter the number of lanes in your gel.

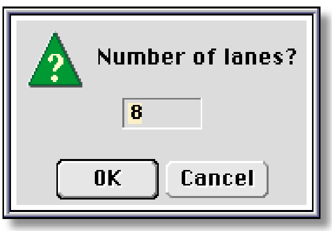


Fig.5-4. **Frame Lanes** dialog box.

Type in the number of lanes in your gel, and press the OK button.

A pop-up box will appear, asking you to confirm the creation of a lane matrix. Click on the Continue button to complete the action.

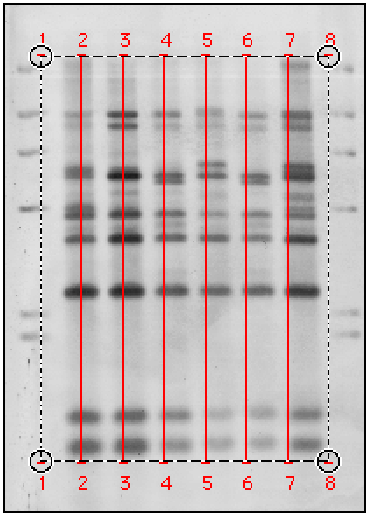


Fig.5-5. The lane matrix.

5.2.b. Adjusting the Placement of the Lane Matrix

Next, you will need to adjust the size and the placement of the lane matrix using the function **Adjust Frame**. **Adjust Frame** will be automatically assigned to the mouse any time **Frame Lanes** is used.

First, adjust the corners

The borders of the lane matrix are denoted with dotted lines and each anchor point (corner) is marked with a circle.

To adjust the lane matrix, place the cursor in one of the circles. While dragging the mouse (with **Adjust Frame** assigned), move the position of the circle so that the first line of the matrix follows along the center of the first lane. (If the lane is not perfectly straight or vertical, there are ways to correct this problem later.) Other lanes in the matrix will move with the first lane.

Repeat this procedure at the other corners of the matrix so that the top and bottom of the matrix correspond to the origin and the dye front of the gel (or the position of the electrodes in IEF), and the left and right borders of the matrix follow the first and last lanes.

Next, create and adjust internal anchors

What if the lanes are not straight, or if the dye front smiles?

Use **Adjust Frame** to create internal anchor points, analogous to the circles at the corners of the frame. These internal anchors can be made on the top, bottom, left and / or right borders of the matrix.

Place the cursor on the lane matrix border (the dotted line) where you want to "bend" the matrix. For example, if your gel smiles, place the cursor along the bottom of the border where the smile is the greatest. Click **Adjust Frame** on the frame border. You will see two new anchors created: one at the position of the cursor and the other directly above it on the top border of the matrix.

These two anchors are connected by a dotted line analogous to the frame borders. Other internal anchor points can be placed along this new dotted line if further internal adjustments to the lane matrix are required.

To adjust the bottom border of the matrix, place the cursor on the newly created anchor point (circle). Drag the anchor point so it corresponds with the smiling dye front. The end points of all the lanes of the matrix will adjust proportionally with the placement of the anchor point.

You can create as many internal anchors as necessary for the lines of the matrix to correspond to the lanes of the gel.

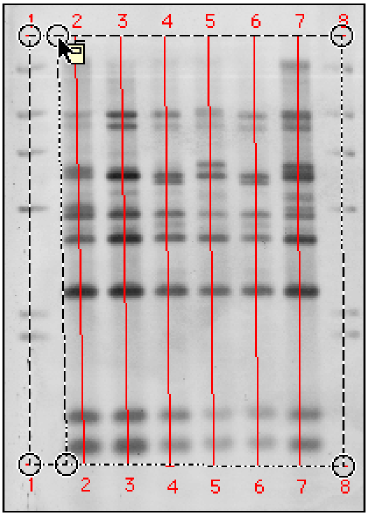


Fig.5-6. Adjusting the Lane Matrix.

If you change your mind about the position of an anchor, you can reposition it as described above. If you decide to remove an anchor point, use the function **Un-Adjust Frame**. Point the cursor at the anchor you wish to remove and click the mouse button. The anchor will disappear and the lanes will “un-adjust,” reflecting the removal of the anchor.

5.3. Single Lane Functions

A number of commands exist for creating and editing single lanes. These functions can be found on the **Lane Tools** palette.

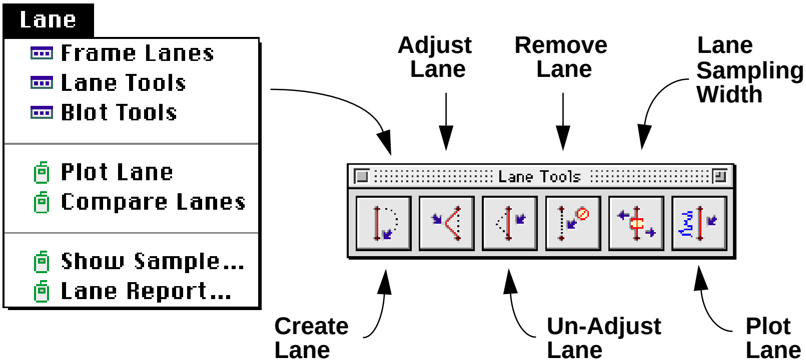


Fig.5-7. **Lane** menu and **Lane Tools** palette.

5.3.a. Manually Defining Lanes

You can manually define single lanes using the **Create Lane** function.

Assign **Create Lane** to the mouse, then click on the lane origin and drag a line from the top to the bottom of the desired lane.

Repeat this procedure to manually create all the lanes on your gel.

5.3.b. Adjusting the Placement of Individual Lanes

You can adjust the position of an individual lane by using the **Adjust Lane** command to move the anchor points (ends) of the vertical lanes, or to create internal anchor points.

First, assign **Adjust Lane** to the mouse.

Then place the cursor on the lane that you want to adjust.

Drag the mouse. You will see the line move with the cursor. When the line is correctly placed, release the mouse button.

Repeat this procedure as many times as necessary on individual lines of the lane matrix until they accurately follow the lanes of your gel.

You can undo any of the lane adjustments with the **Un-Adjust Lane** command. If you remove the anchor points at either end of the lane, you will delete the lane from the gel.

5.3.c. Deleting Lanes

If for any reason you want to delete a defined lane, you can do so with the **Remove Lane** function.

Point at the lane you want to remove and click the mouse. A pop-up box will ask you to confirm the deletion of that lane.

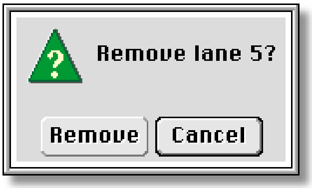


Fig.5-8. Pop-up box confirming the removal of a lane.

5.3.d. Plotting Lane Traces

Immediately after defining a lane (using either **Frame Lanes** or **Create Lane**, as discussed above), the density profile (trace) can be plotted using the **Plot Lane** function. Each point on the plot is generated by taking the average O.D. of the pixels in every horizontal scan line along the defined lane. The number of pixels used in this calculation depends on the specified sample width. The default sample width is 1.95 mm. This value can be changed from either the **Gel Layout** form (see Section 7.3), the **Detect Bands** form (see Section 6.2), or with the **Lane Sampling Width** command (see below).

The trace of a lane can be useful for choosing a suitable background O.D. for background subtraction or for examining peak heights of bands. It is also useful for visualizing the data in a manner similar to other gel scanners and for graphically examining the data for the set of dots/slots in a column. To remove the trace of the lane use the **Clear Overlays** command (**View** → **Clear Overlays**).

5.3.e. Adjusting the Sampling Width of a Single Lane

The **Lane Sampling Width** command (on the **Lane Tools** palette) allows you to manually adjust the sampling width for a single lane. To do so, assign the **Lane Sampling Width** command to the mouse and click on a lane.

The following box will appear, asking you to enter a new sampling width:

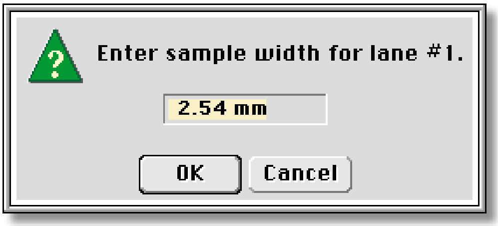


Fig.5-9. Setting the width for a single lane.

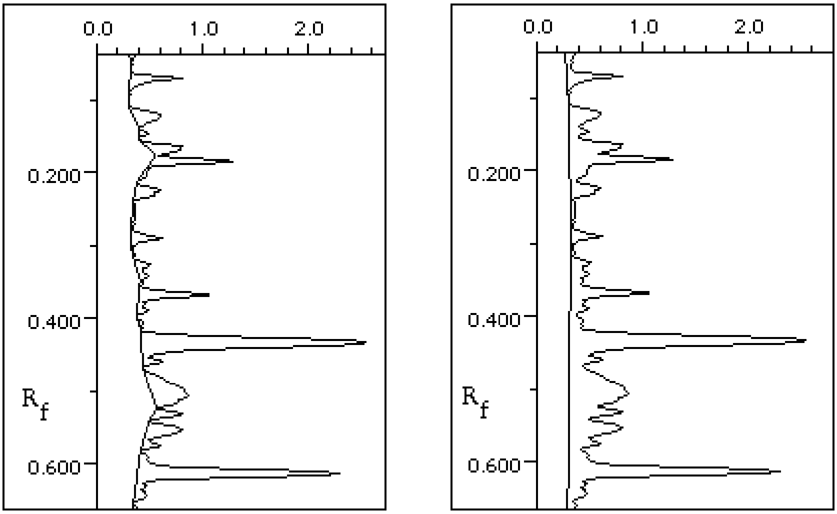
Click on the OK button to change the width.

See Section 6.1 on *Bracket Quantitation* for a discussion of the effects of sampling width on calculating band quantitation.

5.4. Lane-Based Background Subtraction

After defining lanes, we strongly recommend that you perform lane-based background subtraction. A method called Rolling Disk background subtraction is used to calculate the density that is removed. This method removes different densities along the length of the lane while following the contour of the lane's density trace.

Depending upon the size of a user-defined disk, different amounts of background will be subtracted. A large disk will follow the general trends in background density, while a small disk will be sensitive to smaller fluctuations in density. The smaller the disk, the more density will be detected as background (refer to illustration below).



Small Radius **Large Radius**

Fig.5-10. Example of the different amounts of density removed by rolling disk subtraction with a small disk radius, as compared to a large disk radius.

Larger radii result in slower execution and poorer removal of background. Typical values are from 50 to 150.

Try different disk sizes to find the one that gives you the desired background subtraction results.

5.4.a. Subtracting Lane Background

To remove background density on a lane-by-lane basis, use either the **Lane Background** function (on the **Frame Lanes** palette), or the **Gel Layout** form (see Section 7.3).

When you click on a gel with **Lane Background** assigned to the mouse, a trace of the lane nearest the cursor will be displayed, along with the following dialog box:

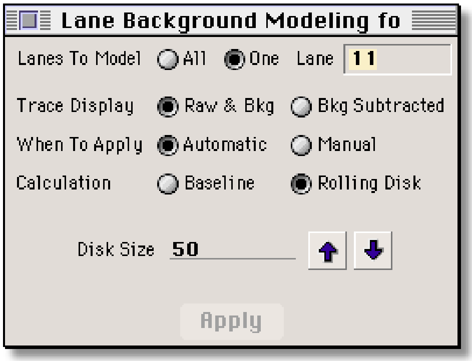


Fig.5-11. Lane-based background subtraction dialog box.

The settings in the dialog box will correspond to the lane at which the cursor was pointing when the command was invoked.

The following information can be specified on the form:

Lanes to Model

If the background subtraction should be applied to all the lanes of a gel, click on the All button.

If you prefer to apply the subtraction to one particular lane, click on the One button and type the number of the lane next to the Lane prompt.

Trace Display

The trace of the lane can be displayed in one of two ways. Selecting the Raw & Bkg option (with the functions Automatic and Rolling Disk activated) will display the original "raw" trace of the lane in white with an orange line representing the background beneath the peaks of the trace.

Changing the size of the disk will change the placement of the background line.

As you decrease the disk radius, you will see the background line “hug” the bottom of the density trace more closely.

Pressing the Bkg Subtracted button will display the lane trace as if all the background beneath the orange line in the Raw & Bkg display were actually removed. This display allows you to visually compare the relative densities of the bands in the lane.

When to Apply

As you change either the calculation method or the disk size, the new settings can be applied to the image automatically every time a change is made by selecting the Automatic button.

Alternatively, you can apply each new setting manually by setting the Manual button and clicking on the Apply button at the bottom of the form whenever the settings are changed.

Calculation Method

Selecting Baseline does not affect the lane.

If you press the Rolling Disk button, the dialog box will reconfigure so you can change the size of the disk.The value of the Disk Size is equal to the number of pixels of the disk radius.

You can enter a numeric value for the disk size or you can use the arrows to change the value in 10% increments.

When you apply the Rolling Disk background subtraction, the lane trace display will change but the image of the gel will *not* reflect the change in background density.

5.5. Compare Lanes

Compare Lanes (**Lane** → **Compare Lanes**) allows you to superimpose in a single graph the density traces of any number of lanes present on any number of open gels. Once a lane is displayed, **Compare Lanes** can zoom in on a particular regions of interest and display it in greater detail.

To run **Compare Lanes**, first assign the command to the mouse and then click on the lane that you wish to see displayed.

Note: A gel must have defined lanes, but does not require defined bands, for **Compare Lanes** to work.

The **Compare Lanes** window will appear. You can add any number of additional lanes by clicking directly on the lanes that you want to include.

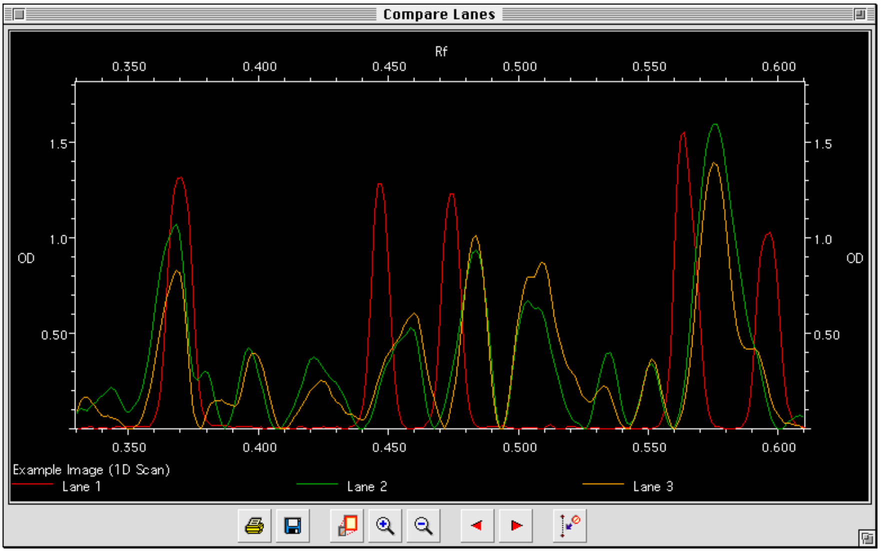


Fig.5-12. The **Compare Lanes** dialog box.

Each lane that you select will be displayed in one of eight colors, and will be identified by color in a legend underneath the graph. If you select more than eight lanes to display, the colors used to designate each lane will begin to repeat, but will still be identified underneath the trace display. There is no absolute limit to the number of lanes that can be displayed simultaneously.

The plots for each lane are determined in the X-axis by Rf values, and in the Y-axis by O.D. values. **Compare Lanes** automatically "best-fits" lanes within the display window in order to maximize the range of O.D. values included in the graph. Rf values are displayed from 0.0 to 1.0.

Changing the Display

The Zoom In and Zoom Out buttons can be used to "move" in and out within the display in order to examine interesting regions in detail. You can also zoom in by dragging across the display with the cursor.

"Zooms" are accomplished by restricting the range of displayed Rf values and graphically stretching that range to fit the original width of the window. When you drag across the display, you redefine the width of the graph to match the horizontal distance that you dragged the mouse (i.e., the bars define the region that will be magnified). Dragging the mouse one half the distance of the graph would change the displayed range from 1.0 Rf to 0.5 Rf, doubling the perceived "magnification."

The Full button returns the field of view to its complete, unadjusted format (Rf Range = 1.0).

The Left and Right buttons can be used to pan left or right along a zoomed in image by shifting the Rf range up or down.

Function Buttons

The Remove Lane button displays a dialog box which allows you to remove any lane that you have selected.

Note: If you remove a lane, the color designations of the remaining lanes will be altered in response. Check the lane legend for an updated color code.

Click on the Print button to produce a graphical report of the **Compare Lanes** display.

6. Defining Bands

Once the lanes of your gel are in place, you will need to define the bands. Band detection is the process by which **Diversity Database** identifies and quantitates the bands on your gel. When you look at a magnified band on-screen, you can see that it is made up of a set of pixels, and that each pixel has its own O.D. value. When a band is quantitated, an O.D. value is calculated for all the pixels within a band.

6.1. Bracket Quantitation

The first step in calculating bracketed band quantitation is to find an average O.D. value for each horizontal row of pixels. The number of pixels included in this value is determined when you specify a sampling width. The wider you set the sampling width, the more pixels are included in the O.D. average.

Since all the pixels that fall within the sampling width are included in quantitation calculations, it is important to avoid including background area. Background pixels decrease the average pixel values in band quantitation. If you need to choose between including background versus leaving out part of the band, it is better to opt for the latter. Since you are taking the average O.D. of all the pixels in a row, it's not critical if a few pixels are omitted from the band.

Once you have completed this step, there will be "N" number of average O.D. values for the band. "N" is determined by the specified band height. The wider apart you set the brackets that define a given band, the greater the number of pixels that will be included in the quantitation.

The final step in calculating band quantitation is integrating under the curve to the baseline, giving you units of O.D. x millimeters.

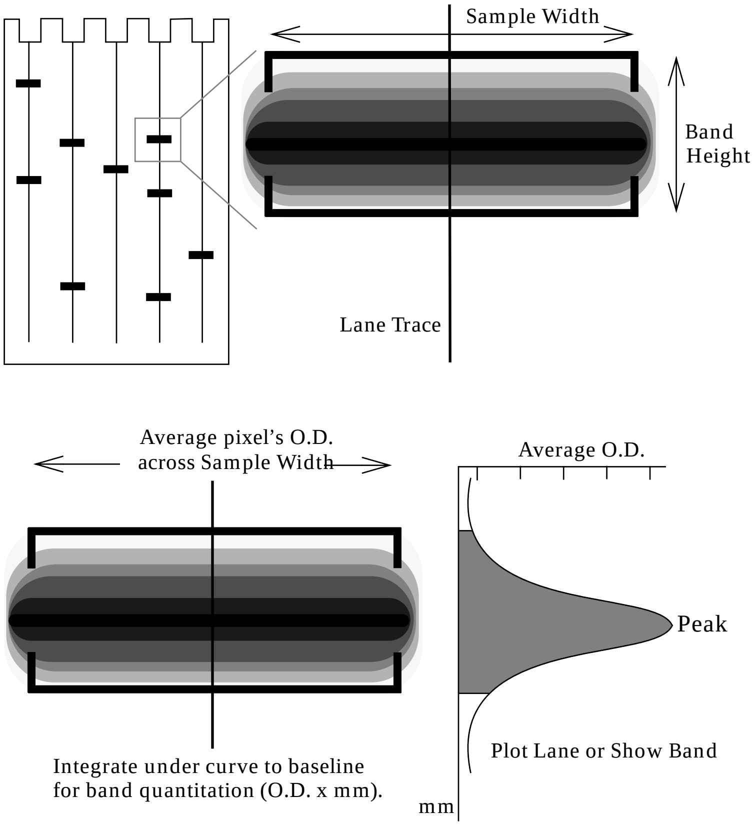


Fig.6-1. Illustration of bracket quantitation.

6.2. Automatic Band Detection

Use the **Detect Bands** function (**Band** → **Detect Bands** or on the **Band Tools** palette) to automatically find the bands.

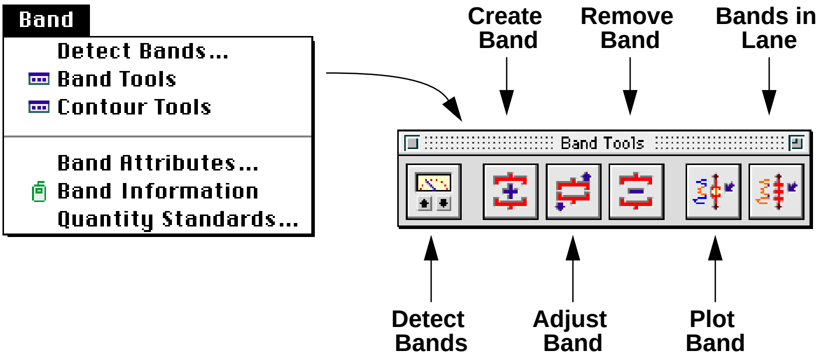


Fig.6-2. **Band** menu and **Band Tools** palette.

Automatic band finding functions should be used before you make manual adjustments to the bands since all manual work will be overwritten by the automatic band decisions.

When you are ready to detect the bands on your gel, open the **Detect Bands** form (see below).

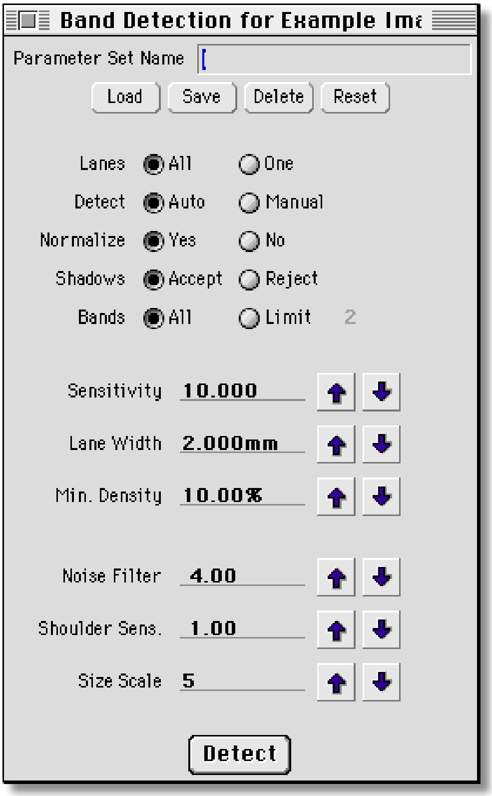


Fig.6-3. **Band Detection** dialog box showing default values.

The **Detect Bands** form allows you to adjust the parameters that affect band detection, enabling you to customize band detection to match your gels. Once determined, custom detection strategies can be saved for use in future sessions.

To change any of the parameter settings, you can either type in a new value or you can use the arrows to increase or decrease the setting by 10%. Since the properties of your gels are unique, we suggest that you experiment with various settings to find those best suited to your gels.

6.2.a. Setting Band Detection Parameters

Lanes to Detect

You can specify whether you want to detect the bands in all the lanes of the gel at once or, if you prefer, to detect the bands in a single lane by selecting All or One next to the Lanes prompt. If you choose One, you will need to specify which lane by typing its number in the text box next to the One button.

It may be necessary to detect one lane individually if it has different characteristics from the other lanes and the detection parameters applied to the whole gel are not optimal for that lane, (e.g., faint bands are not being detected or high background in that lane is being detected as bands). You can re-detect bands in that lane by singly applying a different set of detection parameters.

For example, if a smaller amount of a sample was loaded in one lane, the bands in that lane would be fainter in comparison to the other lanes of the gel. Detection of bands in that lane could be enhanced by increasing the sensitivity and decreasing the minimum density parameters.

The **Band Detection** dialog box allows you to customize band detection parameters to suit the varying properties throughout your image.

When to Detect

If you set the form to Auto detection (next to the When prompt), each time you change any of the detection parameters, band detection will immediately occur. You will not need to click on the Detect button located at the bottom of the form.

If you want to change more than one parameter before re-detecting, choose the Manual option. With the Manual option, you can change parameter settings first, and then apply them by clicking Detect.

Normalization

Normalization is a way for band detection to compensate for differences in the darkness of the lanes on a gel. *It does not normalize for band quantitation.*

If you select the No button next to the Normalize prompt, the band detection parameters are taken as absolute values and applied the same way to every lane of the gel.

If you select Yes, the parameters applied in the band detection process are automatically adjusted according to the darkness of the lane. The darkness of the lane is determined by the darkest band in the lane. For example, all but one of the lanes on a gel contain bands up to 2.00 O.D. In the one light lane, the darkest band is only 1.00 O.D. If normalization is turned on, band detection will be twice as sensitive when processing the light lane, improving the detection of faint bands.

Shadow Rejection

"Shadow bands" are common artifacts seen on PCR gels. These shadows are spaced at tandem repeat intervals and decrease in density they progress further from a "real" band. Two parameters are provided to improve the likelihood of **Diversity Database** calling only the bands of interest: Shadows and Band Limit (see below).

If you select Reject next to the Shadows prompt, the Automatic Band Detector will only find a band if it is darker than the one above it or if it is spaced further than the size of one tandem repeat unit from the previous band. Since shadow bands decrease in density as you move away from the real band, they will not be erroneously detected by the Automatic Band Detector.

If you select Accept next to the Shadows prompt, **Diversity Database** will not consider whether a band is a shadow band when it runs its detection.

Band Limit

If you know that all the lanes of a gel contain a specific number of bands, click the Limit button next to the Bands prompt and type the number of bands which you know to be present. Only that number of bands will be detected in each lane, reducing the need for later editing.

Sensitivity

The higher the sensitivity value, the more bands will be detected.

If the sensitivity is set too high, background density will be erroneously detected as bands. If the setting is too low, real bands may be missed.

The default sensitivity setting is 10.000. If your gel has faint bands (O.D.$\leq$ 0.05), you may wish to increase this value to 20.00.

Lane Sampling Width

The Lane Width of a band determines the width of the region of optical density that will be sampled for quantitative data. The wider you set the sampling width, the more pixels are included in the O.D. average (see Section 6.1 for more information on determining band quantitation).

If you set the width too high, you will include background area in your band quantitation. If you set the width too low, automatic detection will be more likely to mistake random variations for bands. Wide lane widths (i.e., 2/3 of the actual lane) are best for allowing high sensitivity with few false bands.

If you need to choose between including background versus leaving out part of the band, it is better to opt for the latter. Since quantitation uses the average O.D. of all the pixels in a row, it's not critical if a few pixels are omitted from the band.

Minimum Density

When Normalization is turned off, this is the lowest density (O.D.) that will be counted as a band.

Before entering a minimum O.D. value, plot a trace of the lane that includes some faint bands. Choose an O.D. value that is lower than the O.D. of the peak of a faint band but is still above the background.

If faint bands are still being missed after applying this parameter to band detection, double check the O.D. of faint bands on the lane trace and / or consider increasing the Sensitivity setting.

If Normalize is turned on, Min. Density is converted from an absolute O.D. value to a percentage. This percentage is the fraction of the O.D. of the darkest band in the lane that will be detected as a band. For example, if the darkest band in a lane is 2.00 O.D., and the Min. Density is set to 1.00 O.D., when you turn on Normalize, the Min. Density will switch to 50%. In other words, for a band to be detected it must be at least 50% as dark as the darkest band in the lane.

Noise Filter

The Noise Filter can be used to minimize the number of small fluctuations in the image (i.e., noise) that are called bands while still recognizing larger features (i.e., real bands). This distinction becomes increasingly critical the higher the sensitivity parameter is set.

The noise filter value is a window size in pixels. Features below this value will not be recognized as bands. Setting the noise filter to zero turns it off completely. The default value is 4.00.

If band detection calls a doublet a single band, decrease the noise filter setting and / or increase the sensitivity.

You can also try decreasing the Size Scale parameter instead of the Noise Filter to improve the detection of closely-spaced bands. Be careful of decreasing both the noise filter and the size scale, as this may result in the fuzziness around bands being mistakenly detected as separate bands.

Shoulder Sensitivity

Normally, band detection tries to distinguish shoulders as separate bands. When looking at the lane trace, these bands appear as flat or gently sloping sides on darker, better defined bands (i.e., there is no dip on the trace between the two bands).

Increasing the shoulder sensitivity value will result in more shoulders being called as bands while decreasing it will result in fewer. Setting it to zero turns this parameter off, so that no shoulders will be recognized as separate bands.

If band detection calls a doublet a single band, check the lane trace to see if there is a dip between the peaks of the two bands. If there is no dip, increasing the shoulder sensitivity value should help resolve the two bands.

Size Scale

The Size Scale helps in distinguishing between trends in density and random density fluctuations. It is the number of pixels in a vertical row which are taken together to determine trends.

The Size Scale parameter is closely related to the Noise Filter. The default Size Scale setting of 5 is optimal for most gels. It can be set to any whole number greater than or equal to 3.

If a gel has high levels of background noise, a larger Size Scale may be preferable. When noise permits, small values are preferred so that small features will be detected.

You may also choose to increase the Size Scale if your gel only has a small number of thick bands and it is scanned at high resolution.

6.2.b. Saving Band Detection Parameters

Saving Parameters

Once you have found a set of parameters that are optimal for detecting bands in your gels, you may want to save them for future use on other gels. The Save button at the top of the **Band Detection** dialog box allows you to do so.

If you have not already entered a name for your parameter set at the top of the dialog box, you will be prompted to do so.

Loading Parameters

To apply a set of parameters that you saved at an earlier time, press the Load button near the top of the **Band Detection** form.

Select the parameters that you wish to load from the list of previously saved sets that appears. The parameter values of that set will be displayed in the **Band Detection** form.

Deleting Parameters

To remove a set of parameters from your list, first load it, then press the Delete button. A dialog box will ask you to confirm your decision to delete that set of parameters or cancel out of the Delete operation.

Default Parameters

To return all detection parameters to their default values, click on the Reset button at the top of the form. Saved detection parameter files are not affected by this action, and can be reloaded at any time.

6.3. Manual Band Detection

To manually define the bands in each lane, use the three manual band commands found on the **Band Tools** palette.

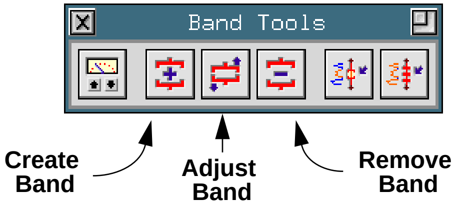


Fig.6-4. **Create**, **Adjust**, and **Remove Bands**.

6.3.a. Defining New Bands

When you use the manual band commands, a trace diagram will be displayed to help you determine band boundaries. Most people will define bands as bounded by their inflection points when they are only working with a gel image. When using the trace profile, they tend to include much more of the band's tail. You can see that the band limits can be defined as either the trough between the peaks on the trace or as the inflection points on the curves defining the peaks. The inflection points correspond to the visual limits of the band.

In Brackets Mode

To define a band manually in Brackets mode (bands can be displayed in either *Brackets* or *Lines*, see Section 6.4.b for more information), first assign **Create Band** to the mouse, then position the cursor so that it points to either the top or bottom of the band of interest.

A density trace of the lane will be displayed when you click on the gel.

Drag the cursor until the area of the band that you wish to define, as represented by a peak on the density trace, has been completely enclosed, then release the mouse button to complete the band definition

Brackets denote the region that you have called a band on the image. The corresponding section of the density trace is highlighted.

In Lines Mode

To define a band manually in Lines mode, first assign **Create Band** to the mouse, then position the cursor so that it points to the center of the band of interest. Click the mouse and the display will update to reflect the placement of the new band at that point on the lane.

6.3.b. Adjusting Bands

After defining bands on your gel, either automatically or manually, you may want to go back and change the placement of the band boundaries.

To reposition a band's boundaries, assign **Adjust Band** to the mouse, point at either the upper or the lower bracket of the band that you want to adjust, and click the mouse button.

If it is not already shown, a density trace will be displayed. The band being adjusted will be highlighted, and bars extending from the bracket to the corresponding region on the density trace will appear.

While holding down the mouse button, move the bracket to the desired location, then release the button.

This procedure can be performed on the upper and/or lower boundaries of any and all bands that need editing.

6.3.c. Deleting Bands

Use **Remove Band** to remove a band from a lane.

Assign **Remove Band** to the mouse, point the cursor at the unwanted band, and click the mouse button.

If your bands are designated with brackets, a trace of the lane will be displayed and a pop-up box will ask you to confirm removal of the band.

Click on Yes to delete the band, or No to cancel the procedure.

If you delete the band, the region will no longer be counted as a band, but its density will still contribute to the total lane density.

6.4. Displaying Band Information

6.4.a. The Band Information Form

You can examine and enter quantitative data for individual bands using the **Band Information** function. To do so, select **Band Information** from the **Band** menu. Point at a band of interest and click the mouse button. The density trace diagram of that lane will be displayed, the band of interest will be highlighted, and the following dialog box will be displayed:

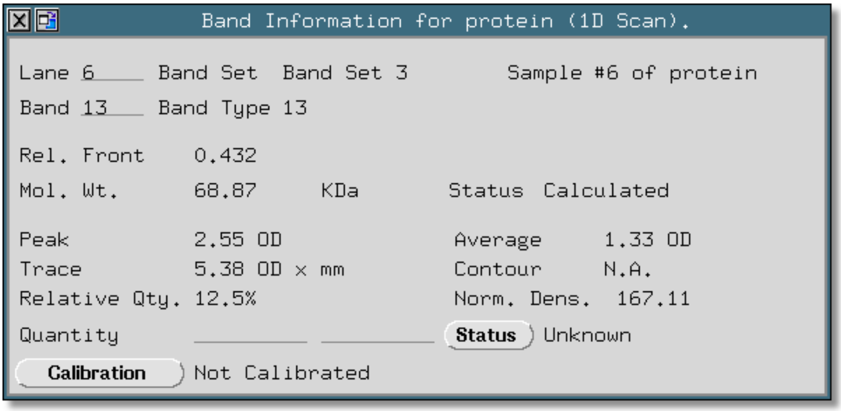


Fig.6-5. **Band Information** dialog box.

The Lane and Band fields at the top of the form allow you to type in the numbers of other bands on your gel. The information displayed in the **Band Information** dialog box will update to reflect the new band, as will the trace overlay and band highlighting in the image window.

Type in known quantitative band values and their units next to the Quantity prompt.

6.4.b. The Band Attributes Form

After bands have been defined, there are several options for displaying them, all of which can be found in the **Bands** menu.

All the various quantitative data, as well as other band information such as band number, Rf and molecular weight, can be displayed directly on the image using **Band Attributes**. Selecting this feature causes the following dialog box to be displayed:

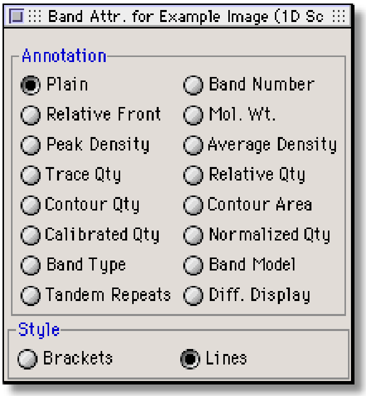


Fig.6-6. **Band Attributes** dialog box.

Click on any of the buttons in the dialog box to display that particular attribute. The information is listed on the image next to each individual band.

The options within the Style section allow you to switch between two styles of band display: Brackets or Lines.

Bands designated with brackets appear in the format described in Section 6.1, *Bracket Quantitation*.

Bands designated with Lines appear on your gel image as single bar running through the center of each defined band. Frequently, in gels with densely packed bands, band brackets will overlap, and may become difficult to distinguish from each other. Generally, lines are easier to read than brackets in gels with closely packed bands.

6.4.c. Other Display Band Functions

Plot Band (on the **Band Tools** palette) displays the density trace of the lane nearest the cursor and highlights the band that you selected on that diagram.

Bands in Lane (on the **Band Tools** palette) displays the density trace of the lane nearest the cursor and highlights all the defined bands.

6.5. Whole Band Integration

If the bands on your gel are irregularly shaped, you may prefer using the contour functions for defining bands. Quantitation of contoured bands is calculated based on the optical density of all the pixels within the band boundary, using the following formula:

> quantity = the sum of [(O.D. of pixel) x (area of pixel)] for all the pixels in the band.

The first step in calculating the quantitation of a contoured band is the multiplication of the O.D. of a pixel by the area of the pixel. This is done for all the pixels that are within the boundary of the contoured band. The area of the pixel is determined by the scan resolution of the gel.

The resulting values have units of O.D. x microns^2. These values are all added together and the result is converted from O.D. x microns^2 to O.D. x mm^2.

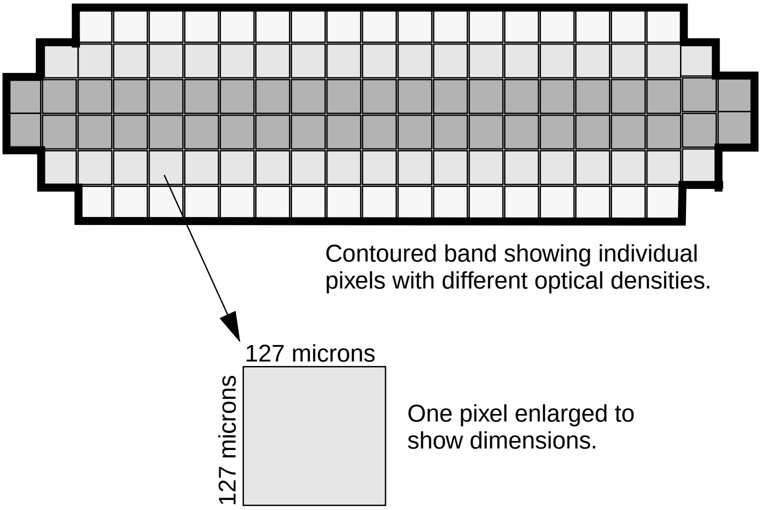


Fig.6-7. Magnified illustration of a contoured band scanned at 127 x 127 microns.

If the bands on your gel are irregularly shaped (for example, dumbbell-shaped) you may prefer using the contour functions for defining bands. Contouring will draw a "lasso" around a band and use the O.D. of all the pixels within the boundary to determine band quantitation.

6.5.a. Contours

The functions needed to contour bands are found in the **Contour Tools** palette.

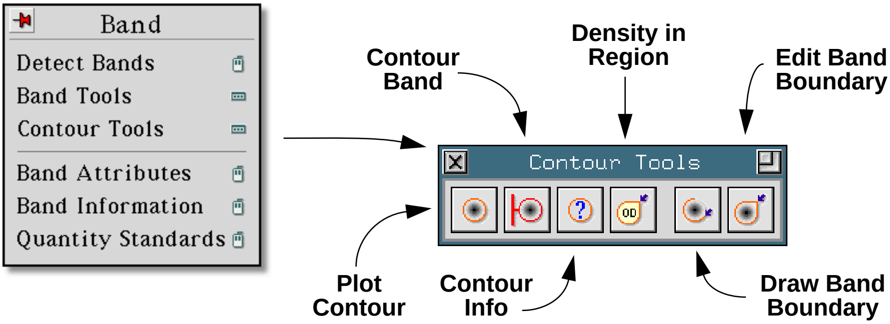


Fig.6-8. **Band** menu and **Contour Tools** palette.

Plotting Contours

Before using any of the contour features, we suggest that you increase the magnification of the image so that you can see all the pixels of the band. This allows you to position the cursor more accurately.

Assign the function **Plot Contour** to the mouse and place the cursor on the edge of the band.

Clicking the mouse button will display a contour enclosing pixels whose O.D. is equal to or greater than that of the pixel at the cursor.

If the contour does not satisfactorily encircle the band, reposition the cursor and click again. A new contour will be drawn in place of the old.

To access information about the size, density, and average density of the contour, use **Contour Information**. Point the cursor at the contour and click the mouse. A pop-up box with information about the contour will be displayed.

Converting a Contour into a Band

When you are satisfied with the contour's definition of the area of your band, you can redefine the contour as a band with the function **Contour Band**.

Contour Band converts any currently displayed contour into a band, and assigns that band to the nearest lane. The contour boundary will change color from yellow to red, indicating that the "lasso" is now a band.

Once a contour is converted, all the functions in the **Bands** menu can be applied to it.

Calculating the Density in an Arbitrary Region

Use **Density in Region** to specify a region about which you would like density and area information.

Drag the mouse button to which this function is assigned. Move the cursor as if it were a pen, to enclose a region of interest.

When you close the border, it changes color and a pop-up box with information about the enclosed area will be displayed.

Drawing Band Boundaries

If you prefer to draw the boundary of the band manually, you can do so using the function **Draw Band Boundary**.

Assign this function to a mouse button, and place the cursor at the edge of the band.

Drag the cursor along the perimeter of the band. The cursor will act like a pen, drawing a line that defines the boundary of the band.

Sometimes, you may find that your control of where the cursor draws is not as precise as you would like. It may be necessary to backtrack and retrace the path of the band boundary. When you backtrack small distances, the previous path will be erased and the new one will replace it. For this reason, it is important to magnify the image before drawing a contour. If you try to draw a very small contour, the system will think that you are backtracking, and erase the boundary.

When the cursor crosses over the boundary you are drawing, the contoured band is complete.

Once the lasso is complete, the **Draw Band Boundary** command continues to update the lasso. Each time the mouse crosses its old path, the new path is used instead.

Editing Band Boundaries

If your contoured band outline needs minor adjustments to follow the perimeter of the band, you can fix it using **Edit Band Boundary**.

First, assign the function to a mouse button, then drag the cursor as if it were a pen to "redraw" parts of the band boundary.

6.6. Blots

6.6.a. Defining a Blot Matrix

If you are working with a dot blot or a slot blot, you already know the number of lanes and the number of bands in each lane of your image. Rather than creating lanes and detecting bands, you can skip these steps by telling **Diversity Database** the number of lanes and bands in your blot. **Blot Dimensions** (on the **Blot Tools** palette in the **Lane** menu) lets you create such a matrix in a single step.

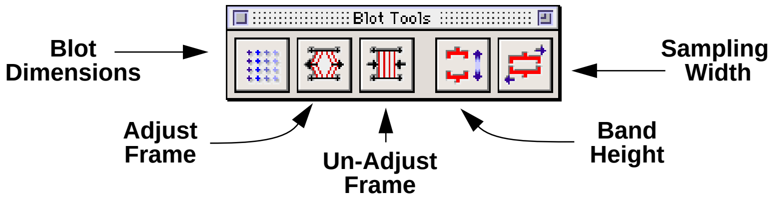


Fig.6-9. The **Blot Tools** palette.

First, select **Blot Dimensions** from the palette.

Click the mouse button in the window in which you wish to create the dot/slot blot.

A dialog box will ask you to enter the number of rows (horizontal) and the number of columns (vertical) in the blot.

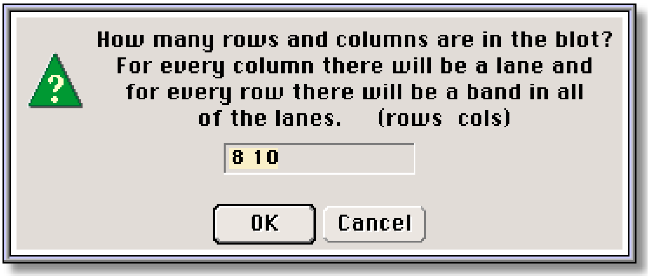


Fig.6-10. Dialog box for setting blot dimensions.

Enter the number of bands per lane followed by the number of lanes on the blot. Separate these two values with a space (no characters, punctuation, etc.).

Click on the OK button to execute the action.

Another pop-up box will ask you to confirm the creation of a blot with the settings you specified.

If you choose to continue, the dialog box will disappear and a blot matrix will be overlaid on the image of your gel.

6.6.b. Adjusting the Blot Matrix

Next, you will need to adjust the position of the blot matrix so that it lines up with the regularly spaced bands on your dot/slot blot. Use **Adjust Frame** (see Section 5.2.b) to move all the lanes and bands simultaneously (move one lane and the others will move proportionally along with it).

The position of the cursor determines which side of the matrix is going to move. Place the cursor at the corner of the matrix to move that corner and the two adjacent sides. If the cursor is on one side of the matrix, only that side will move.

Click and drag the cursor to reposition the blot matrix.

If necessary, the **Adjust Lane** (see Section 5.3.b) and **Adjust Band** (see Section 6.3.b) commands can be used to "fine-tune" the placement of lanes and bands within the matrix.

6.6.c. Setting Band Height and Width

You may want to adjust the width or the height of the dot samples in order to maximize the area being sampled.

The sample width should be set to encompass as much of the bands as possible without including background regions. Including background area will decrease the average pixel values that comprise the quantitation.

Use the functions **Band Height** and **Sampling Width** to change the width and height of the dot samples.

Band Height changes the vertical distance between the brackets for every defined band. When you select this function, a dialog box will ask you to specify the height in millimeters that should be used for all bands.

Fig.6-11. Setting the band height.

After entering a value, click on the OK button.

If your bands are designated with brackets, the brackets will now be redisplayed at the distance you specified.

You can experiment with various heights until you find one that includes the maximum area of the band without including background.

Sampling Width alters the width of all the bands in your blot.

Use this function to display a dialog box in which you can specify a new sample width (in millimeters).

After entering the new value, click on the Apply button.

You can also use the **Lane Sampling Width** function to change the width of individual lanes in the blot. To do so, assign **Lane Sampling Width** to the mouse and click on the particular lane in the blot matrix that you wish to change. A dialog box will appear, asking you to enter the new sample width for the lane you selected.

You can repeat this procedure, trying different sample widths, in order to find the optimal width that includes the maximum area of the band without including background noise.

7. Building a Database

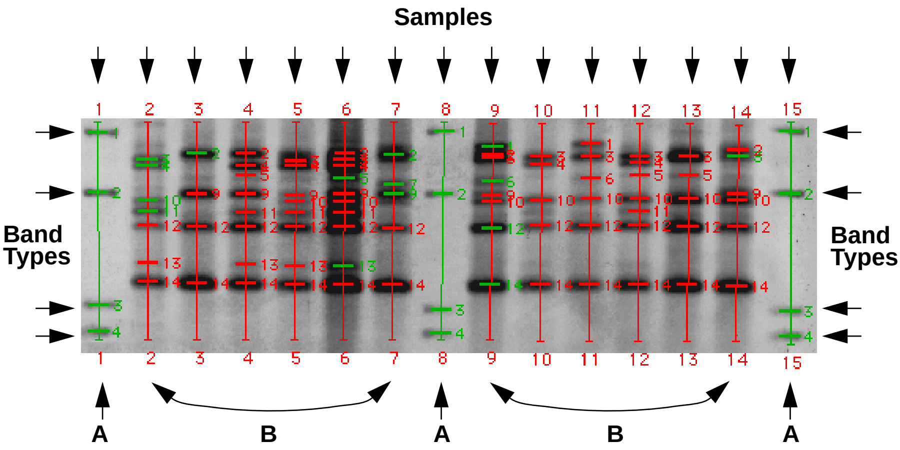
Once you have defined the lanes and bands on your gel, you are ready to enter the gel into a database file. Database files store all the information saved in your gel and allow you to perform comparative analysis on lanes, matched bands, or multi-lane samples within a single gel, a user-defined set of gels, or all the gels in your database. Database files can also be used as flexible gel cataloguing systems, allowing the user to store up to 50,000 lanes in a single database and perform similarity searches on those lanes in under 30 seconds.

Diversity Database uses several terms which have specific meanings as they relate to a database file. It is worth taking a moment to explain these terms before continuing.

Sample: Typically, a sample is a lane on one of your gels, although **Diversity Database** also has the capability to handle multiple lanes per sample (multiple probes) provided that each lane belongs to a different Band Set. A database sample can include as many lanes as necessary to define your biological sample.

Band Set: Band Sets are the basis by which **Diversity Database** compares lanes. They are used to assign standard bands to lanes and serve as the standard of comparison for experimental bands. Lanes which are similarly detected (i.e., same enzyme and probe, etc.) should be placed in the same Band Set. Observed bands within a band set are known as Band Types.

Band Type: Band Types (i.e., observed bands) are essentially matched bands within a Band Set. Adding a Band Type to a Band Set records the position of the band, as well as molecular weight or isoelectric point information (if available). If other lanes in the Band Set contain bands in with same position, with the same molecular weight, etc. as a previously observed band, they are identified as belonging to the same Band Type. Comparisons are based on different lanes (i.e., Samples) sharing the same classified bands (i.e., Band Types).



A: The Standard Band Set **B: The Experimental Band Set**

Fig.7-1. Samples, Band Sets & Band Types.

Each of these terms will be described in greater detail in later sections. For now, let us move on to the database file itself.

7.1. The Summary Form

7.1.a. Creating a New Database

The **New Database** command (in the **File** menu) creates a new database file and displays a new **Database Summary** form on-screen. The new database file will be unsaved and will not contain any gels.

7.1.b. The Database Summary form

The **Database Summary** form is the on-screen representation of the database file. It accesses the six main dialog boxes that are used to run a database file: the **Gels** list, the **Gel Layout** form, the **Samples** list, the **Band Sets** form, the **Searches** form, and the **Populations** form.

It also allows you to save or rename your database, and to write a brief description of it. The **Summary** form will remain on-screen for as long as the database file is loaded. Dismissing the form will close the database file and each member gel. Experienced users may prefer to use the palette version of the **Summary** form, which is accessible by clicking on the resize button.

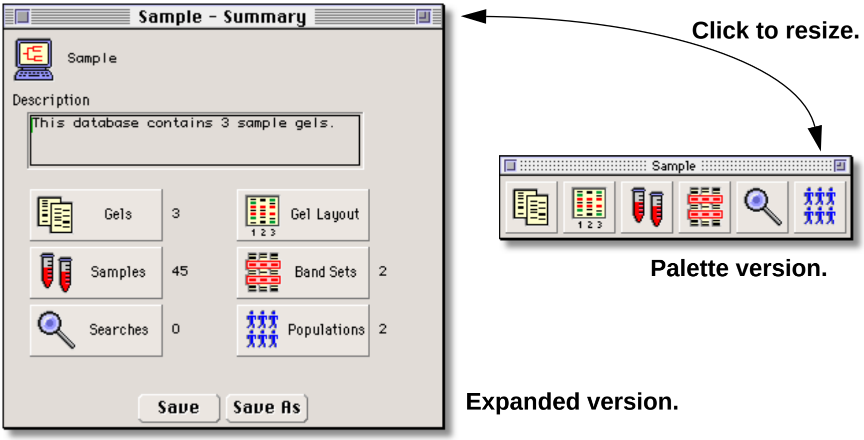


Fig.7-2. **Database Summary** form.

The numbers next to the buttons on the expanded **Summary** form indicate the number of members of each type included in the database. For example, the “3” next to the **Gels** button in the Figure above indicates that the database contains 3 gels, the number “45” next to the **Samples** button indicates that those 3 gels contain 45 individual samples, etc.

7.2. The Gels List

Clicking on the first button on the **Summary** form displays the **Gels** list. All gels currently included in the database will be listed there. The name of each gel is accompanied by its scan date, and any user-defined category or descriptive information.

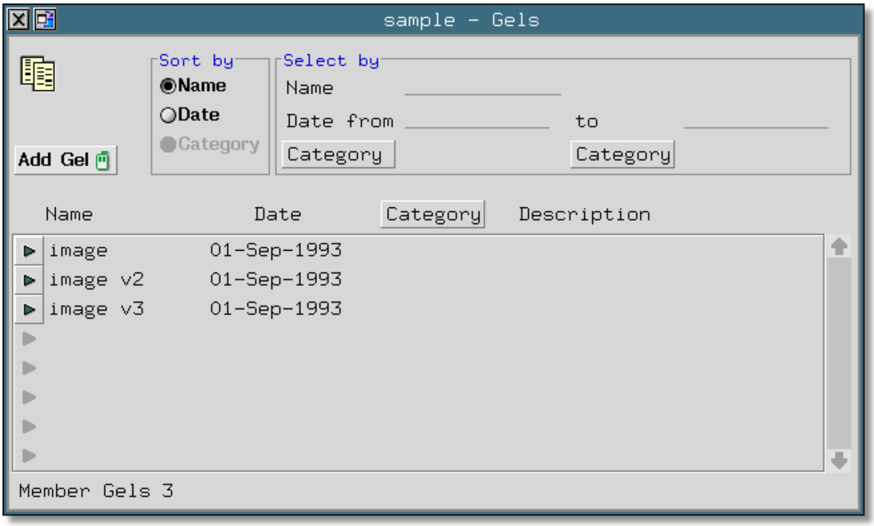


Fig.7-3. The **Gels** list.

Use the triangular "pop-up" icons on the left of the gel name to open the gel, open the **Gel Layout** form (see Section 7.3) for that gel, or to exclude the gel from the database (Show Image, Show Layout, and Exclude, accordingly, see below). Excluding a gel will delete all Sample and Band Set information about the gel from the database.

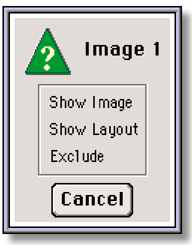


Fig.7-4. The **Gels** pop-up form.

To add a new gel to the database, assign the **Add Gel** command to the mouse, open the gel that you wish to add, and click on the gel. The name of the gel you selected will appear in the **Gels** list.

7.2.a. Sorting and Selecting Gels

A database may include hundreds of gels. Two facilities exist for controlling the display of gels within the **Gels** list: the Sort By and Select By fields.

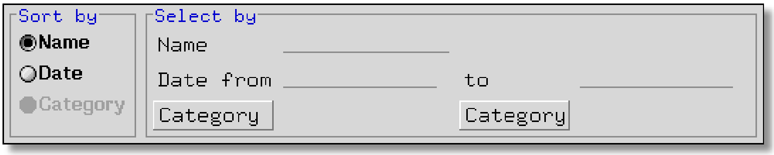


Fig.7-5. The **Sort By** field.

The Sort By field allows you to sort all the gels currently displayed in the **Gels** list by name, date, or user-defined categories. The Select By field allows you to restrict the field of displayed gels by name, date, or user-defined category. To define a new category, click on one of the Category buttons. The following form will appear:

Fig.7-6. The **Select Category** pop-up box.

To create a category, simply click on the Edit button. The following form will appear:

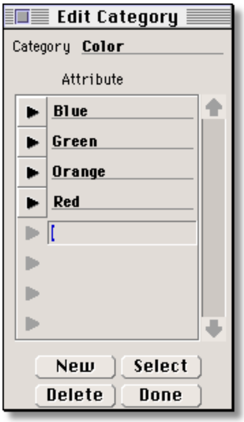


Fig.7-7. The **Edit Category** form.

Type the name of the new category next to the Category prompt (e.g., "Color") and list attributes of that category in the Attributes field (e.g., "Red," "Green," "Blue," etc.). The form will automatically sort your attributes alphabetically within the Attributes field. Categories and attributes can be defined to reflect any characteristics of your gel that would be useful to sort by. Typical categories might be "Enzyme," "Primer," "Probe," "Type," "Who," etc.

Once you've created a category and attribute field, you can choose to apply it to the **Gels** list with the Select Category pop-up.

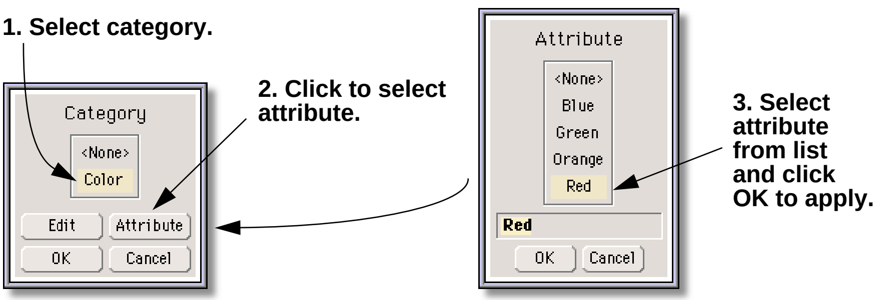


Fig.7-8. Selecting a category & attribute.

Click on a Category button to access the Select Category form. Select the category to be applied from the available categories on the list (or select "None"). Click on the Attribute button to specify an attribute. Click OK to apply your selection to the **Gels** list.

Note: The Sort By and Select By fields are general sorting and display tools that appear on several forms in **Diversity Database**, as do Category buttons. Their behavior throughout the software is consistent with the description given above.

7.3. The Gel Layout Form

Comparing samples across multiple gels needs more textual information than previously required in **Diversity One™**. This information can be entered and edited on the **Gel Layout** form.

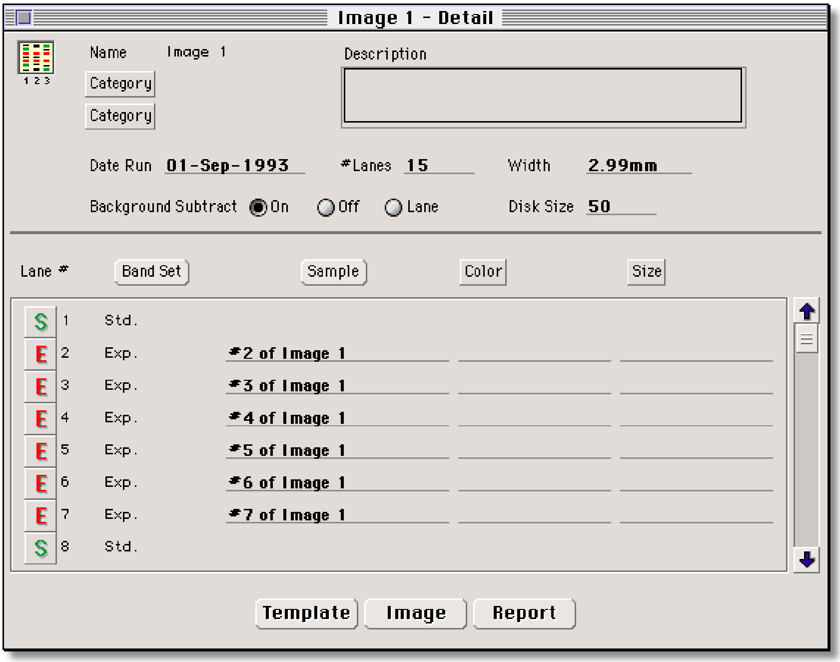


Fig.7-9. The **Gel Layout** form.

The **Gel Layout** form is accessible from the **Summary** form (Section 7.2), through the **Gels** list (Section 7.3), or with the **Layout** command in the **Gels** menu (see below).

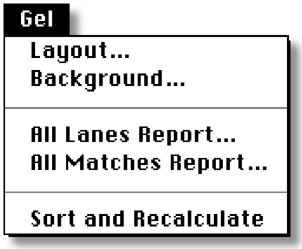


Fig.7-10. The **Gels** menu.

The following information about your gel can be specified at the top of the **Gel Layout** form:

- Categories / Attributes information for the whole gel
- A brief description of the gel
- The gel run date
- The number of lanes
- The lane width (see Section 6.2.a)
- Lane-based background subtraction and disk size (see Section 5.4)

Each lane in your gel has its own line on the **Gel Layout** form. The following information can be specified for a lane:

- Band Set type (Standard or Experimental)
- Lane number
- Band Set name
- Sample name
- Categories / Attributes information for individual lanes

The pop-up buttons on the left side of the form offer a number of choices pertaining to the individual lanes on your gel. Clicking on one will cause the following form to appear:

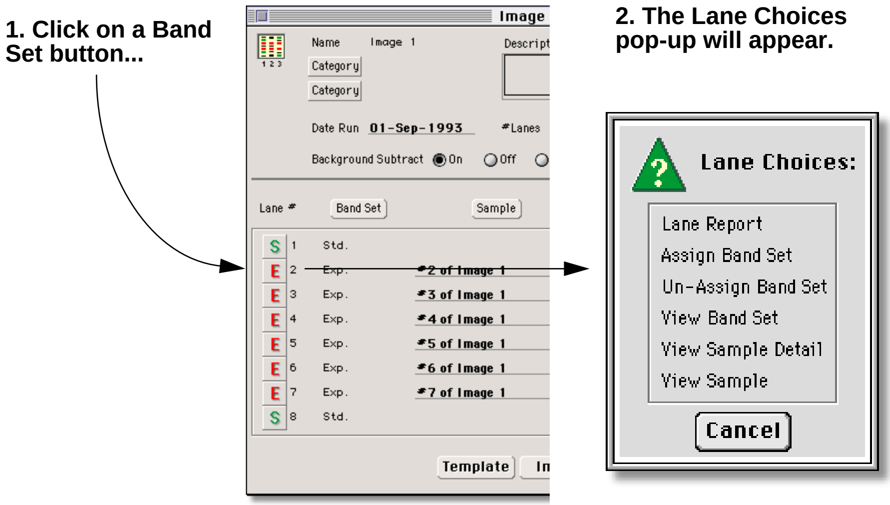


Fig.7-11. The **Lane Choices** pop-up form.

Choosing Lane Report will open a customizable lane report form that can be printed or exported (see Section 9.1.c).

Choosing Assign Band Set allows you to select the specific Band Set that is to be applied to that lane (see Section 7.5 on creating Band Sets).

Un-assign Band Set allows you to remove the Band Set that is currently applied to that lane.

View Band Set displays the **Band Set** form for the Band Set that is currently applied to that lane.

View Sample Detail displays the **Sample Detail** form for the current lane (see Section 7.4).

View Sample opens an image window containing the current Sample and displays it with the current Band Types overlaid on the image.

7.3.a. Creating a Template

If the gels you run have common detection methods or descriptive information, you can save a lot of repetitive typing by creating a gel "template." Templates store the information that you enter for one file and allow you to apply it to another. To create a template, open the **Gel Layout** form for a gel that you consider to be typical of your research and click on the **Template** button. The following form will appear:

Fig.7-12. The **Template Options** list.

Select Make New from the list, and your new gel template will appear, based on the gel you selected. To apply a pre-existing template to a different gel, open the **Gel Layout** form, click on the Template button, and select Apply Template from the Template Options list. Edit Template allows you to make changes to pre-existing template.

7.4. The Samples List

Each sample present in the database can be reviewed and edited from the **Samples** list.

Fig.7-13. The **Samples** list.

The form offers you Sort By and Select By fields (see Section 7.2.a) that allow you to display specific groups of samples. Click on the Search button at the bottom of the form to perform the search.

Clicking on the pop-up button to the left of a sample name accesses the **Sample Detail** form (see below) which allows the user to enter extensive information about a particular sample type.

Fig.7-14. The **Sample Choices** pop-up box.

The **Sample** pop-up form also allows the user to display the Sample in its own image window or to delete the Sample from the database.

Samples are automatically entered into the database as you add gels to it, and samples can be added by typing directly into the Names list, but the list can grow out of date if gels are unlinked from the database.

Click on the Cleanup button if you have extraneous samples in your list, and **Diversity Database** will remove all samples that are not linked to a current gel in the database.

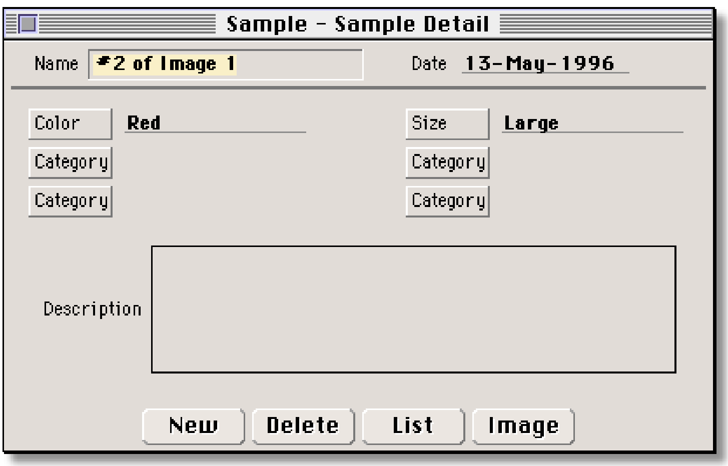


Fig.7-15. The **Sample Details** form.

Click on the New button to create a new, empty, **Sample Detail** form.

Click on the Delete button to remove the current sample from the **Sample** list.

Clicking on the List button to open the **Sample** list.

Click on the Image button to open an image window of your sample lane.

7.5. The Band Set Form

Band Sets are the basis by which **Diversity Database** compares samples. Lanes which are similarly detected (i.e., same enzyme and probe, etc.) should be placed in the same band set. Lanes which are *not* similarly detected belong in *separate* band sets.

Band Sets are easiest to think of as a combination of **Diversity One's Allele Sets** and **Mr pI Template** with added functionality. They are used to assign standard bands to lanes and serve as the basis of comparison for experimental bands.

There are two primary types of Band Sets: *standard* and *experimental*. Each standard lane in your gel should belong to a standard Band Set and each band in the standard lane should be identified in the Band Set. The more standard lanes you run in your gel, the more accurately **Diversity Database** will model your experimental lanes.

Experimental lanes belong in experimental Band Sets. Experimental Band Sets are created by identifying all of the experimental Band *Types* that occur in a gel. It is important to choose a gel that is typical of your research from which to create your experimental Band Set so that all the Band Types in which you are interested are initially included. This saves you from having to add new Band Types to your Band Set at a later date.

Band Types are manually added to a Band Set by clicking on one band belonging to the set using the **Add Band Type** command, or by typing new entries on a **Band Set** form. Once created, a saved Band Set can be applied to new gels as you enter them into the database with a minimum of editing. This will be discussed in detail later.

For the moment, consider the following gel:

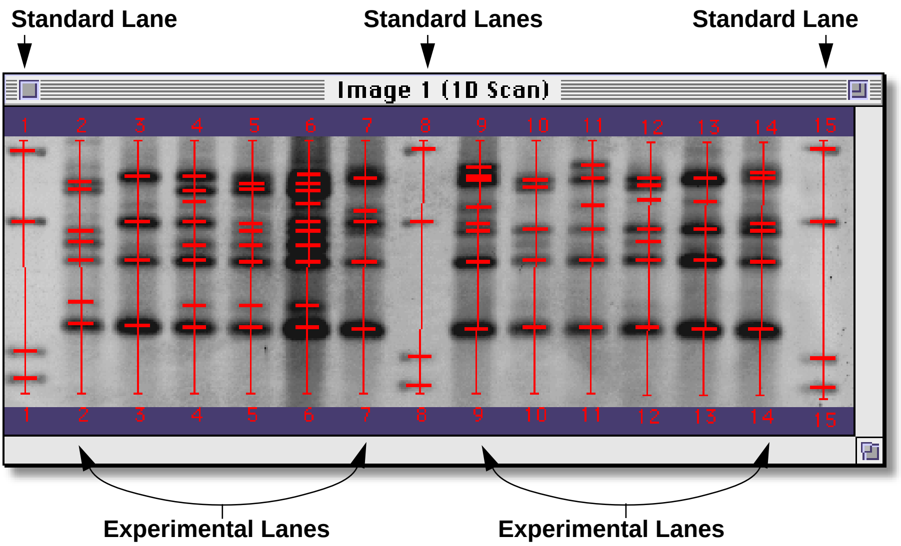


Fig.7-16. Standard and experimental lanes on **Image 1**.

Image 1 contains 15 lanes: 3 standard lanes (lanes 1, 8, and 15) and 12 experimentals. We will use **Image 1** as an example to illustrate how to create standard and experimental Band Sets.

7.5.a. Creating a Standard Band Set

The first step in analyzing **Image 1** is to create a standard Band Set. To do so, click on the Band Set button on the **Database Summary** form. A pop-up box will appear. Select New Band Set from the list and the following form will appear:

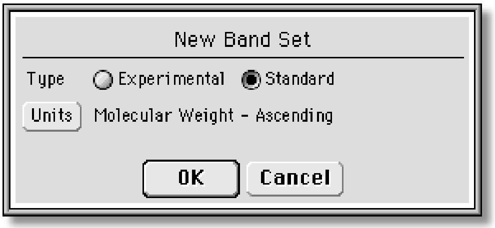


Fig.7-17. Creating a standard Band Set.

Select Standard next to the Type prompt, and specify your units: in this case Molecular Weight - Ascending (You also have the option of choosing Base Pairs, Isoelectric Point, Normalized Rf, or any other units which you specify.). A new standard **Band Set** form will appear:

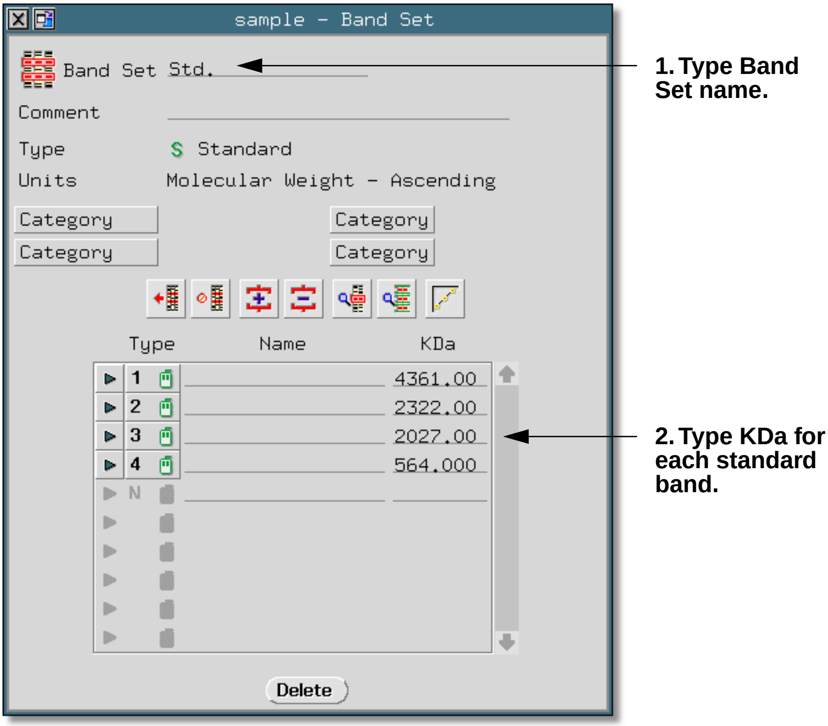


Fig.7-18. The standard **Band Set** form.

Type the name of the Band Set at the top of the form (*Std.*) and any comments or category/attribute information you want to associate with the Band Set. Enter the known molecular weight values for your standards underneath the KDa prompt. The four molecular weight values entered in Figure 7-17 represent the four standard Band *Types* in the standard lanes of **Image 1**.

The **Standard Band Set** form contains a tool palette which allows you to apply a standard Band Set to a gel and edit the Band Types. The functions on the palette are described below:

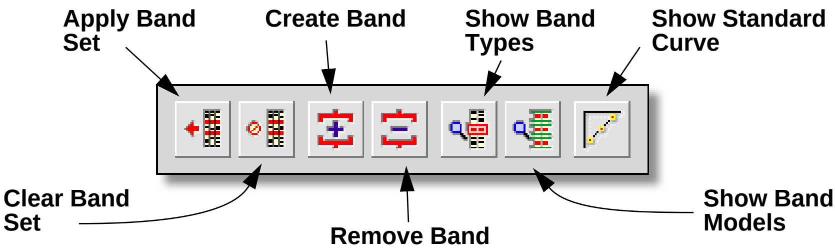


Fig.7-19. **Standard Band Set** tools.

Apply Band Set applies the current Band Set to the sample (lane) on which you click the mouse.

Clear Band Set removes the current Band Set to the sample on which you click the mouse.

Create Band allows you to create new bands manually (see Section 6.3.a).

Remove Band allows you to remove new bands manually (see Section 6.3.c).

Show Band Types displays all the currently assigned Band Types

Show Band Models displays the currently assigned Band Types along with the Band Set modeling lines (see below).

Show Standard Curve displays the molecular weight standard curve for the standard lane on which you click the mouse.

To assign the standard Band Set to the standard lanes in **Image 1**, assign **Apply Band Set** to the mouse and click on lanes 1, 8, and 15.

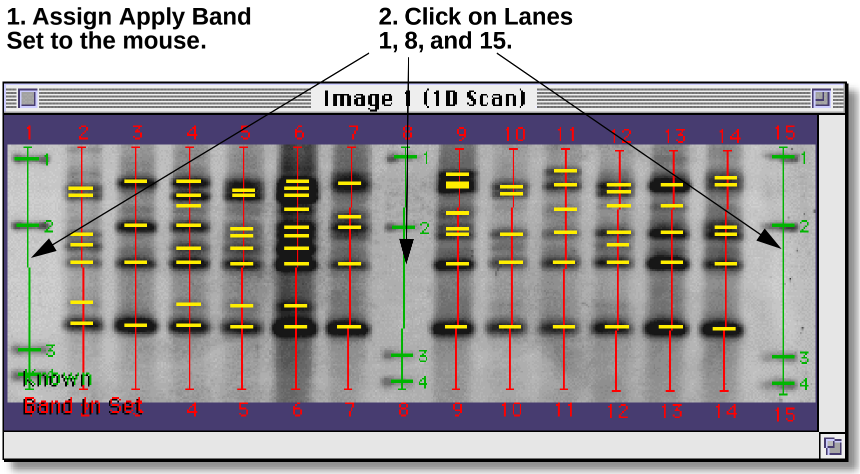


Fig.7-20. Assigning the standard Band Set.

Once created, standard lanes will appear in green. **Image 1** now has correct molecular weights assigned to its standard lanes, and we can begin to create the experimental Band Set.

7.5.b. The Experimental Band Set Form

Once you have created a standard Band Set, the next step is to produce an experimental. An experimental Band Set is a collection of experimental Band Types. Band Types are analogous to matched bands; an experimental Band Set contains all the matched experimental bands in your gel. When creating a new band set it is important that you choose a gel that is very representative of your research, as the Band Set you create will be used as a template to model all subsequent gels in your database containing that Band Set.

To create the experimental Band Set for **Image 1**, click on the Band Set button on the **Database Summary** form. A pop-up box will appear. Select New Band Set from the list and the following form will appear:

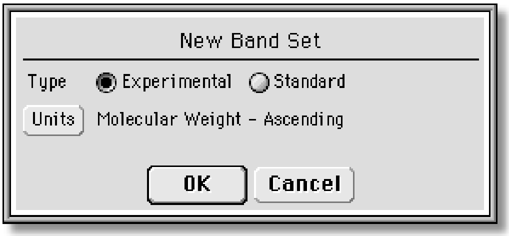
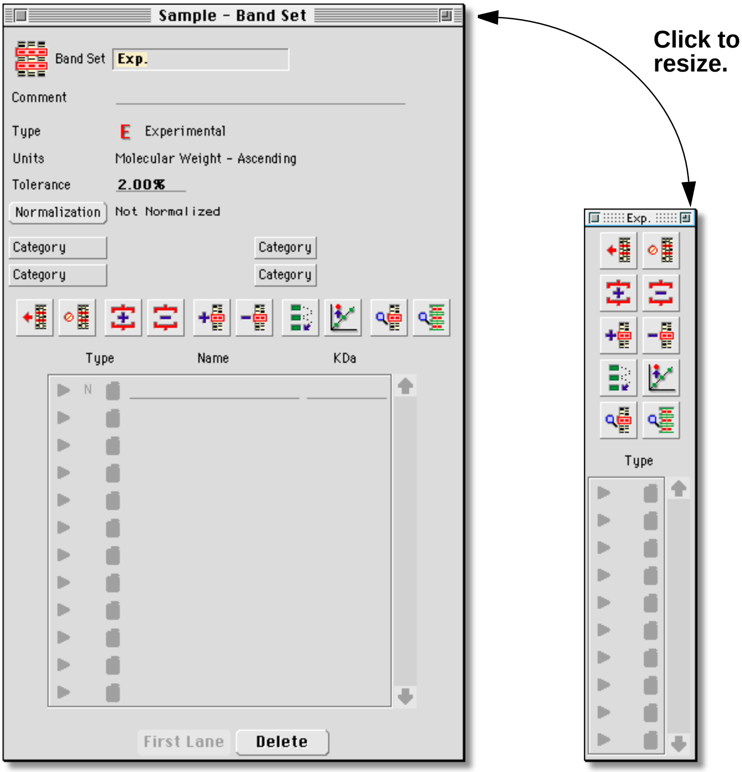


Fig.7-21. Creating an experimental Band Set.

Select Experimental next to the Type prompt, and specify your units: in this case Molecular Weight - Ascending. A new experimental **Band Set** form will appear:



Expanded version. **Palette version.**

Fig.7-22. Experimental **Band Set** form.

Type the name of the Band Set at the top of the form (*Exp.*) and any comments or category / attribute information you want to associate with the Band Set.

The Band Set Type and Units are displayed in the top half of the form, as is the matching Tolerance and Normalization button. Tolerance allows you to specify the minimum spacing that the Band Set model expects to find between Band Types; the Normalization button allows you to pick a specific band type to normalize your gel against (see Section 8.4 on *Differential Display* for more information on band normalization).

Clicking on the resize button will reconfigure the **Band Set** form to its smaller, palette version, which displays only the tool icons and mouse-assignable Band buttons (see below). We recommend that experienced users of **Diversity Database** who are familiar with the larger version try the palette version.

The **Experimental Band Set** form contains a tool palette which allows the user to apply the Band Set to a gel and edit the Band Types. The functions on the palette are described below:

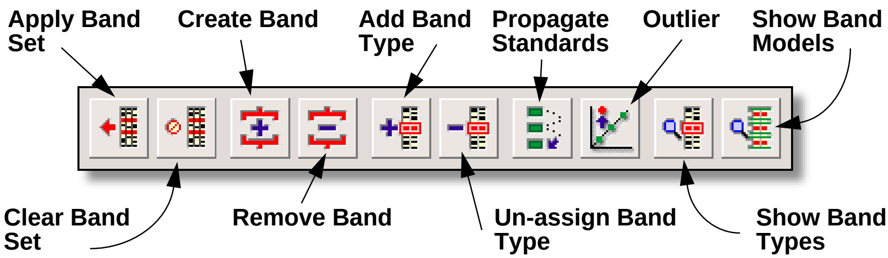


Fig.7-23. Experimental Band Set tool palette.

Apply Band Set applies the current Band Set to all the un-assigned samples (lanes) in your gel.

Clear Band Set removes the current Band Set to the sample (lane) on which you click the mouse.

Create Band allows you to create new bands manually (see Section 6.3.a).

Remove Band allows you to remove new bands manually (see Section 6.3.c).

Add Band Type allows you to create new Band Types manually by clicking on a band in the gel.

Un-assign Band Type allows you to manually un-assign Band Types by clicking on a band in the gel.

Propagate Standards applies all the known Band Types to the bands in a lane, based on a few manually-assigned bands and the Band Set model.

Outlier excludes an assigned band from the Band Set model.

Show Band Types displays all the currently assigned Band Types

Show Band Models displays the currently assigned Band Types along with the Band Set modeling lines (see below).

These functions will be explained more fully as we create the experimental Band Set for **Image 1**.

7.5.c. Creating the Experimental Band Set

Once you have created a new Band Set form, you will need to specify the Band Types in your gel. This is done by clicking on observed bands with the **Add Band Type** command assigned to the mouse. Each Band Type added to a Band Set records the position of the band, as well as molecular weight or isoelectric point information (if available). If other lanes contain bands in the same position as an observed band, they are identified according to the name (or number) of the Band Type already in the set. Comparisons are based on different lanes sharing the same classified bands.

To assign the new Band Set to **Image 1**, you'll need to assign the **Add Band Type** command to the mouse and select a lane from your gel.

The first time you click on a lane in a new gel, the following pop-up form will appear:

Fig.7-24. Apply Band Types to the whole lane.

Yes is the default selection and is the appropriate choice for most gels. Choosing Yes means that the software will automatically assign Band Types to all the bands in the lane that you selected. Since those assignments will serve as the basis for all subsequent modeling, it is important that you pick a lane with most, if not all, of the Band Types in your gel, with minimal distortion and clean, easily identifiable bands. Once chosen, the lane will be defined as the “First Lane” in your database, and will be available for visual comparison with other lanes on other gels as you attempt to assign Band Sets to them (more on that topic later).

In **Image 1** we chose lane 4 to be the first lane, clicked on it with the **Add Band Type** command assigned to the mouse, and allowed **Diversity Database** to assign its Band Types.

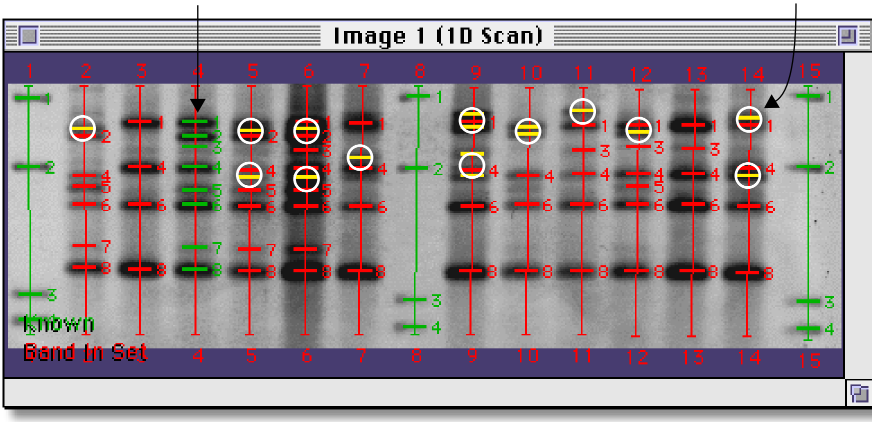


Fig.7-25. Assigning Band Types to lane 4.

Known bands appear in green. Bands which **Diversity Databases’s** modeling can confidently identify as belonging to a specific Band Type are labelled in red, with the number of the Band Type to which they belong appearing immediately to their right. Yellow bands are bands which the Band Set model can not accurately call. The Band Set model is purposefully conservative in its Band Type calling to avoid incorrect labelling, so a number of yellow band may appear, particularly on your first gel. As you can see from the above Figure, **Diversity Database** can find a majority of your Band Types automatically. The remainder you will have to identify manually.

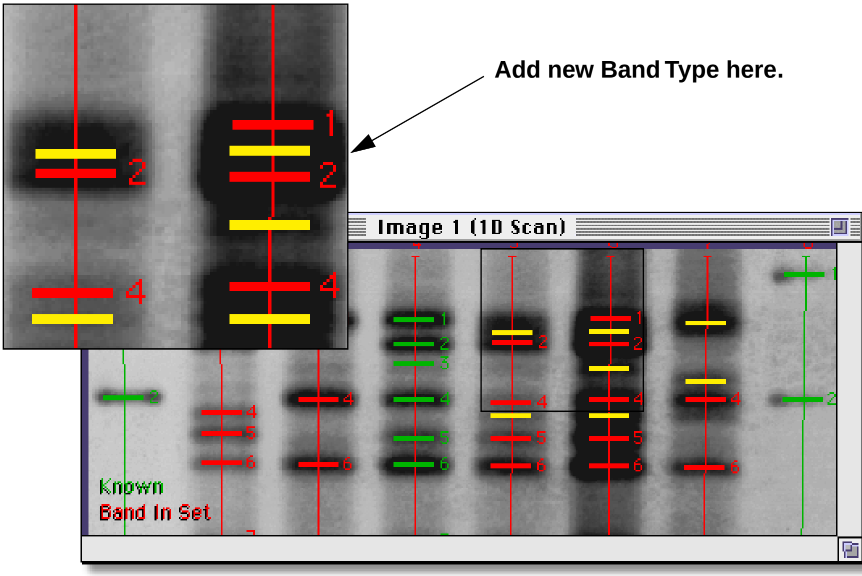


Fig.7-26. Zoom in to examine bands.

In Figure 7-25, above, we have zoomed into the upper left corner of **Image 1** to manually add the remaining Band Types. As you can see, an unknown band is present between the known Band Types 1 and 2. This means that your gel contains a Band Type that was not present in the first lane, but is present in other lane, and therefore belongs in the Band Set. Click on the band with **Add Band Type** to add it to the Band Set. The new band will become Band Type 2, and all other Band Types will renumber accordingly.

Sometimes it will not be as clear whether a band belongs to a new Band Type or to an existing one, and additional editing will be required before you can add the new type. For example:

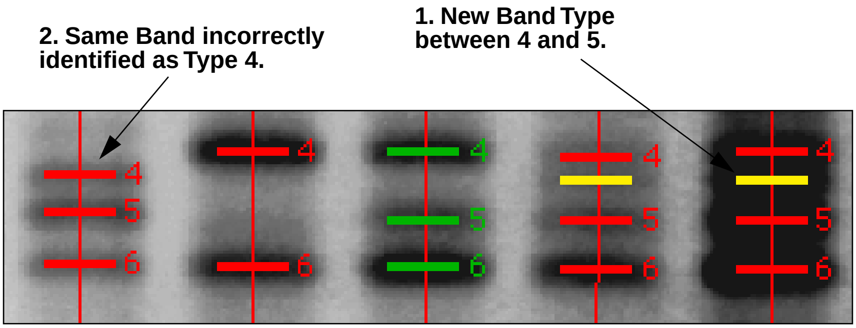


Fig.7-27. More unknown Band Types.

In the region of **Image 1** displayed above, we can clearly see a new band between Types 4 and 5 on the right side of the gel (**1**), and yet, if you follow a visual line across the gel to the left side, that same new Band Type is identified as Band Type 4 (**2**). In a situation like this, the first step is to improve the band modeling around the region in which we are interested. We do this by assigning known bands.

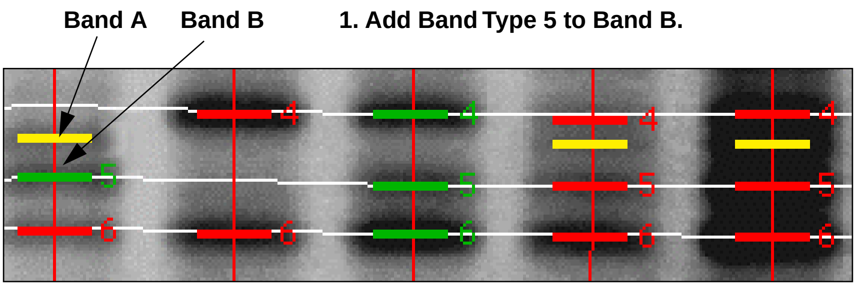


Fig.7-28. Adding a known Band Type.

Declaring Band B to be Type 5 improves the Band Set modeling for the region, even though the Band B had already been correctly identified as belonging to Band Type 5 (see Fig. 7-26, above). The white lines represent the modeling lines used by the Band Set model. Band A is now labelled as an unknown, and it is clear that a new type exists between Types 4 and 5.

To assign a band to a particular Band Type (as opposed to a new type) it is necessary to use the Band buttons on the **Band Set** form.

Assign Band Type
5 to the mouse.

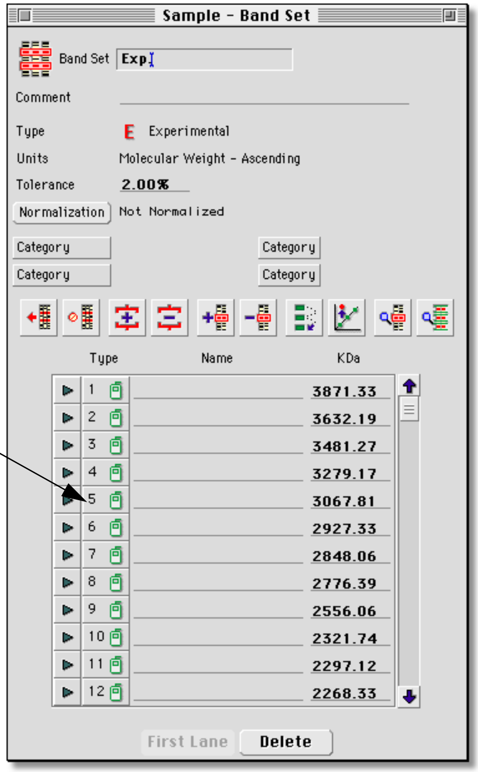


Fig.7-29. Assigning Types from the **Band Set** form.

Underneath the Type prompt are listed mouse-assignable buttons for each of the Band Types in your Band Set. Clicking on a button will assign that Type number to the mouse. Clicking on a band will then declare that band to be of the known type that you specified, and the Band Set model will adjust accordingly.

To display the Band Set modeling lines seen in Fig. 7-27, it is necessary to use the **Show Band Models** command from the **Band Set** form tool palette. Clicking on that icon will immediately display the modeling lines in the current gel window. To remove the modeling lines, use the **Clear Overlays** command (see Section 5.1), or click on the **Show Band Types** icon (also in the **Band Set** tool palette) to return the gel to show Band Types only.

To create the new Band Type in Fig. 7-27, assign **Add Band Type** to the mouse and click on the unknown band.

General Hints on Adding Band Types

Band Types should only be added to areas of the gel that are modeled well. If the gel has standard lanes, then the best choice is to add the type to a band in a lane next to a standard lane. Otherwise, find a band that is in an area of the gel where its neighboring bands are assigned correctly, and the white band set lines intersect these bands.

Completing the Band Set

Repeat the steps listed above until each Band Type in your gel has been assigned to the Band Set. Your gel should have no yellow (unknown) bands visible, and your modeling lines (use **Show Band Models**) should be relatively straight and evenly spaced. On well-modeled gels, the modeling lines will intersect at or near the middle of the Band Types across the gel. If your gel contains serious distortions, and the modeling lines are not even, you should consider using a different gel from which to build your Band Set. Poor modeling decisions on your first gel will influence the modeling on all the subsequent gels in your database, so take your time while creating the set, and if necessary, choose a different gel on which to base the modeling.

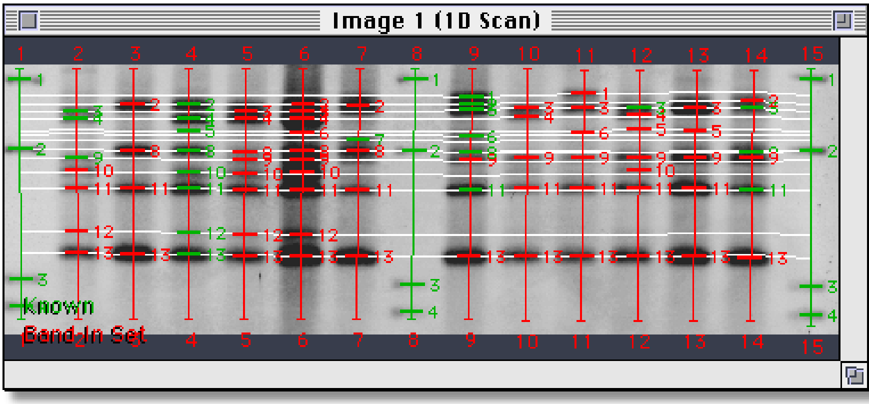


Fig.7-30. Well-modeled Band Set.

7.5.d. Applying an Existing Band Set to a New Gel

Once your Band Set is complete and you've saved the database, you should be ready to apply your new band set to any number of other gels in the database. To do so, first open the gel to which you want to apply the existing Band Set. Then, open the **Standard Band Set** form for that gel and apply it to the standard lanes in your gel following the same method previously outlined in Section 7.5.a.

With your standard lanes identified, you should be ready to apply the experimental Band Set.

First, open the **Band Set** form that you wish to apply:

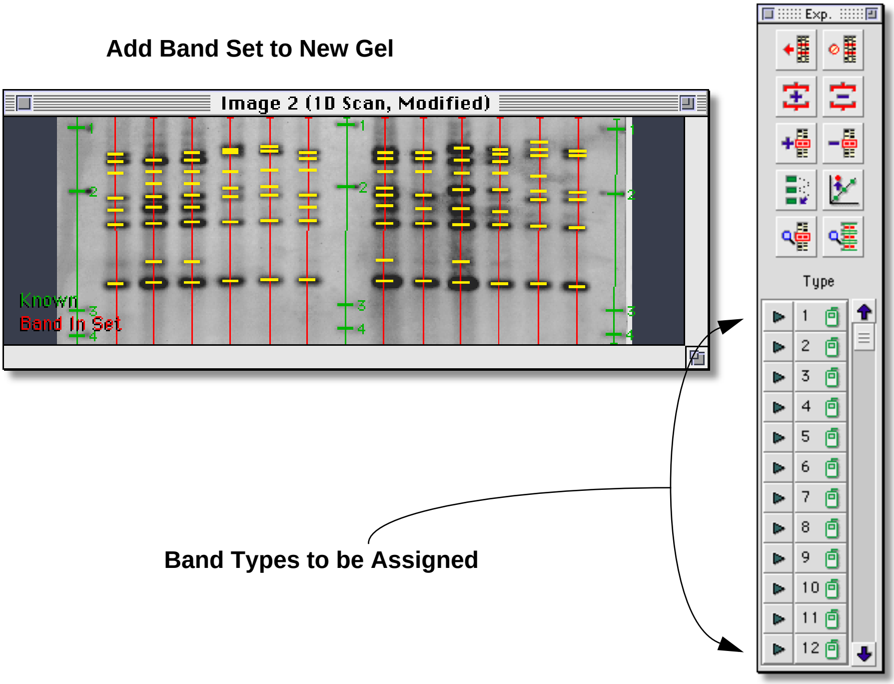


Fig.7-31. Adding an existing Band Set to a new gel (Image 2).

Next, assign the **Apply Band Set** button to the mouse and click on any un-assigned lane. The experimental Band Set will be applied to all of the remaining lanes in your gel.

Diversity Database will also automatically assign Band Types to each of the un-assigned lanes in your gel, based on the modeling information recorded when you created the Band Set.

Unknown bands appear in yellow.

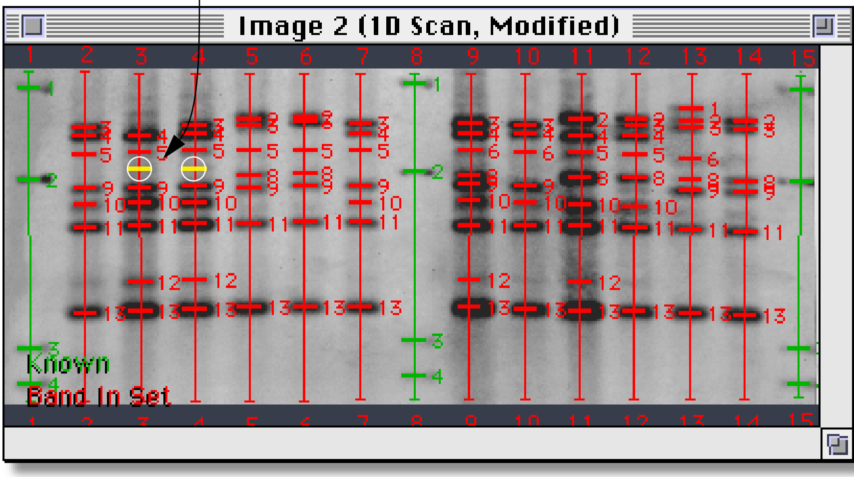


Fig.7-32. Band Set model finds majority of Band Types.

As you can see from the image above, **Diversity Database** can correctly call most of the bands in your gel without requiring you to do any editing. However, depending on the degree of distortion in a new gel, or on differences between the new gel and the gel from which the Band Set was modeled, it is not unusual to find unknown, or incorrectly called bands in your gel. Further editing will be required to include those bands in the band set.

For example:

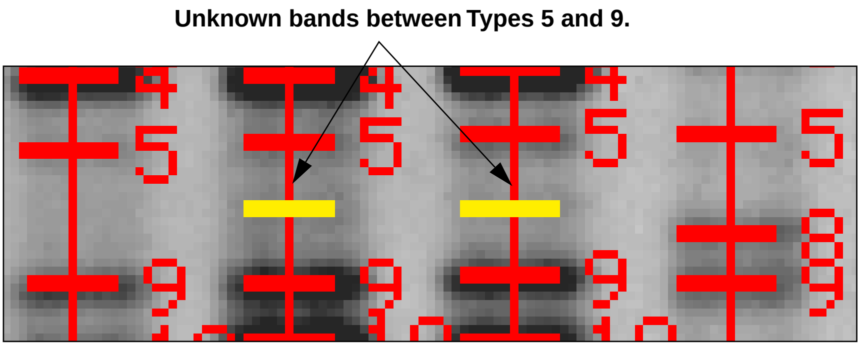


Fig.7-33. Unknown Band Types.

In the region of **Image 2** displayed above, you can clearly see an unknown band between Types 5 and 9 on the right side of the gel. In a situation like this, the first step is to improve the band modeling around the region in which you are interested. You do this by assigning known bands.

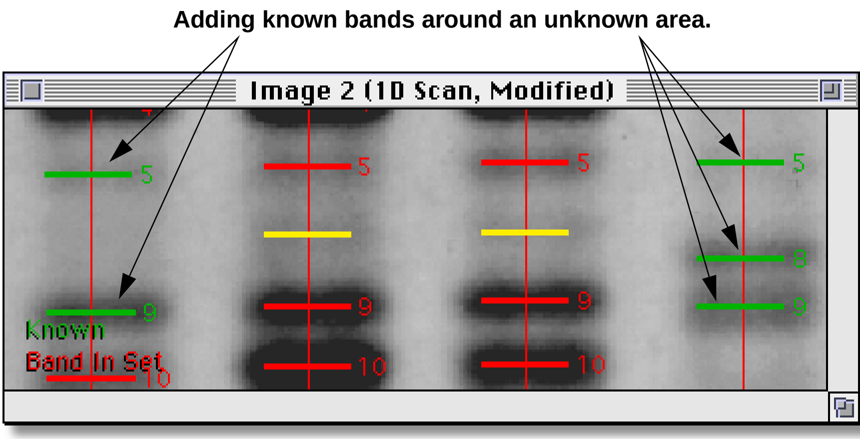


Fig.7-34. Known bands improve the modeling.

As a general rule, you should not assign a Band Type to a unknown band unless you are absolutely certain of its identity. Instead, try assigning band types to red, model-assigned bands that are surrounding the unknown band. This will cause a less severe distortion on the Band Set model. However, as with Figure 7-33, adding known bands will not always cause unknown bands to be correctly assigned. Using the **Show Band Models** command reveals that the unknown bands in **Image 2** are directly in between two Band Types (7 & 8) predicted by the model:

Unknown bands are between predicted Band Types 7 & 8.

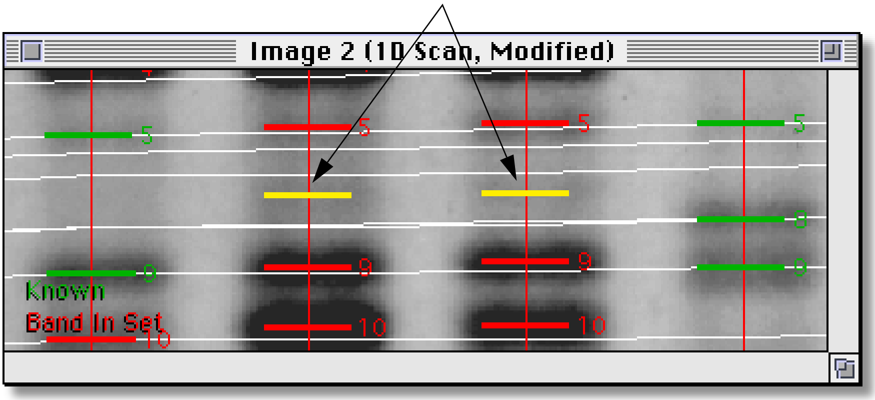


Fig.7-35. Use **Show Band Model** to display modeling lines.

At this point, you should consider the possibility that this band belongs to a new Band Type, not included in the model. If so, assign **Add Band Type** to the mouse and click on the unknown band. The new type will be added to your Band Set, and all the other gels in your database using that type will be updated to reflect the change.

If you are sure that the band it isn't a new type, then assign the type to the mouse and click on the unknown bands (in this case Type 7).

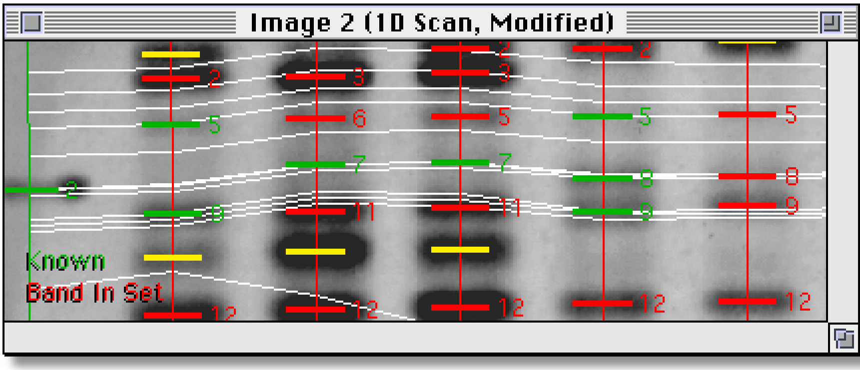


Fig.7-36. Manually assigning Type 7 to unknown bands.

As you can see in Figure 7-35, assigning Type 7 to the unknown bands created serious distortions in the Band Set model, causing correctly modeled bands to be incorrectly labeled. This suggests that the unknown bands belonging to a new type. However, if you are convinced that the unknown bands belong in Type 7, you should consider using the **Outlier** command on them.

The **Outlier** command tells the Band Set model to disregard known bands that you click on when it determines its modeling. Clicking on the two Type 7 bands with **Outlier** returns **Image 2** to a reasonable model.

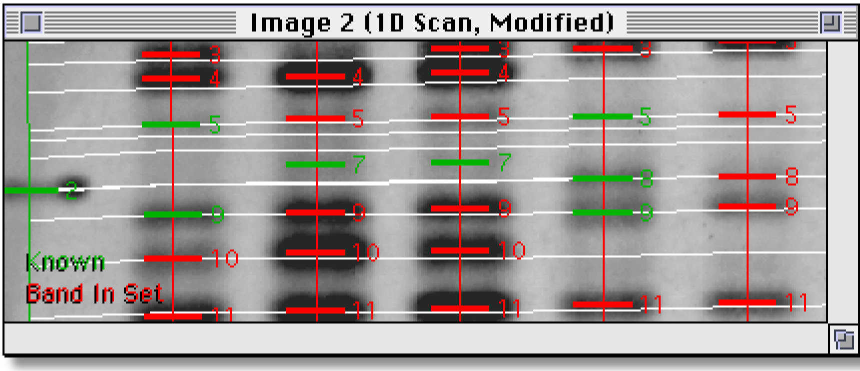


Fig.7-37. Manually assigning Type 7 to unknown bands.

7.5.e. Other Editing Techniques

Using Normalized Rf

If you have a sample that you can reliably produce in-house, you may consider using it in your gels instead of commercially prepared standards. Such samples typically contain more known Band Types than commercial standards and provide better modeling. Such gels can be analyzed by **Diversity Database** without Molecular Weight information, using units of "Normalized Rf."

Normalized RF sample lanes depend on the reproducibility on their Band Types to provide a positional standard for comparing other experimental lanes. Normalized Rf units are based on the Rf position of the Bands in the sample lane from which you create the Band Set.

To create a Normalized Rf Band Set, click on the Band Set button on the **Database Summary** form. A pop-up box will appear. Select New Band Set from the list and the following form will appear:

Fig.7-38. Creating a Normalized Rf Band Set.

Select Experimental next to the Type prompt, and specify your units: in this case Normalized Rf - Descending. A new experimental **Band Set** form will appear.

To assign the new Band Set to your gel, you'll need to assign the **Add Band Type** command to the mouse and select the sample lane that you intend to use as a standard for all your subsequent gels.

When you click on the lane in a new gel, the following pop-up form will appear:

Fig.7-39. Apply Band Types to the reference sample lane.

Yes is the appropriate selection. Choosing Yes means that the software will automatically assign Band Types to all the bands in the sample lane. Once chosen, the lane will be defined as the “First Lane” in your database, and will be available for visual comparison with other lanes on other gels as you attempt to assign Band Sets to them24.

Diversity Databases will attempt to model the rest of your gel based on the sample lane that you selected. Ideally, you will have more than one lane containing the reference sample lane in your gel. To improve the modeling for the rest of your gel, you should apply the known Band Types in your Band Set to those lanes using the **Propagate Standards** command.

Propagate Standards is a feature that not only simplifies assigning band types, but it often allows the software to do some optimizations that will significantly speed up the modeling.

Choose a sample lane that need to have Band Types assigned. Start by assigning one or two bands in the lane that are easy to identify, then try assigning types to the top and bottom bands in the lane. Once this is done, use the **Propagate Standards** command to assign the remaining bands in the lane.

Propagate Standards assigns the bands in the lane to band types one at a time. The command will only assign types to bands if it is very sure that it is the correct assignment.

Once your reference sample lanes have been modeled, add any unknown Band Types to the gel using the methods outlined in Section 7.5.c.

Assigning Band Sets from the Gel Layout Form

The **Gel Layout** form can be used to directly assign existing Band Sets to the lanes in your gel (see Section 7.3). If you commonly run gels using the same Band Sets in the same lanes, you should consider creating a Template from one of your gels and applying it to new gels as you enter them into the database.

7.6. The Search Form

The **Search** form allows you to find samples in your database according to a wide range of user-defined criteria. Searches use one of two primary search strategies- by Sample (Lane) Similarity or by Band Set membership. In either case, for lanes to be comparable, they must use the same Band Set. Later population analysis of your sample (cluster analysis, etc.) is dependent upon sample comparability.

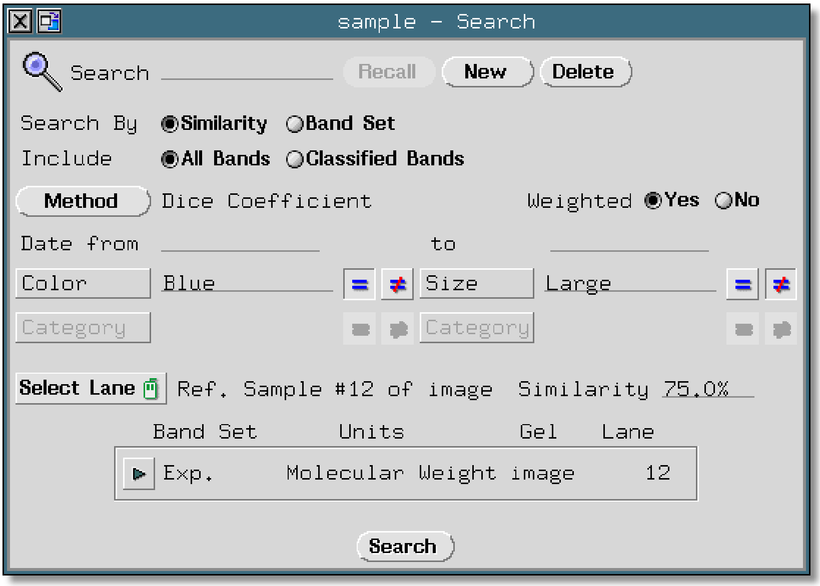


Fig.7-40. The **Search** form.

To create a new search, specify Similarity or Band Set next to the Search By prompt, and All Bands or Classified Bands next to the Include prompt.

Similarity searches allow you to select a lane from a gel by clicking on it with the Select Lane button and specifying the degree of similarity by which other lanes must match the lane you chose (i.e., at least 75% similar, 85% similar, etc.).

Band Set searches allow you to specify a particular Band Set (from the Select Band Set button), and search for all Samples belonging to that Band Set that conform to the Date and Category/Attribute information which you specify in the middle of the form.

Including All Bands in a search means that all the bands included in a Sample, including unknown bands that don't belong to any Band Type, will be used for the purposes of comparison. Including Classified Bands only, will insure that **Diversity Database** ignores bands that don't belong to Band Types when making its comparisons.

The method of searching can be either Dice Coefficient or Jaccard Coefficient (see Section 7.7.a below for a detailed description of each method). Either can be specified by clicking on the Method button.

Selecting Yes or No next to the Weighted prompt will indicate whether **Diversity Database** should use band density as a similarity consideration when comparing Band Types.

Once you are happy with the search parameters that you've specified, click on the Search button at the bottom of the form to generate a new population. If you want to save those parameters for future use, simply enter a name next to the Search prompt at the top of the form and your "Search" will be saved. To load an existing Search, click on the Recall button and select the Search from the list that appears. To remove an existing Search, click on the Delete button.

7.7. The Populations Form

The result of a search is a set of comparable samples which **Diversity Database** calls a "population". Each population appears in its own window and contains a number of different icons. Each icon represents a different tool for analyzing the population. The list of tools includes an **Image Report**, a **Population Report**, **Phylogenetic Trees**, a 3-D **Principle Component Analysis**, a **Sample Report**, **Band Frequency** analyses, and a **Similarity Matrix** report. The population can be calculated using one of three available methods: Dice Coefficient, Jaccard Coefficient, or Euclidean Squared Distance. Each analysis tool and the three search methods are described below:

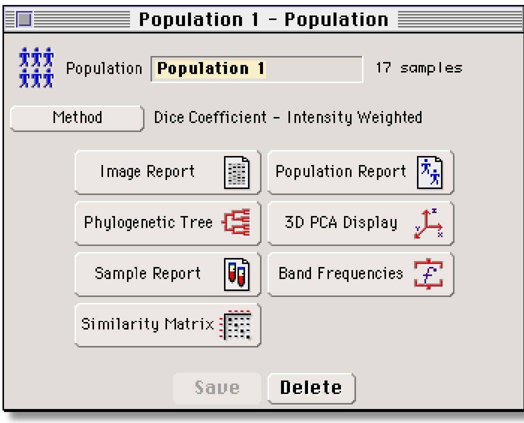


Fig.7-41. The **Populations** form.

7.7.a. Population Search Methods

Diversity Database offers three different methods for computing similarity values between samples. For simplicity, this discussion will assume one lane for each sample, and that the search was performed using the Classified Bands option. To compute a similarity value, a vector is constructed that represents the bands found in the lane. The sample vector that is constructed depends on the type of search that was made. If the search was done on Classified Bands only, then the vector **S** contains B elements (**S** = [s_1, s_2, s_3, ..., s_B]) where B is the number of band types in the lane's band set. The values for s_1, s_2, s_3, ..., s_B have the following values:

Weighting Off Search

s_i = 1 if the i'th band type is found in the lane.

= 0 if it is not found.

Weighting On Search

If the band set has a normalizing band type, then:

s_i = The normalized density of the band assigned to the i'th band type.

= 0 if the lane does not have a band assigned to the i'th type.

Otherwise:

si = The quantity of the band assigned to the i'th band type.

= 0 if the lane does not have a band assigned to the i'th type.

Let **S** and **T** be the vectors representing two samples with lanes that are in the same band set. The following similarity and distance calculation methods are available:

Dice Coefficient

$$sim = 200 \times \frac{\sum_{i=1}^{B} Min(s_i, t_i)}{\sum_{i=1}^{B} (s_i + t_i)}$$

$$dist = 100 - sim$$

Jaccard Coefficient

$$sim = 100 \times \frac{\sum_{i=1}^{B} Min(s_i,t_i)}{\sum_{i=1}^{B} (s_i + t_i) - \sum_{i=1}^{B} Min(s_i,t_i)}$$

$$dist = 100 - sim$$

Euclidean Distance Squared:

This metric is only available from the **Population** form.

$$dist = \sum_{i=1}^{B} (s_i - t_i)^2$$

$$sim = 100 \times \frac{B - dist}{B}$$

7.7.b. Population Analysis Tools

Image Report

The **Image Report** displays a series of lane images of your population, sorted in order of decreasing similarity from the Reference Sample for Similarity-searched populations, and alphabetically for Band Set-searched populations. For more information on the **Image Report**, see Section 9.1.c.

Population Report

The **Population Report** displays all the members of your population, sorted in order of decreasing similarity from the Reference Sample for Similarity-searched populations, and alphabetically for Band Set-searched populations. For more information on the **Population Report**, see Section 9.1.c.

Phylogenetic Tree

Phylogenetic trees are schematic representations of lane similarity. There are eight methods available for producing phylogenetic trees; each is described below.

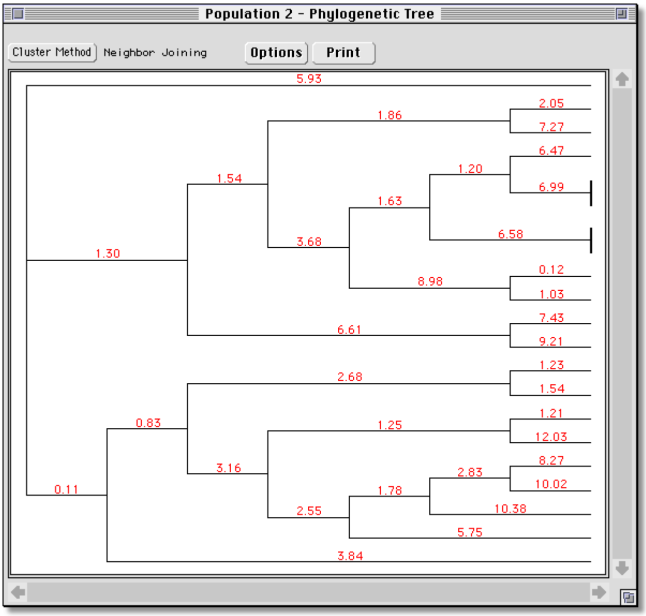


Fig.7-42. The **Phylogenetic Tree** form.

Phylogenetic trees appear in resizable windows like the one pictured in Figure 7-39. Clicking on the Cluster Method button allows you to pick between the eight varieties of phylogenetic trees available in **Diversity Database**.

Choosing a new method will automatically reconfigure the tree displayed in the window. Each method calculates the similarity of lanes in slightly different ways.

Neighbor Joining[1]

The phylogenetic tree that is computed is based on minimizing the total branch length at each stage of clustering. The method also finds branch lengths between nodes. The approximate distance between any two samples in this tree can be found by adding the branch lengths that connect the samples.

Other Methods[2,3]

The following methods are base on the algorithm below:

1: Begin with N clusters - one cluster for each sample.

2: Compute the similarity matrix for the samples.

3: Convert the similarity matrix into a distance matrix, **D**, using the appropriate distance formula.

4: Join the two clusters with the minimum distance into one cluster. Compute the similarity value for this cluster.

5: Recompute the distance matrix **D** using the cluster that was formed in Step 4.

Steps 4 and 5 are repeated until there is only one cluster. The difference between the methods below is based on the definition of minimum distance in Step 4, and on the method of computing the new distance matrix in step 5. In the discussion that follows:

p and **q** are indices indicating two clusters that are to be joined into a single cluster;

---

1. For a complete explanation of the calculations and assumptions used to generate the dendrogram, please refer to Saitou and Nei, *Mol. Biol. Evol.* **4** (4): 406-425, (1987).

2. For a complete explanation of the calculations and assumptions used to generate these dendrogram, please refer to Sneath and Sokal. *Numerical Taxonomy*, San Francisco: W. H. Freeman & Company, 1973.

3. Vogt and Nagel, *Clinical Chemistry* **38** (2): 182-198, (1992).

k is the index of the cluster formed by joining clusters p and q;

i is the index of any remaining clusters other than cluster p, q, or k;

$\mathbf{n_p}$ is the number of samples in the p'th cluster;

$\mathbf{n_q}$ is the number of clusters in the q'th cluster;

n is the number of clusters in the k'th cluster formed by joining the p'th and q'th cluster ($n = n_p + n_q$);

$\mathbf{d_{pq}}$ is the distance between cluster p and cluster q.

Single Linkage:

(also called Nearest Neighbor or Minimum Method)

$$d_{ki} = min(d_{pi}, d_{qi})$$

Complete Linkage:

(also called Furthest Neighbor or Maximum Method)

$$d_{ki} = max(d_{pi}, d_{qi})$$

Single and Complete linkage are good algorithms for indicating outlier clusters.

UPGAMA:

Unweighted, pair - group method using arithmetic averages, (also called Weighted Average Linkage.

$$d_{ki} = \left(\frac{n_p}{n}\right) \cdot d_{pi} + \left(\frac{n_q}{n}\right) \cdot d_{qi}$$

WPGAMA:

Weighted, pair - group method using arithmetic averages, (also called Average Linkage).

$$d_{ki} = 0.5 \cdot d_{pi} + 0.5 \cdot d_{qi}$$

WPGAMA is a special case of UPGAMA which favors the most recent member clusters in forming new clusters.

Centroid:

$$d_{ki} = \left(\frac{n_p}{n}\right) \cdot d_{pi} + \left(\frac{n_q}{n}\right) \cdot d_{qi} - \frac{(n_p \cdot n_q)}{n^2 d_{pq}}$$

Median:

$$d_{ki} = 0.5 \cdot d_{pi} + 0.5 \cdot d_{qi} - 0.25 \cdot d_{pq}$$

These two methods are similar to UPGAMA and WPGAMA respectively, but the distance formula contains an additional third term. Centroid and Median methods are not monatomic hierarchical clustering algorithms. In other words, the similarity value between cluster k and any other cluster may be greater than the similarity between cluster p and cluster q. This condition occurs if the centroids (or medians) of the different clusters have approximately the same distance as the distances between the samples that make up the cluster.

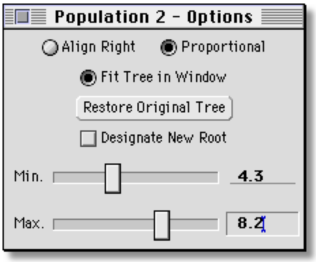
Ward's Method:

Attempts to minimize the information by describing a set of N samples using a fewer number of clusters.

$$d_{ki} = \frac{n_p + n_i}{n + n_i} d_{pi} + \frac{n_q + n_i}{n + n_i} d_{qi} - \frac{n_i}{n + n_i} d_{pq}$$

From our experience, we have found that Ward's method, UPGAMA and WPGAMA give the most plausible clusters and are affected the least by samples that are outliers.

Clicking on the Options button will access one of the following forms:



Neighbor Joining Only

All Other Methods

Fig.7-43. Two forms of the **Options** dialog box.

The form on the left has controls which are specific for the Neighbor Joining method. The form on the right is applicable to all the other tree methods. Both forms share certain features.

Each form allows you to choose between aligning the tree to the right or proportionally align it; Right alignment uses a fixed branch length for displaying the distance between nodes, proportional alignment differs for Neighbor Joining and other tree modes. In Neighbor Joining mode, proportional alignment shows branch length sizes proportional to the min./max. values (see below) set in the **Options** form. Other modes plot the nodes at locations determined by their similarity values.

The Fit in Window option scales the tree so that the entire tree will fit onto a single printed page. With Fit in Window turned off, the entire tree will appear, with correct distances between nodes preserved, and the printed tree can be tiled across multiple sheets of paper.

The Neighbor Joining **Options** form allows you to designate a new root node (the node at the very top of the tree), enabling you to look at the relationships between lanes from a different perspective. You do so by clicking on the Designate New Root button. This will highlight all the nodes of the tree on-screen. Choose the one you want to serve as the new root by clicking on it. The rearranged dendrogram will then be displayed. If you wish to return to the original form of the tree, click on the Restore Original Tree button.

The Min. and Max. sliders are active only for proportionally aligned Neighbor Joined trees. They allow you to adjust the Minimum and Maximum distance values between nodes on branches, so that you can adjust the display of your tree to highlight specific regions of data. Distances below the Min. or above the Max. values will be collapsed to fit on the form.

The Clusters option allows you to define the number of clusters (0 to 18) that will be identified in your tree for any non-Neighbor Joined tree. Entering a value greater than zero in this field will cause the tree to re-display into separate clusters, identified by letters. Populations grouped into clusters will maintain those clusters when displayed in the **3-D PCA Display** form (see below).

3-D PCA Display

PCA (Principal Component Analysis) is a method used for reducing the number of variables in a complex set of data. In **Diversity Database**, we use PCA to reduce the dimension of the sample vectors which generally have a high number of dimensions, to 3 dimensions. The samples can then be represented as a 3-D point (x,y,z) and plotted on a 3-D set of axis. This is very useful to visually inspect the similarity of the samples since dissimilar samples will appear further apart on the plot than highly similar samples. In addition, if a tree was computed and the number of clusters set, the display will assign a color to each cluster and paint each sample point in the appropriate color. The display can then be used as a visual indication of the compactness of each cluster and the dissimilarity of each cluster.

Clicking on the 3-D PCA Display button on the **Populations** form will produce the following form:

Note: If you have not defined clusters in the **Phylogenetic Tree** form, this form will display all its data points in the same set.

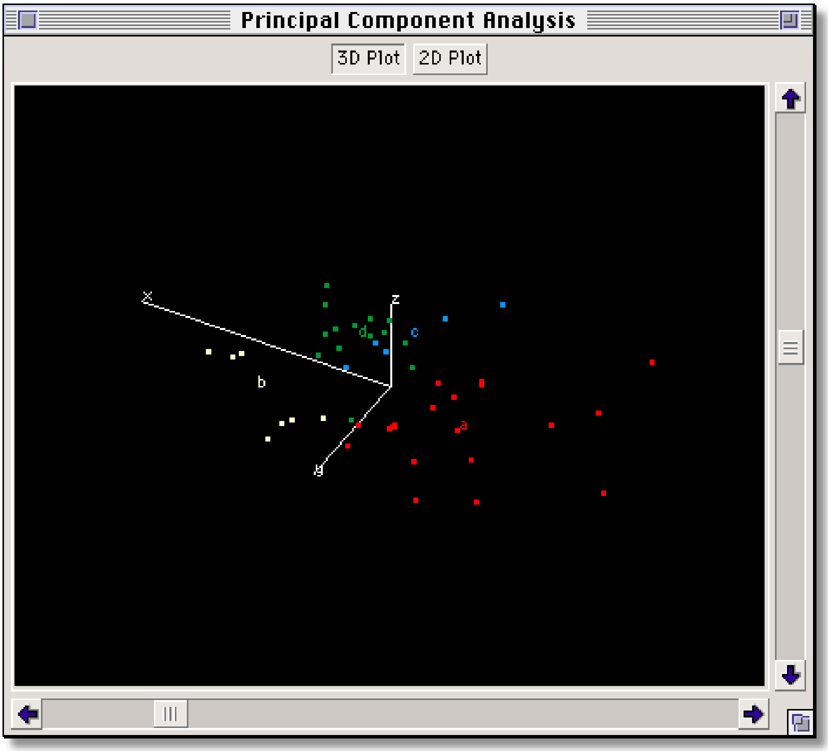


Fig.7-44. The **3-D PCA Display** form.

Each cluster is represented with its own color. Each cluster's centroid is designated with a letter. Each point in the display is a Sample in your population. Up to 18 clusters can be defined for a given population. In the above example, 4 clusters have been defined.

Clicking on the scroll bars to the right and bottom of the screen will rotate the display axes continuously through 3 dimensions, to allow you to better visualize distinctions in the clusters.

If you are interested in a particular point of data, click on it. The **Sample Detail** list for that point will open (see Section 7.4) displaying any and all Category/Attribute information for that Sample, and allowing you to access the **Samples** list and open a Sample Image window to view the lane directly.

Sample Report

The **Sample Report** displays the samples in your population, sorted in order of decreasing similarity from the Reference Sample for Similarity-searched populations, and alphabetically for Band Set-searched populations. For more information on the **Sample Report**, see Section 9.1.c

Band Frequencies

Band Frequency displays the weighted and unweighted band frequency data for each of the Band Types in your population. For more information on the **Band Frequency** report, see Section 9.1.c.

Similarity Matrix

The **Similarity Matrix** contains all of the similarity values relating any two samples in the population. If there are N samples in the population, then the similarity matrix is an N by N matrix that is computed using one of the similarity calculation methods listed above. The matrix has the following properties:

The diagonal elements always have values of 100. This is because a sample is always 100% similar to itself.

The matrix is symmetric($\mathbf{M}_{ij} = \mathbf{M}_{ji}$).

For more information on the **Similarity Matrix**, see Section 9.1.c

8. Other Advanced Analyses

Diversity Database contains several other types of analysis tools that can be performed on 1-D gels. These include quantity calibration, VNTR analysis, differential display, and matched band reporting. Each is explained below:

8.1. Calculating Quantity Standards

Diversity Database includes functions that allow you to create a calibration curve for determining absolute band quantitation.

Band quantitation can be calculated from O.D. (as described in previous sections). The quantitation derived from these calculations is given in units of O.D. x mm (for bracket quantitation) and O.D. x mm^2 (for contours). These values are useful for comparing bands to one another. For example, you can determine that there is twice as much of Band X in Lane 4 as there is in Lane 5. This does not, however, tell you how many micrograms there are of Band X in Lane 4.

The **Quantity Standards** function lets you calculate the absolute quantitation of bands on your gel and save the calibration curve data in a file that can be applied to other gels. To determine a calibration curve, the quantitation of at least two bands must be known. The greater the number of known bands and the wider the range of their values, the more accurate the calibration curve will be.

8.1.a. The Calibration Data Dialog Box

Select **Quantity Standards** and a pop-up box will appear, asking you to create a new curve or load a previously defined calibration curve (if any have been saved).

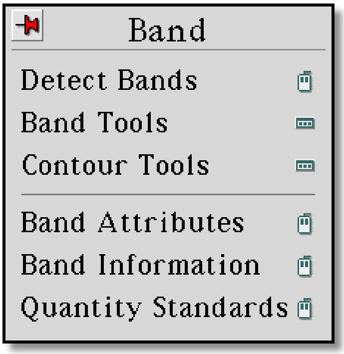


Fig.8-1. **Band** menu.

If you choose to load an old curve, the **Quantity Standards** form containing saved calibration curve values will be displayed. Select the set of values that are appropriate to your gel.

If you create a new curve, a blank dialog box will be displayed instead:

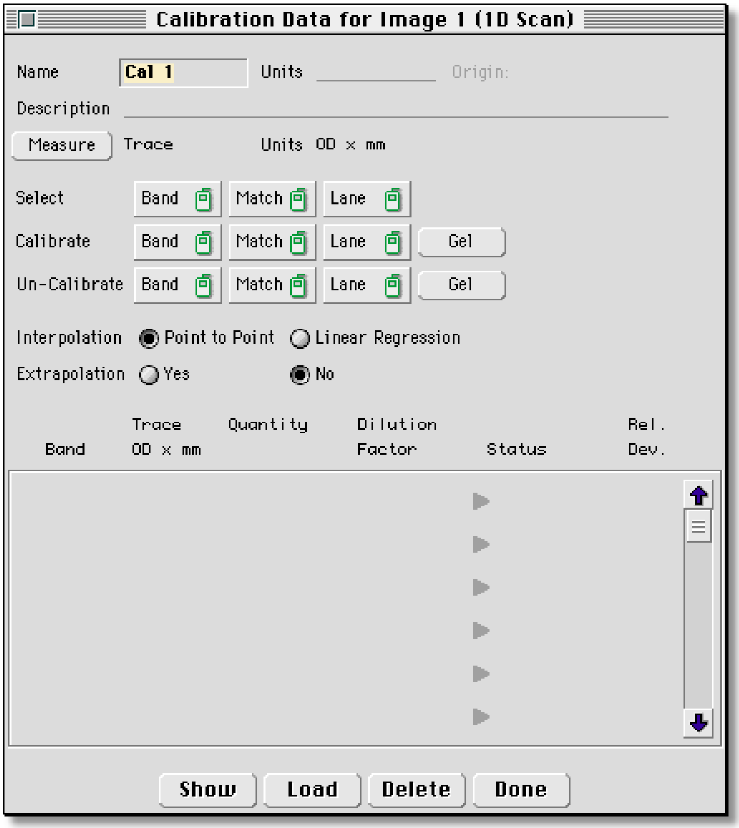


Fig.8-2. **Quantity Standards** dialog box.

At the top of the dialog box, enter a name for the calibration file and specify the quantitation value units (e.g., µg).

You can enter a longer name or extra descriptive information next to the Description prompt.

The calibration curve is generated by plotting the known values you provide against O.D.-based quantitation. There are several O.D.-derived values that can be used for the Y-axis of the plot. Press the button labelled Measure to display a list from which to make your choice.

Selecting Bands with Known Quantitation

The next section of the dialog box includes several mouse-assignable functions. These are used to specify which bands on the gel are to be used to generate the calibration curve as well to which bands the calibration curve should be applied.

If the standard bands are located in multiple lanes and match groups, you will probably want to select them one at a time. To do so, assign the **Band** function (next to the Select prompt) to a mouse button. Point at each of the bands whose quantitation you know and click the mouse.

If all of the standard bands are part of one match group, assign the **Match** function (next to the Select prompt) to a mouse button. Point at the match containing the standard bands and click.

If all of the standard bands were run in one lane, assign the **Lane** function (next to the Select prompt) to the mouse. Move the cursor so that it points to the lane containing the standard bands and click the mouse.

When you use any of these functions to select bands, the bands change color, indicating that they have been selected.

In the lower part of the **Calibration Data** dialog box, the lane and band numbers of the selected bands appear under the Band header. The values determined from their O.D.'s are also displayed.

Entering Known Quantity Values

After selecting standard bands, enter their quantity values in the text box on the line beneath the Quantity header.

Place the cursor on the line so that it is highlighted and enter any known values.

When you move the cursor off the line, the text below the Status header changes from Unknown to Known.

After three values have been entered, relative deviations are automatically calculated and displayed under the Rel. Dev. header. These values are based on the difference between the known value that you entered and the values that would have been calculated from this calibration curve.

If the deviation value is too high, you may want to exclude some bands from the calibration curve. Click on the arrow button next to the status of the problem band. A pop-up box offers several different status options from which to choose. To remove a band from the calibration data file and omit it from calibration curve calculations, click on the Remove option. All the information regarding that band will be deleted from the **Calibration Data** file.

Alternatively, you can select Outlier from the pop-up list. This retains the information about the band in the calibration file but does not include it when calculating the calibration curve.

Applying the Calibration Curve to Bands on the Gel

Once you have selected and entered values for the known bands, you are ready to choose the band(s) whose quantitation you want to calculate with this calibration curve. This is done using the mouse-assignable functions found next to the Calibrate prompt.

To calculate quantitation for individual bands, assign **Band** to a mouse button. Click on the band of interest.

If you want to calculate quantitation for all the bands in a match group, assign **Match** to a mouse button. Point the cursor at any of the bands in the Match Group and click the mouse.

For calculating the quantitation of all the bands in a lane, use the **Lane** function. Point the cursor at a lane of interest and click the mouse.

To calculate the quantitation of all the bands on a gel, click on the Gel button next to the Calibrate prompt.

Bands to which a calibration curve is applied are highlighted after they are selected.

To “un-do” the calibration of a band, match, lane or gel, use the corresponding command next to the Un-Calibrate prompt.

8.1.b. Generating Standard Bands via a Dilution Series

One way to get a set of bands of known quantity is by starting with a stock solution and making several dilutions from it. A different dilution can be loaded onto each lane of a gel, resulting in a group of bands whose quantitation can be calculated from the known quantity used to make up the stock solution and the dilution factor.

Diversity Database will perform these calculations for you if you provide it with the known quantity and the dilution factor (e.g., for a dilution factor of 1 to 10, type **`1/10`** or **`0.1`**).

In the **Calibration Data** dialog box, next to the band that came from the undiluted stock solution, enter the known quantity.

Beneath the Dilution Factor header, type: **`Stock`**

Next to each of the other bands in the dialog box, enter the dilution factor on the line below the Dilution Factor header.

The quantity of each band will be automatically calculated.

8.1.c. Other Calibration Curve Specifications

In the **Calibration Data** dialog box, select the method with which the calibration curve should be calculated. Next to the Interpolation prompt, choose either Point to Point or Linear Regression.

Point to Point produces a curve in which each point is connected to the next, regardless of the shape of the resulting curve.

Linear Regression (using the method of Least Squares) produces a curve which is the "best fit" of the values you provided.

Also, indicate whether or not the curve should be extrapolated. If you choose not to extrapolate, the highest and lowest values you entered will be the ends of the curve and you will be unable to calculate estimated quantitation for any bands that lie outside that range. If you want to extrapolate, you can calculate a quantitation value for any band on the gel. But bands whose quantitation are calculated from the extrapolated region of the calcurve may be less accurate.

8.1.d. Displaying the Calibration Curve

To display the calibration curve, click on the Show button at the bottom of the **Calibration Data** dialog box. A graph of known quantity vs. measured density is displayed in a separate window. Points used in calculating the curve are shown enclosed in circles. Points that you declare Outliers are enclosed in squares.

To change the status of a point on the graph, point the cursor at it and click the mouse. The status will toggle back and forth between Known and Outlier.

To see the legend for the different types of points on the graph, click on the Titles button.

To print a hard copy of this graph, click on the Print button located under the graph.

To dismiss the window displaying the graph, click on the Done button.

8.1.e. Quantitating Bands Using “Imported” Calibration Curves

A calibration curve can be applied not only to the bands on the gel from which it was created but also to bands on other gels.

Make sure that both the gel you want to quantitate and the gel from which the calibration curve was made are currently loaded.

Load the gel you want to quantitate and select **Quantity Standards**.

When the dialog box is displayed, click on the Load button.

From the list displayed, choose the calibration curve you want to use to quantitate bands on the gel.

When you click on your selection, the values associated with it will be displayed in the **Calibration Data** dialog box. Under the Band header will be stated “Import” reminding you that these bands are not found in the gel currently displayed.

If the quantity of one or more bands on the gel is known, you may want to use this information to double-check that the imported calibration curve accurately quantitates the bands on this gel.

To check these known bands against the calibration curve, go to the Select prompt and assign **Band** to a mouse button.

Click on the known band. Its lane number, band number, and density will be displayed in the **Calibration Data** dialog box.

When you enter the band's value, its status changes to Check. This indicates that it is a check point and is not included in the calculation of the calibration curve.

If you display the graph by clicking on the Show button, these check points are enclosed in diamonds. These check points should fall on or very near the calibration curve. If that is not the case, we recommend that you do not use the imported standards but re-run the gel with its own standards, since the selected calibration curve will probably not give you accurate quantitation data for this gel.

8.2. Variable Number Tandem Repeats

If your experiments involve the use of microsatellites, VNTR's or other repeated elements, you may want to calculate the number of times the repeated element occurs in a band in order to distinguish between various alleles. **Diversity Database** provides a tool for performing this calculation.

8.2.a. Calculating VNTR's

All the commands necessary for calculating VNTR's from your gel can be found in the **Match** menu.

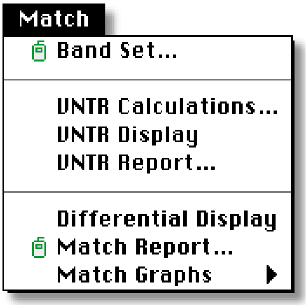


Fig.8-3. **Match** menu.

The mouse-assignable function **Calculate VNTR** displays a dialog box in which you enter information about the repeated element.

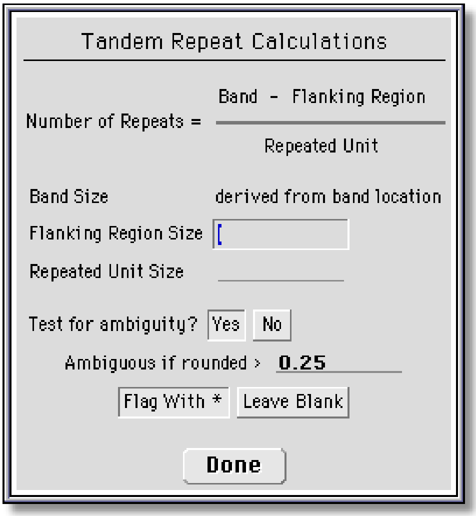


Fig.8-4. **Tandem Repeat Calculations** dialog box.

At the top of the dialog box is the equation used to calculate the number of times an element is repeated in a band.

The band size is determined by the position of the band on the gel.

In the text box next to the Flanking Region Size prompt, enter the size (in base pairs) of the part of the fragment that does not include any repeated elements. This would include primer length and the length of any sequences that fall between the end of the primer region and the beginning of the stretch of repeats.

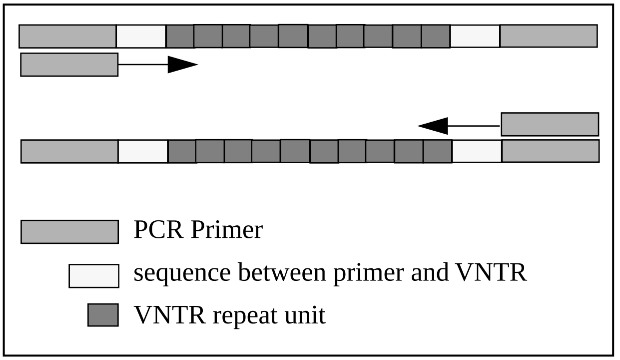


Fig.8-5. Diagram of components of DNA fragment.

Next to the Repeated Unit Size prompt, enter the size (in base pairs) of the repeated element (e.g., if you are working with (CA) repeats, enter "**2**").

The number of repeats calculated in this operation may not be a whole number, due to the limitations of exact band size determination. Since the number of times an element is repeated must be a whole number, the calculated value is rounded to the nearest whole number. In the **Tandem Repeat Calculations** dialog box you have the option to "flag" those numbers that deviated from the nearest whole number. This flag serves to draw your attention to those values that may warrant further review.

Click on the Yes or No button next to the Test for ambiguity? prompt to indicate whether or not you wish to test for ambiguity.

Next, specify what constitutes ambiguity. How much should the calculated value deviate from the nearest whole number for it to be flagged? Enter this value in the box next to the Ambiguous if rounded prompt.

Finally, specify whether the tandem repeat numbers that meet the ambiguity criteria should be marked with an asterisk (*) or whether they should be left unmarked.

When you have finished entering all the information in the dialog box, click on the Done button at the bottom. The tandem repeat calculation is then performed on all the bands on the gel.

To display the tandem repeat numbers, click **Show VNTR**.

8.2.b. Printing VNTR Data

The **VNTR Report** displays a table of the number of times a repeat unit occurs in a band. For each band, the table displays the band's molecular weight, the Raw Repeat # and the Rounded Repeat # and the band's name, if it is available. For more information on the **Match Report**, see Section 9.1.d.

8.3. Displaying Match Data

Once the quantitation of the bands has been determined, you will probably want to compare the quantitation of bands across the gel. To do so, bands need to be matched together.

8.3.a. Match Report

Match Report generates a report on one match (Band Type) that the user selects by clicking on it with the command assigned to the mouse. For more information on the **Match Report**, see Section 9.1.b.

8.3.b. Histograms of Matched Bands

Use the functions in the **Match Graphs** submenu to display histograms of matched bands.

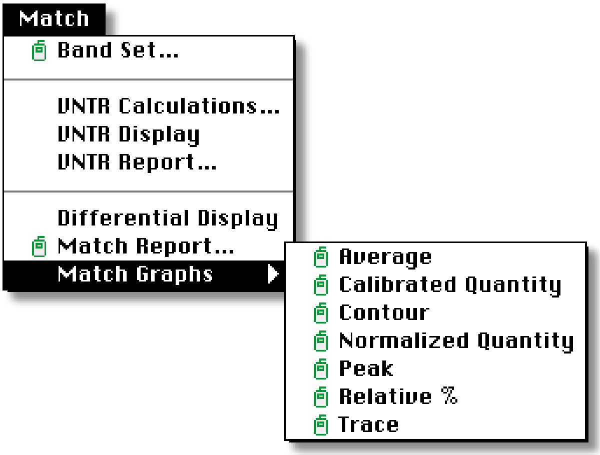


Fig.8-6. **Match Graphs** menu.

Assign a **Match Graphs** command to the mouse and click on a match.

The match will be displayed, as will the histogram that you chose.

Each bar of the histogram will be labelled on the X-axis with the lane number, and (where space permits) the band number. The Y-axis is labelled with quantitative values.

Select either **Average**, **Calibrated Quantity**, **Contour**, **Normalized Quantity**, **Peak**, **Relative%**, or **Trace** for the Y-axis of the plot.

Average displays a histogram of the average densities of the bands in a match group.

Calibrated Quantity displays a histogram of the absolute quantitation of matched bands based on a calibration curve (for more information on calculating absolute band quantitation, refer to Section 7.5).

Contour displays a histogram of the density of each contoured band in a matched group. This function is only valid for contoured bands.

Normalized Quantity displays a histogram of the normalized quantity of the bands in a match group.

Peak displays a histogram of the peak densities of the bands in a match group.

Relative % displays a histogram in which each bar represents the percentage that the bands in your match group represent out of either the total density of their respective lanes, or the sum density of the bands in their lanes (see Section 2.3.b *Display Preferences*, for more information on Relative % calculations).

Trace displays a histogram of the integrated densities of the bracketed bands in a match group.

The histogram display reserves space on the left side to label the Y-axis and show its units. If the bands in the match group span the entire window from the left edge to the right, the histogram will not include bars for the bands in the left most lanes because of the space required for labeling the axis.

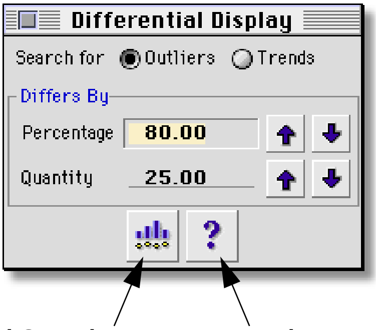
To avoid this problem, decrease the magnification of the image using **Zoom Out** until there is blank space between the left side of the window and the first band of the match group.

Re-display the histogram. If one or more bars are still not included, continue decreasing the magnification until you can see a bar lining up with each lane that contains a band in the match group in which you are interested.

8.4. Differential Display

Differential Display is a technique using mRNA and PCR amplification to study sequence expression. Differential display gels are analyzed with Band Sets and Band Types, just as any other gel in **Diversity Database** would be, with one important distinction. Each gel must be normalized to a specific Band Type that is present in all of your differential display samples. Once you've chosen a normalizing Band Type, open the experimental **Band Set** form for that gel and click on the Normalization button. A list of all the Band Types present in that set will appear. Select a type from the list, and **Diversity Database** will automatically apply the normalization to all gels in the database containing that band set.

You can now access the **Differential Display** dialog box:



Normalized Quantity **Help**

Fig.8-7. The **Differential Display** dialog box.

There are two types of searches which you can perform on your gel, an Outlier search or a Trend search. Both are determined by the Percentage and Quantity values that you specify in the Differs By field.

The Outlier mode searches each band type for bands with normalized quantity that differ from the mean normalized quantity for its Band Type. An outlier is defined as a band with normalized quantity (nq) that satisfies on of the following criteria:

ratio = 1.0 + (percentage / 100)

1: nq > ratio * mean and
nq > mean + quantity

2: nq < (1 / ratio) * mean and
nq < mean - difference

A Trend search looks for increasing or decreasing expression, represented by trends in normalized quantity across a Band Type. In Trend mode, a linear regression of (nq) vs. lane # is computed for each Band Type. The left most and right most lanes of a Band Set are determined. The normalized quantities for these lanes are calculated from the regression model. If:

abs(left most_nq - right most_nq) $\geq$ difference and
Max(left most_nq, right most_nq) /
Min(left most_nq, right most_nq) $\geq$ ratio

then the Band Type is flagged as a trend.

Bands identified either as outliers or as belonging to a trend are displayed in white. If a lane does not have a band assigned to a band type that is flagged as an outlier or a trend, an empty box will be drawn at the expected location of the band.

The **Normalized Quantity** command can be used to display a histogram of normalized quantities and the mean for any Band Type that you select. The **Help** command is available to users who wish an explanation of the differential display commands while running the software.

9. Data Output

9.1. Reports

9.1.a. Scan Report

Scan Report allows you to create a single-page report of a scanned gel. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of a gel for documentation purposes.

The report includes the following information:

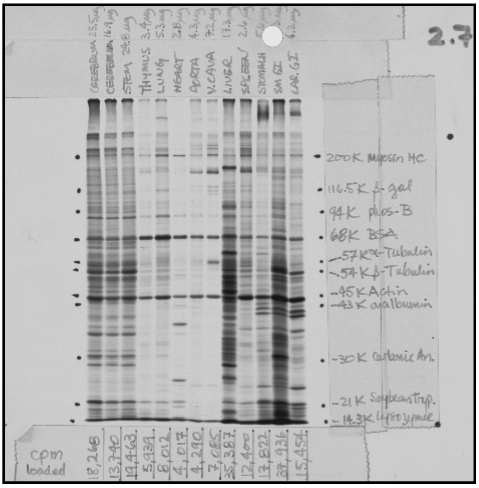
- An image of the scan.
- The report date.
- The scan title and a brief description.
- Its directory location.
- The scan date.
- The scanner type.
- The scan area and number of pixels.
- Pixel size.
- The O.D. range, scan color, and memory size.
- Gel background information.
- Relevant lane and band information.

To produce a scan report of a particular gel, select the **Scan Report** command from the **File → Print** submenu and apply it to the gel you wish to print.

An example of a **Scan Report** document can be found on the next page.

Note: TIFF images read by **The Discovery Series** software do not contain all the tagged information that would normally be included in an image file (for example, scanner type, scan date, scanning color, etc.). For this reason, **Scan Report** may report as "Unknown" information not included in imported TIFF files.

Scan Report **Report date: March 28, 1995**



protein 1D Scan

Description **for tutorial guide whole gel image**
Directory **C:\PDI\user.idb\1D_Demo**

Scan date	**February 15, 1994**	Pixel Size(um)	X:**63.5**, Y:**63.5**
Scanner	**Desktop**	Data Range	**1.00**
Scan area	X:**41.9**, Y:**48.2**	Scan Color	**unknown**
Scan pixels	X:**660**, Y:**763**	Memory Size	**493.06 Kb**

Gel Background
Method **None**

Lane Information

Lane	Bkgd	Radius	Width	Sens.	Min.Den.	Filter	Shldr.	Size
1-8	None	--	2.0	10.0	0.1	0.0	1.0	5

Fig.9-1. Example of the **Scan Report** format.

9.1.b. Lane & Match Reports

There are four distinct lane and match reports: **Lane Report** (in the **Lane** menu), **All Lanes Report** (in the **Gel** menu), **Match Report** (in the **Match** menu), and **All Matches Report** (also in the **Gel** menu).

Lane Report generates a report on one lane that the user selects by clicking on it with the command mouse-assigned.

All Lanes Report generates a report on all the lanes in the current gel.

Match Report generates a report on one match (Band Type) that the user selects by clicking on it with the command mouse-assigned.

All Matches Report generates a report on all the matches (Band Types) in the one of the Band Sets assigned to your gel (a pop-up list will allow you to select the Band Set).

Each of these commands uses the same pop-up box to specify the type of data to be printed in a given report.

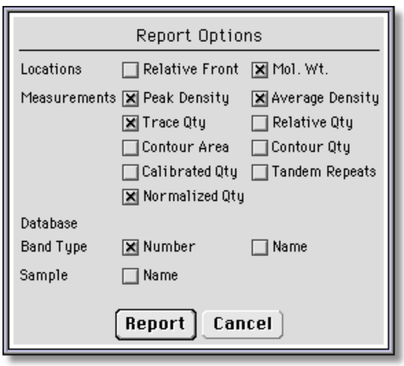
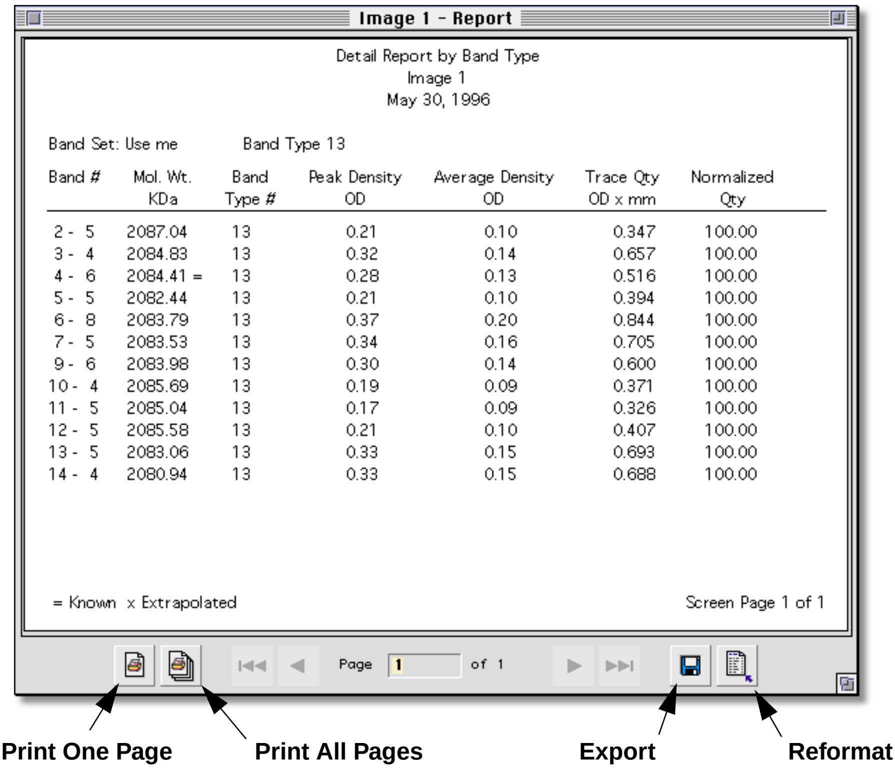


Fig.9-2. **Report Options** form.

Clicking on the Report Option boxes allows you to customize the data that appears on your report.

Note: If you change your mind about the data that is to be included in a report, the form can quickly be reformatted to include new data or exclude current data, so it is not crucial that you select the exact data that you expect to appear in the final version of the report.

Clicking on the Report button will dismiss the form and generate the following report:



Detail Report by Band Type
Image 1
May 30, 1996

Band Set: Use me   Band Type 13

Band #	Mol. Wt. KDa	Band Type #	Peak Density OD	Average Density OD	Trace Qty OD x mm	Normalized Qty
2 - 5	2087.04	13	0.21	0.10	0.347	100.00
3 - 4	2084.83	13	0.32	0.14	0.657	100.00
4 - 6	2084.41 =	13	0.28	0.13	0.516	100.00
5 - 5	2082.44	13	0.21	0.10	0.394	100.00
6 - 8	2083.79	13	0.37	0.20	0.844	100.00
7 - 5	2083.53	13	0.34	0.16	0.705	100.00
9 - 6	2083.98	13	0.30	0.14	0.600	100.00
10 - 4	2085.69	13	0.19	0.09	0.371	100.00
11 - 5	2085.04	13	0.17	0.09	0.326	100.00
12 - 5	2085.58	13	0.21	0.10	0.407	100.00
13 - 5	2083.06	13	0.33	0.15	0.693	100.00
14 - 4	2080.94	13	0.33	0.15	0.688	100.00

= Known x Extrapolated   Screen Page 1 of 1

Fig.9-3. Report viewer dialog box.

All four reports share this same format of the report viewer. In fact, all reports with the exception of the **Scan Report** use some version of the basic report viewer pictured above. Features of the report viewer include:

Print One Page & Print All Pages

Clicking on either of these icons will call the following print form:

PC Version

Mac Version

Fig.9-4. Print Report form.

Print One Page sends only the current page to the printer; **Print All Pages** sends all the pages in the report to the printer. When you are happy with the parameters specified on the **Print Report** form, click on the Go button to send the report.

Export

Clicking on this icon will call the following export form:

PC Version

Mac Version

Fig.9-5. Export Report form.

Specify the field separators that you wish to use to separate the data in the text export file, depending on the database package that you intend to export to (commas for Lotus 1-2-3™, tabs for Excel™). click on the Go button to save the exported file to disk.

Reformat

Clicking on this icon will reopen the **Report Options** form and allow you to change the contents of your report. Click on the Report button to re-generate your lane or match report.

9.1.c. Population Reports

The following reports are all available from the **Populations** form (see Section 7.7.b).

Image Report

The **Image Report** displays a series of lane images of your population. When you click on the **Image Report** icon, you will be presented with the following formatting options:

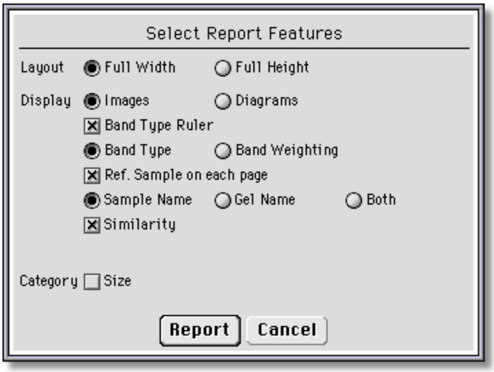


Fig.9-6. **Image Report** options form.

Clicking on the Report Option boxes allows you to customize the data that appears on your report.

Clicking on the Report button will dismiss the form and generate the following report:

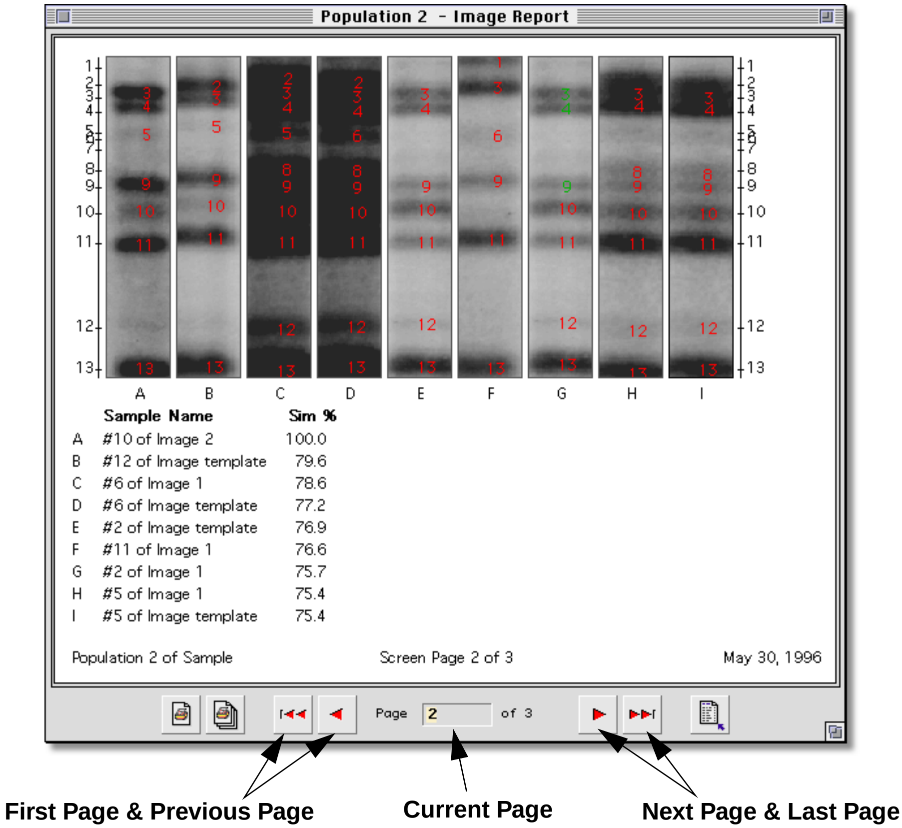


Fig.9-7. **Image Report** viewer.

The **Image Report** viewer shares the same printing and reformatting icons as the report viewer in Figure 9-3, as well as the same page scrolling icons. The icons allow you to page through a report, or you can skip directly to a specific page of a report by typing into the text field next to the Page prompt.

The **Image Report** viewer has a special feature that allow you to click on the image of a gel lane and open that lane's gel directly. This is a useful feature if you decide to edit a lane based on what you see in the **Image Report** (an unassigned, or incorrect Band Type, etc.). Once opened, single-lane windows can be expanded, saved, etc. They are identical to normal Gel windows. Multiple-lane samples, however, call special Sample windows, which are uneditable, and unsavable, but do display all the lanes from your sample.

Clicking on the **Print One Page** or **Print All Pages** icons will call the following print form:

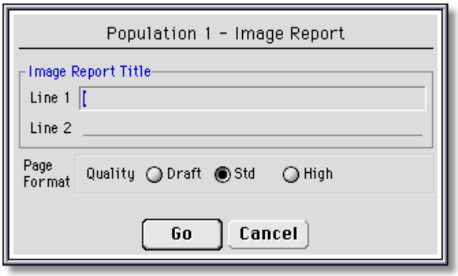


Fig.9-8. **Image Report** print form.

The **Image Report** print form is similar to other report print form except that also contains space for you to enter an title for the report. Click on the Go button to send the report to the printer.

Population Reports

The **Population Report** displays all the members of your population. When you click on the **Population Report** icon, you will be presented with the following formatting options:

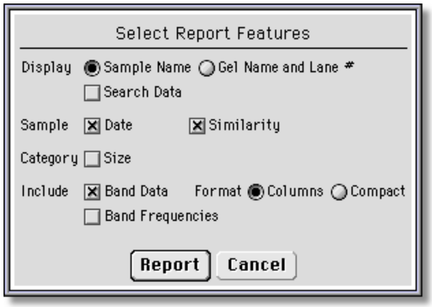
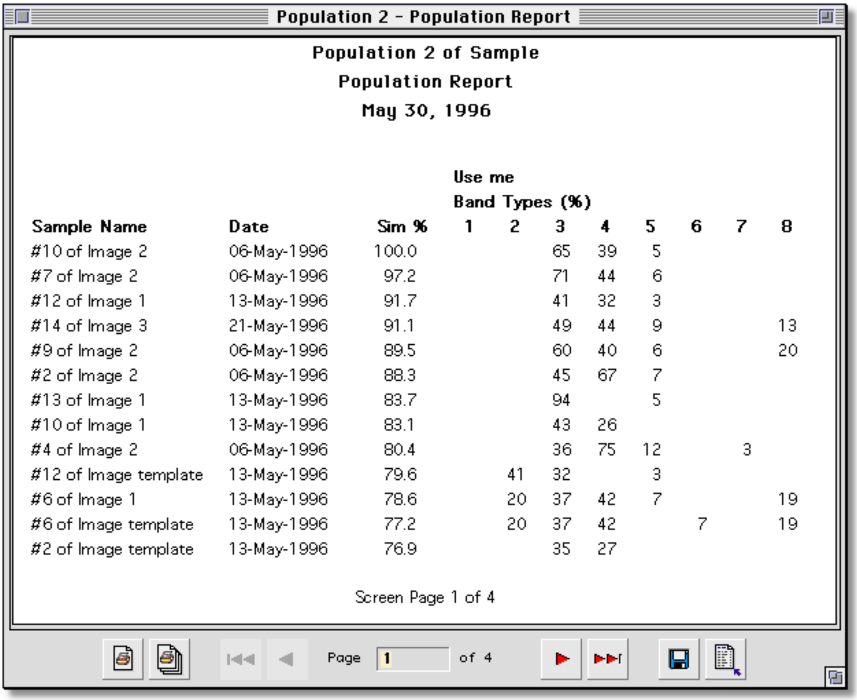


Fig.9-9. **Population Report** options form.

Clicking on the Report Option boxes allows you to customize the data that appears on your report.

Clicking on the Report button will dismiss the form and generate the following report:



Population 2 of Sample
Population Report
May 30, 1996

Use me
Band Types (%)

Sample Name	Date	Sim %	1	2	3	4	5	6	7	8
#10 of Image 2	06-May-1996	100.0			65	39	5			
#7 of Image 2	06-May-1996	97.2			71	44	6			
#12 of Image 1	13-May-1996	91.7			41	32	3			
#14 of Image 3	21-May-1996	91.1			49	44	9			13
#9 of Image 2	06-May-1996	89.5			60	40	6			20
#2 of Image 2	06-May-1996	88.3			45	67	7			
#13 of Image 1	13-May-1996	83.7			94		5			
#10 of Image 1	13-May-1996	83.1			43	26				
#4 of Image 2	06-May-1996	80.4			36	75	12		3	
#12 of Image template	13-May-1996	79.6		41	32		3			
#6 of Image 1	13-May-1996	78.6		20	37	42	7			19
#6 of Image template	13-May-1996	77.2		20	37	42		7		19
#2 of Image template	13-May-1996	76.9			35	27				

Screen Page 1 of 4

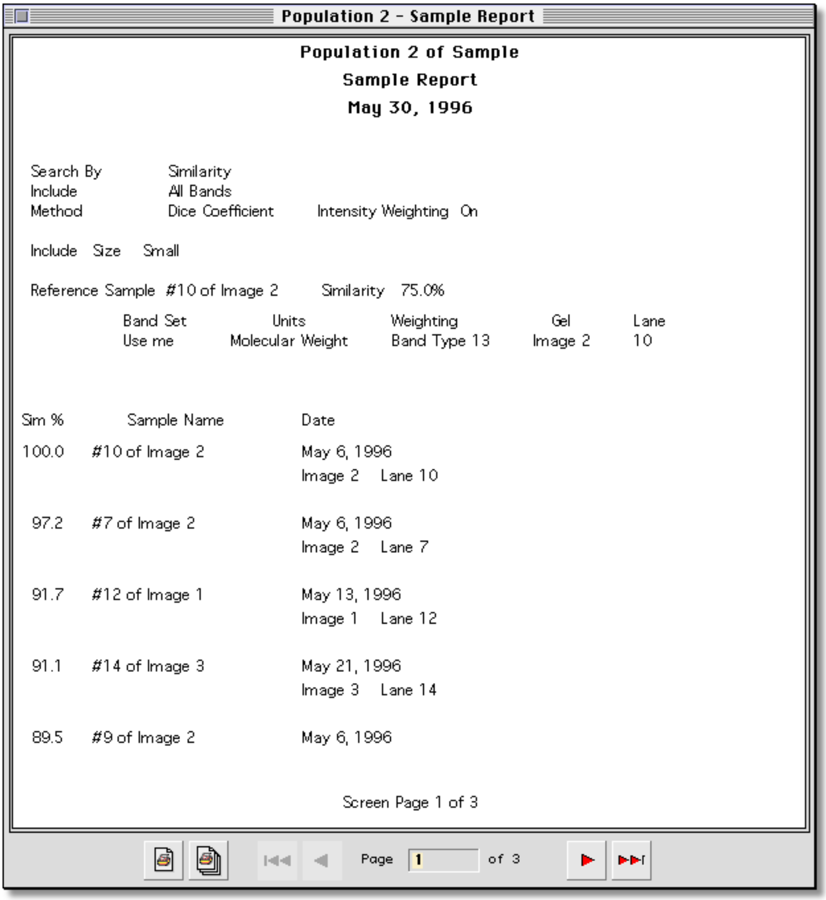
Fig.9-10. **Population Report** viewer.

The **Population Report** viewer shares the same control icons as do the previous report viewers.

Note: The remaining population reports all share print and export forms that are functionally identical. From this point on, print and export form will be mentioned only if they differ meaningfully from those forms previously discussed.

Sample Report

The **Sample Report** displays the samples in your population. When you click on the **Sample Report** icon, the following form will be generated:


Population 2 - Sample Report

Population 2 of Sample
Sample Report
May 30, 1996

Search By	Similarity		
Include	All Bands		
Method	Dice Coefficient	Intensity Weighting	On

Include Size Small

Reference Sample #10 of Image 2 Similarity 75.0%

Band Set	Units	Weighting	Gel	Lane
Use me	Molecular Weight	Band Type 13	Image 2	10

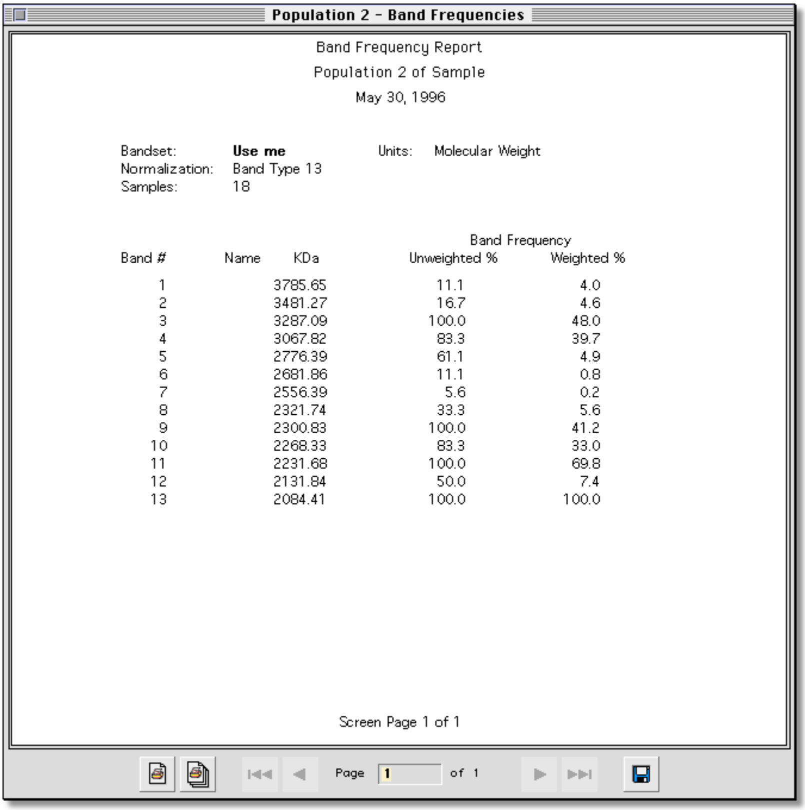
Sim %	Sample Name	Date
100.0	#10 of Image 2	May 6, 1996 Image 2 Lane 10
97.2	#7 of Image 2	May 6, 1996 Image 2 Lane 7
91.7	#12 of Image 1	May 13, 1996 Image 1 Lane 12
91.1	#14 of Image 3	May 21, 1996 Image 3 Lane 14
89.5	#9 of Image 2	May 6, 1996

Screen Page 1 of 3

Fig.9-11. **Sample Report** viewer.

Band Frequency Report

Band Frequency displays the weighted and unweighted band frequency data for each of the Band Types in your population. When you click on the **Band Frequency** icon, the following form will be generated:


Population 2 - Band Frequencies

Band Frequency Report

Population 2 of Sample

May 30, 1996

Bandset: **Use me**
Units: Molecular Weight
Normalization: Band Type 13
Samples: 18

Band #	Name	KDa	Band Frequency Unweighted %	Weighted %
1		3785.65	11.1	4.0
2		3481.27	16.7	4.6
3		3287.09	100.0	48.0
4		3067.82	83.3	39.7
5		2776.39	61.1	4.9
6		2681.86	11.1	0.8
7		2556.39	5.6	0.2
8		2321.74	33.3	5.6
9		2300.83	100.0	41.2
10		2268.33	83.3	33.0
11		2231.68	100.0	69.8
12		2131.84	50.0	7.4
13		2084.41	100.0	100.0

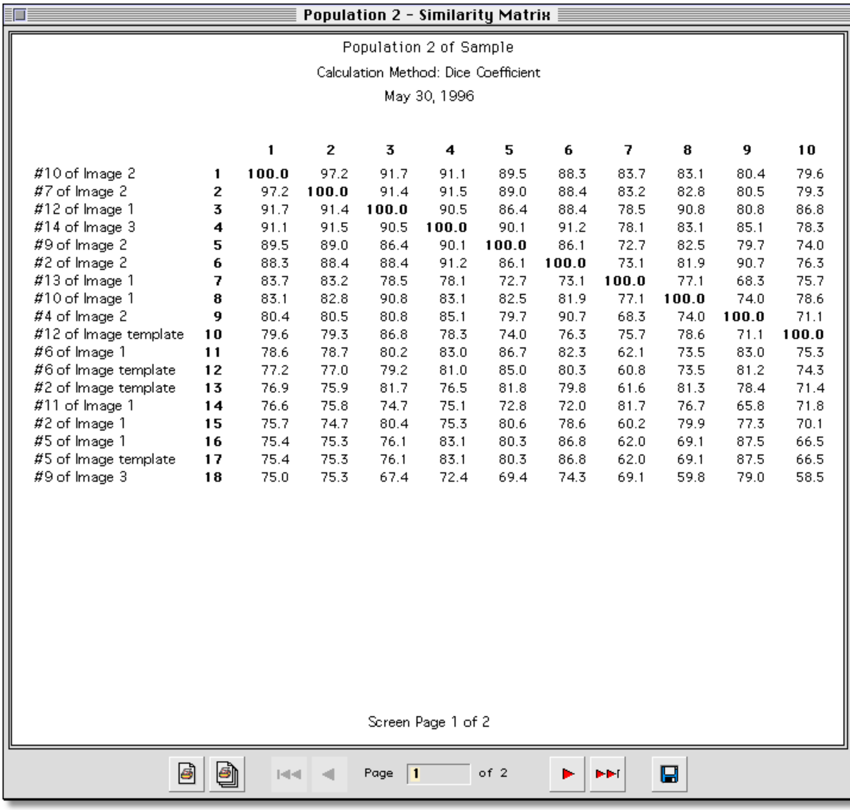
Screen Page 1 of 1

Page 1 of 1

Fig.9-12. **Band Frequency** viewer.

Similarity Matrix

The **Similarity Matrix** contains all of the similarity values relating any two samples in the population. When you click on the **Band Frequency** icon, the following form will be generated:



Population 2 - Similarity Matrix

Population 2 of Sample

Calculation Method: Dice Coefficient

May 30, 1996

		1	2	3	4	5	6	7	8	9	10
#10 of Image 2	**1**	**100.0**	97.2	91.7	91.1	89.5	88.3	83.7	83.1	80.4	79.6
#7 of Image 2	**2**	97.2	**100.0**	91.4	91.5	89.0	88.4	83.2	82.8	80.5	79.3
#12 of Image 1	**3**	91.7	91.4	**100.0**	90.5	86.4	88.4	78.5	90.8	80.8	86.8
#14 of Image 3	**4**	91.1	91.5	90.5	**100.0**	90.1	91.2	78.1	83.1	85.1	78.3
#9 of Image 2	**5**	89.5	89.0	86.4	90.1	**100.0**	86.1	72.7	82.5	79.7	74.0
#2 of Image 2	**6**	88.3	88.4	88.4	91.2	86.1	**100.0**	73.1	81.9	90.7	76.3
#13 of Image 1	**7**	83.7	83.2	78.5	78.1	72.7	73.1	**100.0**	77.1	68.3	75.7
#10 of Image 1	**8**	83.1	82.8	90.8	83.1	82.5	81.9	77.1	**100.0**	74.0	78.6
#4 of Image 2	**9**	80.4	80.5	80.8	85.1	79.7	90.7	68.3	74.0	**100.0**	71.1
#12 of Image template	**10**	79.6	79.3	86.8	78.3	74.0	76.3	75.7	78.6	71.1	**100.0**
#6 of Image 1	**11**	78.6	78.7	80.2	83.0	86.7	82.3	62.1	73.5	83.0	75.3
#6 of Image template	**12**	77.2	77.0	79.2	81.0	85.0	80.3	60.8	73.5	81.2	74.3
#2 of Image template	**13**	76.9	75.9	81.7	76.5	81.8	79.8	61.6	81.3	78.4	71.4
#11 of Image 1	**14**	76.6	75.8	74.7	75.1	72.8	72.0	81.7	76.7	65.8	71.8
#2 of Image 1	**15**	75.7	74.7	80.4	75.3	80.6	78.6	60.2	79.9	77.3	70.1
#5 of Image 1	**16**	75.4	75.3	76.1	83.1	80.3	86.8	62.0	69.1	87.5	66.5
#5 of Image template	**17**	75.4	75.3	76.1	83.1	80.3	86.8	62.0	69.1	87.5	66.5
#9 of Image 3	**18**	75.0	75.3	67.4	72.4	69.4	74.3	69.1	59.8	79.0	58.5

Screen Page 1 of 2

Page 1 of 2

Fig.9-13. **Similarity Matrix** viewer.

9.1.d. VNTR Report

The **VNTR Report** displays a table of the number of times a repeat unit occurs in a band. For each band, the table displays the band's molecular weight, the Raw Repeat # and the Rounded Repeat # and the band's name, if it is available.

The Raw Repeat # is the value calculated using the information that you provided in the Tandem Repeat Calculation dialog box, and is likely to include fractional values. The Rounded Repeat # is the Raw Repeat # rounded to the nearest whole number.

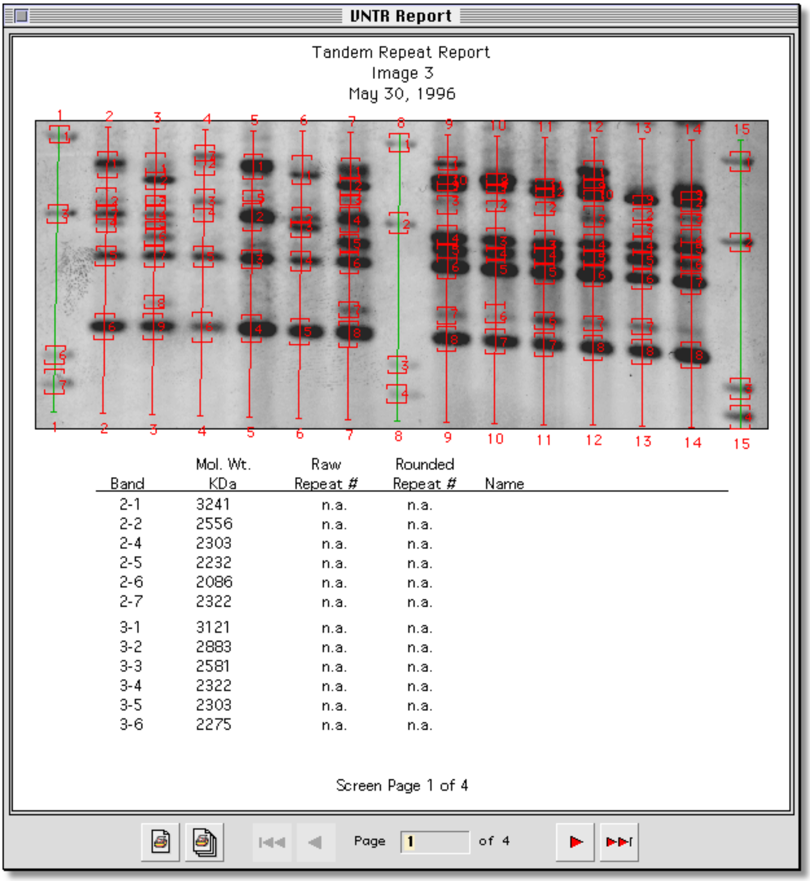
An asterisk (*) will appear to the right of some of the Rounded Repeat #'s if you asked to test for ambiguity and opted for ambiguous numbers to be flagged in the Tandem Repeat Calculations dialog box.

An asterisk appears next to all Rounded Repeat #'s that varied from the Raw Repeat # by more than the ambiguity value that you specified.

Besides the tabular data, you have the option of including the image of your gel in the VNTR report.

When you select **VNTR Report** from the menus, a pop-up box will ask if you want to display the image of the gel in the report. Click on the Yes or the No button to indicate your selection, or click on Cancel if you would prefer not to continue with the report.

The report data (with or without an image) will be displayed on-screen. Using the buttons at the bottom of the window in which the report is displayed, you can review all of the data before sending the report to the printer.



Band	Mol. Wt. KDa	Raw Repeat #	Rounded Repeat #	Name
2-1	3241	n.a.	n.a.	
2-2	2556	n.a.	n.a.	
2-4	2303	n.a.	n.a.	
2-5	2232	n.a.	n.a.	
2-6	2086	n.a.	n.a.	
2-7	2322	n.a.	n.a.	
3-1	3121	n.a.	n.a.	
3-2	2883	n.a.	n.a.	
3-3	2581	n.a.	n.a.	
3-4	2322	n.a.	n.a.	
3-5	2303	n.a.	n.a.	
3-6	2275	n.a.	n.a.	

Fig.9-14. **VNTR Report** viewer.

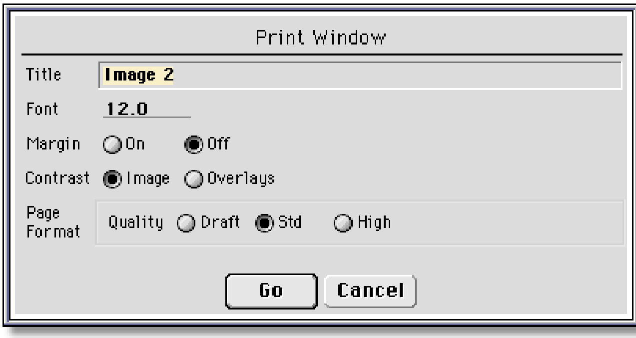
9.2. Printing Images

To print a hard copy of an image, go to the **File** → **Print** submenu and assign **Print Window** to the mouse.

Move the cursor into the window in which the image is displayed and click on the **Print Window** button.

The following dialog box will be displayed:

Mac Version



PC Version

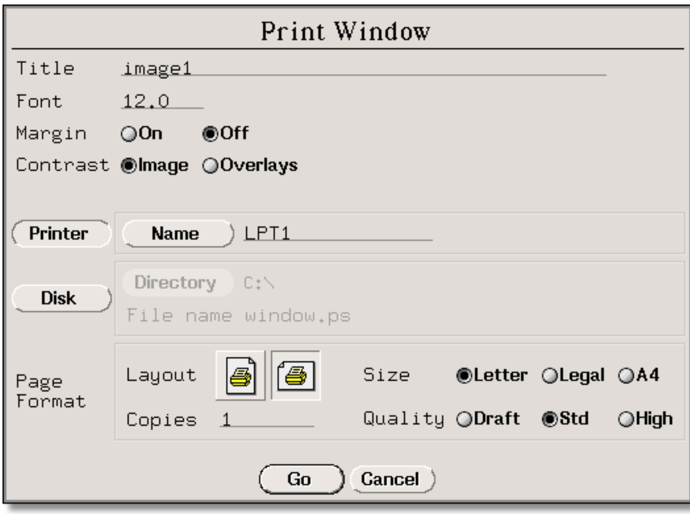


Fig.9-15. **Print Window** dialog box.

At the top of the form, enter the title that you want to appear above the image when you print. Then, specify a font size in the text box next to the prompt. You can use font sizes ranging from 6 to 64, but we recommend you use the default setting of 12.0.

Choose whether or not a margin should appear around the border of the image by clicking either the On or Off button next to the Margin prompt.

9.2.a. PC Only

If you want to display overlays on the printout of an image, click on the Favor Overlays button. This will optimize the use of the printing range so that you can read overlays on even the darkest regions of your image. If you are not going to display any overlays, click on the Favor Image button (next to the Contrast prompt). This allows the full range (black to white) to be used to print the image and will generate the clearest, sharpest image possible.

There are three possible destinations for the data: a printer, the hard disk, or a floppy disk. Click on the Printer or Disk button to select your destination. The Disk button is used for both hard and floppy disks.

Sending the Image to the Hard Disk

If you choose to send the image to the hard disk, enter a name for the file next to the File name prompt. Select the directory to which it will be sent by clicking on the Directory button.

After entering the information in the dialog box, click on the Go button to send the image to the hard disk.

Sending the Image to a Floppy Disk

To send the data to a floppy disk as a PostScript file, insert a DOS formatted floppy into the disk drive,

Access the Select Directory form through the Directory button and choose the drive in which you inserted your floppy next to the Drive prompt.

Click on the Go button to save the file to your floppy disk.

Sending the Image to a Printer

When you click on Printer, the Name button and text box will become active.

To select a printer, click on the Name button. A list of available printers will be displayed. Select one by highlighting it with the cursor and clicking the mouse. Your selection will appear in the text box next to the Name button.

Near the bottom of the Print Window dialog box are several options regarding how the page is formatted.

Select the paper orientation by clicking on the appropriate icon next to the Layout prompt.

Indicate how many copies should be printed in the text box next to the Copies prompt.

Specify the paper size by clicking on the Letter, Legal or A4 button next to the Size prompt.

9.2.b. PC and Mac

Select the printing resolution by clicking on one of the buttons next to the Quality prompt. Draft produces a printed image in less time, but the quality is lower. High produces prints with better contrast and detail at the expense of extra printing time. Std produces an image of intermediate quality.

When you are satisfied with all the print parameters, click on the Go button to send the image to the printer.

9.3. The Print Status Form (PC Only)

Any command that sends a file to your printer will call the **Print Status** form (**File** → **Print** → **Status**):

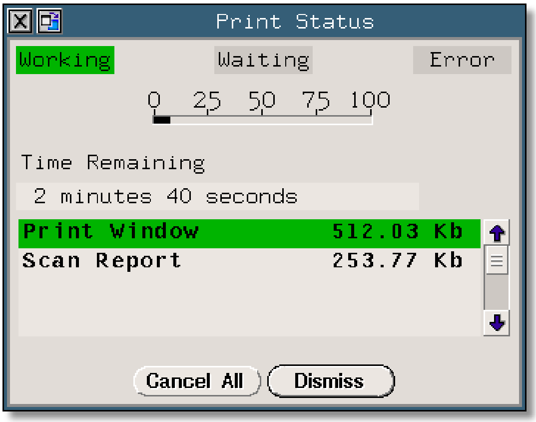


Fig.9-16. **Print Status** form.

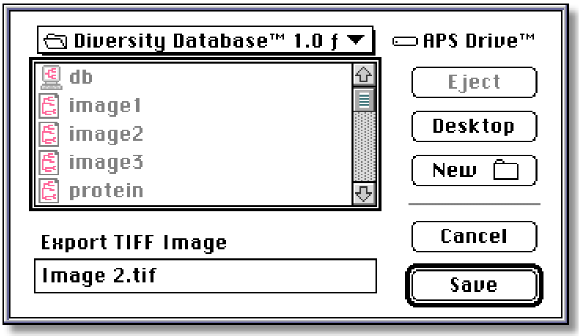
This form reports the order in which files are sent to the printer, and the status of each file as it is being printed.

9.4. Exporting

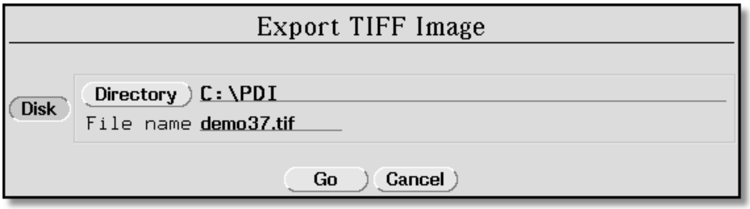
9.4.a. Exporting TIFF Images

To export a TIFF image, assign the **TIFF Image** to a mouse button (**File** → **Export** → **TIFF Image**) and click on the image that you want to export.

The following dialog box will appear:



Mac Version



PC Version

Fig.9-17. **Export TIFF Image** dialog box.

Select the directory in which you wish to save your TIFF export, name the file, and click on the Save/Go button to complete the action.

The Discovery Series applications generate 8-bit grayscale TIFF images compatible with most Macintosh and PC applications.

9.5. Thermal Printing

The **Thermal Print** option allows users of the **NightHawk** system (see Section 3.5 for more information on the **NightHawk**) to print **NightHawk** and traditional **Discovery Series** image files on a thermal printer. Thermal prints are a convenient and cost-effective way to photodocument ethidium bromide gels. Note: this function is only available to current **NightHawk** users.

Appendix A. Keyboard Functions

Key	Normal	Shift	Alt	Control
Left	Pan Left		Middle Left	
Up	Pan Up		Upper Mid.	
Right	Pan Right		Middle Right	
Down	Pan Down		Lower Mid.	
Help	Help			
F1	Entire Image	Contrast		
F2	Display In Box	Grab		
F3	View Tools	Density Tools		
F4	Clear Overlays	New Window		
F5	Show Band Type	Show Band Model		
F6	Band Attributes	Toggle Brackets		
F7	Scan	Image Tools		
F8	Gel Detail	Show Summary		
F9	Frame Lanes	Lane Tools		
F10	Band Tools	Contour Tools		
F11		View Sample		
F12	Save	Save All		
KP 1	Lower Left			
KP 2	Lower Middle			
KP 3	Lower Right			
KP 4	Middle Left			
KP 5	Middle			
KP 6	Middle Right			
KP 7	Upper Left			
KP 8	Upper Middle			
KP 9	Upper Right			

Key	Normal	Shift	Alt	Control
b				Contrast
c				Copy
d				Delete
e				Density Tools
g				Grab
h				Help
i			Get Info	Zoom In
o			Open	Zoom Out
p			Print	Previous Image
q			Quit	
r				Rename / Move
s			Save	
t				Plot Cross-Sect.
v				Version
w			Close	
y			Save All	

Appendix B. Using Discovery Series™ Files on Multiple Platforms

It is possible to transport gel data between **Discovery Series** software applications on different platforms, either over a network, or via disk. There are different protocols depending on which platform (PC or Macintosh) you are transferring from, and which you are transferring to.

B.1. Macintosh to PC

To transfer a file from a **Discovery Series** application running on a Macintosh to a **Discovery Series** application running on a PC, you need to tag the file name with the DOS suffix appropriate to the image file type (i.e., 1-D Scan or DNA Scan).

For 1-D gels (**Diversity Database, Quantity One**) use the suffix: **.1sc**

For Database files (**Diversity Database**) use the suffix: **.ddb**

For DNA Scans (**DNA Code**) use the suffix: **.dsc**

Any Mac file transferred to a PC will need to follow the general DOS naming conventions, specifically, file names should be limited to 8 characters (not including the suffix) and "." should not be used in the file name except at the end of the name proceeding the 3 character suffix.

B.2. PC to Macintosh

PC versions of **Discovery Series** files can be read directly by Macintosh applications, with no required modifications, provided that you set the PC Exchange control panel to recognize the specific file type that you are transferring.

To do so, select PC Exchange from your Macintosh **Control Panels** folder.

When the PC Exchange form appears, click on the Add... button.

The following dialog box will appear:

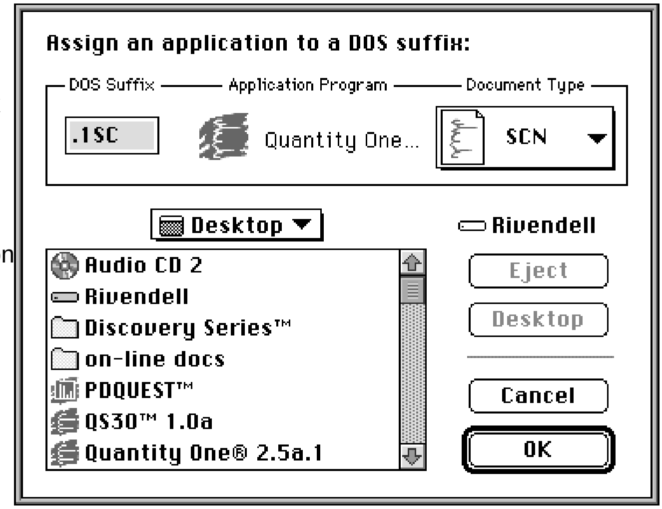


Fig.B-1. Setting a DOS suffix.

1. Type in the DOS suffix for your **Discovery Series** application under the DOS Suffix header:

For 1-D gels (**Diversity Database, Quantity One**) type: **.1sc**

For Database files (**Diversity Database**) type: **.ddb**

For DNA Scans (**DNA Code**) type: **.dsc**

2. Select your application file from the hard disk.

3. Click on the Document Type menu and select the SCN document for your application.

4. Click OK when you are done.

Your PC Exchange preferences file will be modified to remember the file suffixes that you specified. Hereafter, your Macintosh will automatically correctly interpret and load all files of the type that you specified on the PC Exchange form.

Index

A

B

C

D

E

F

G

H

I

J

K

L

M

N

O

P

Q

R

S

T

U

V

W

Z